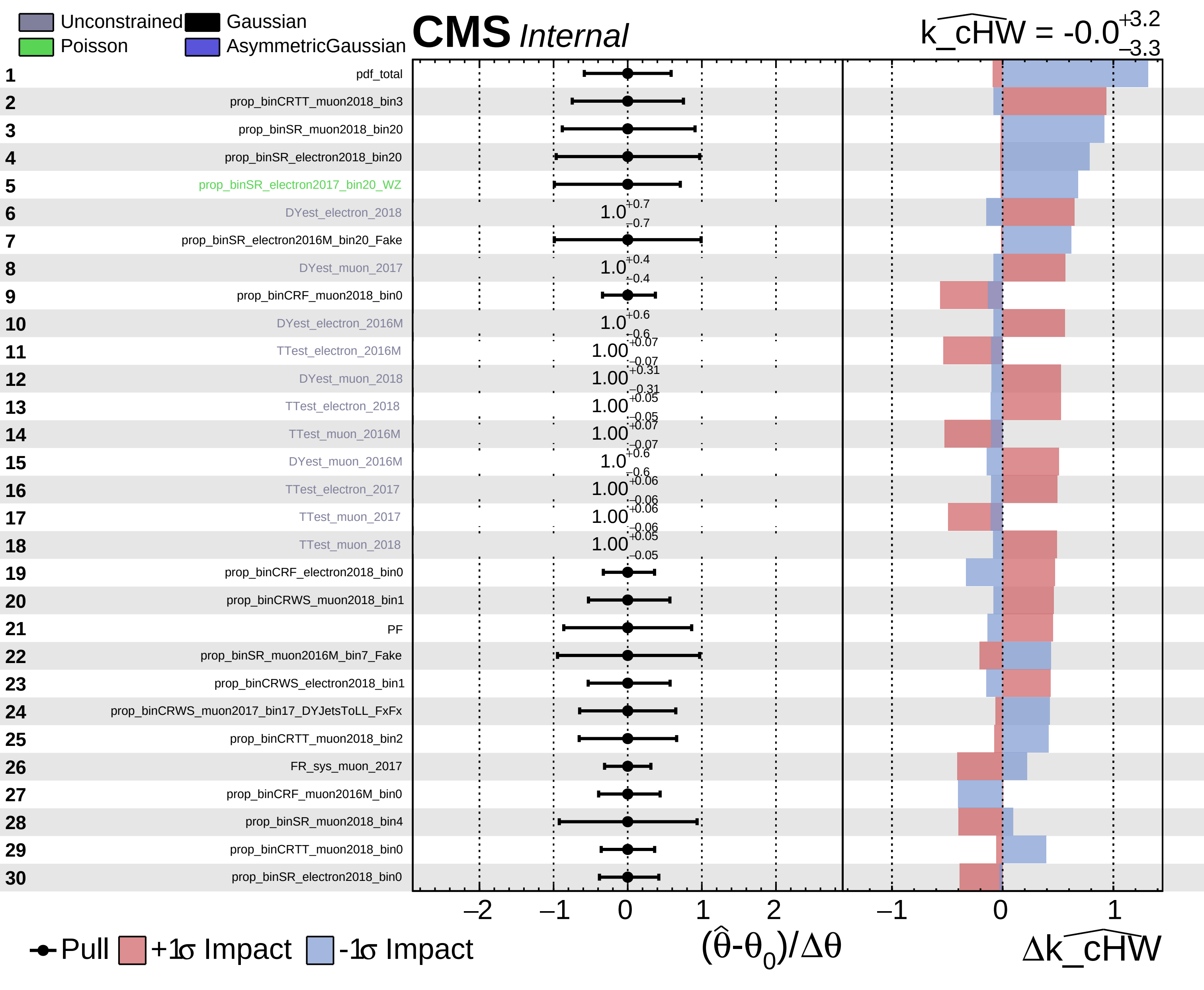
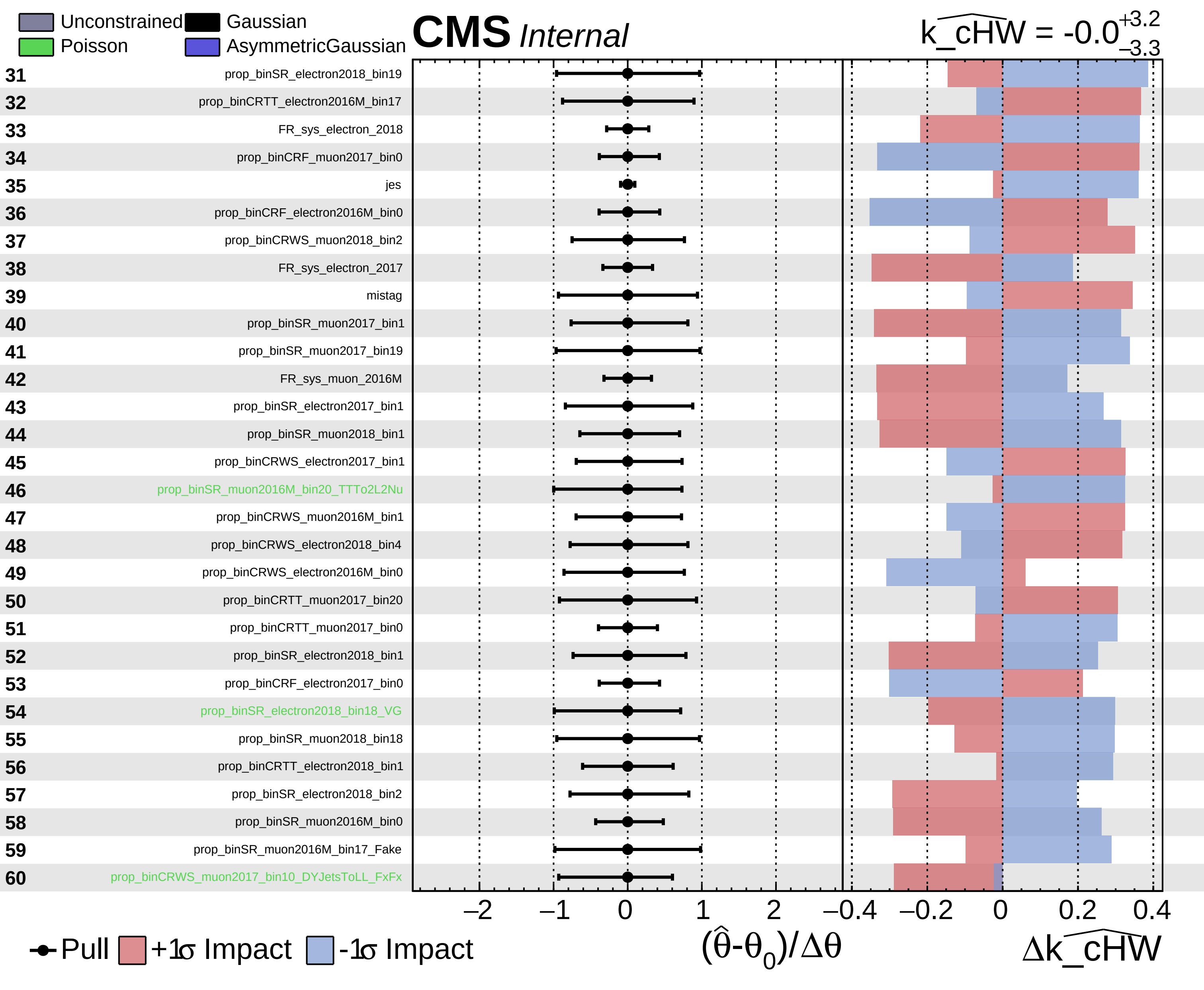
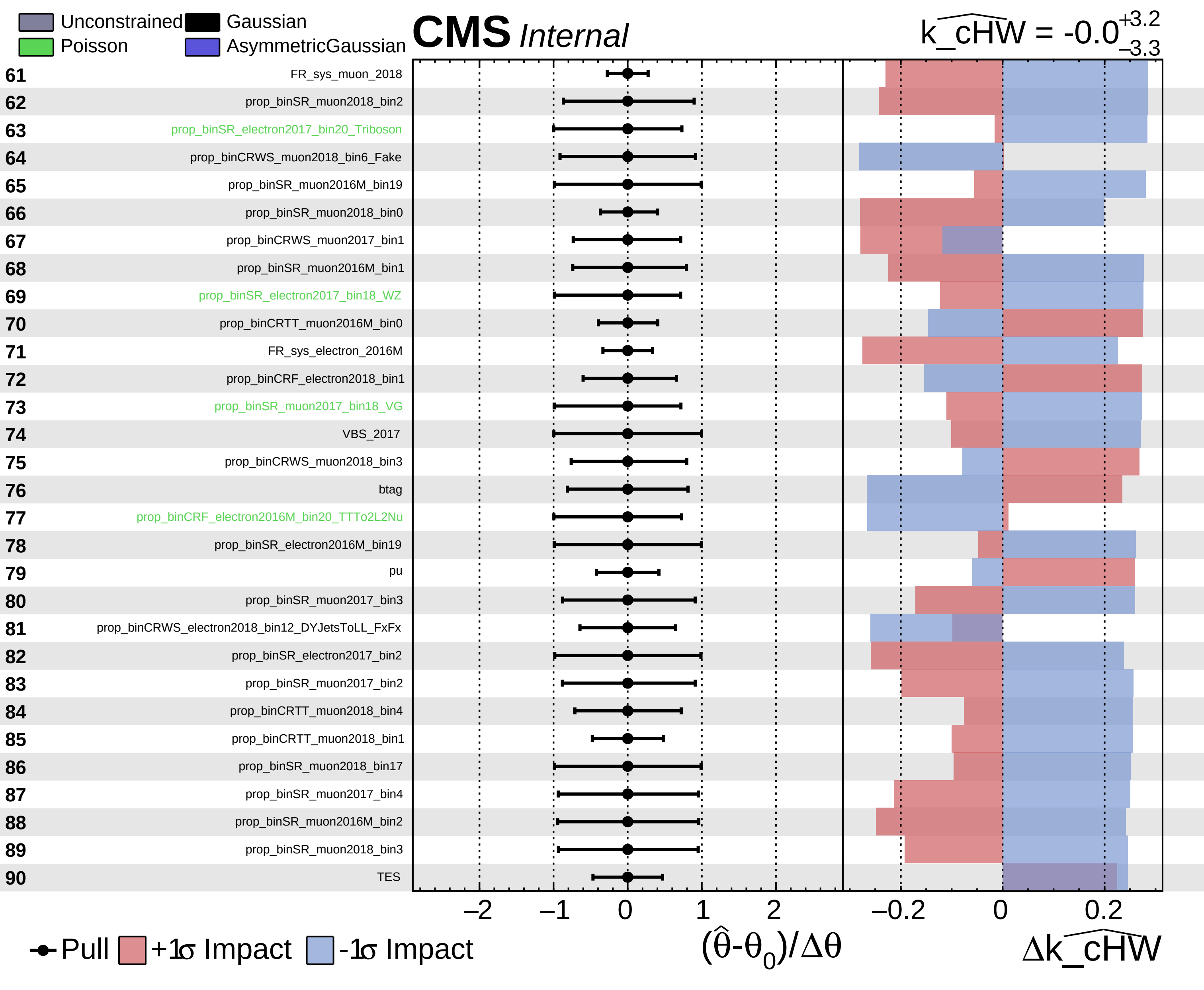
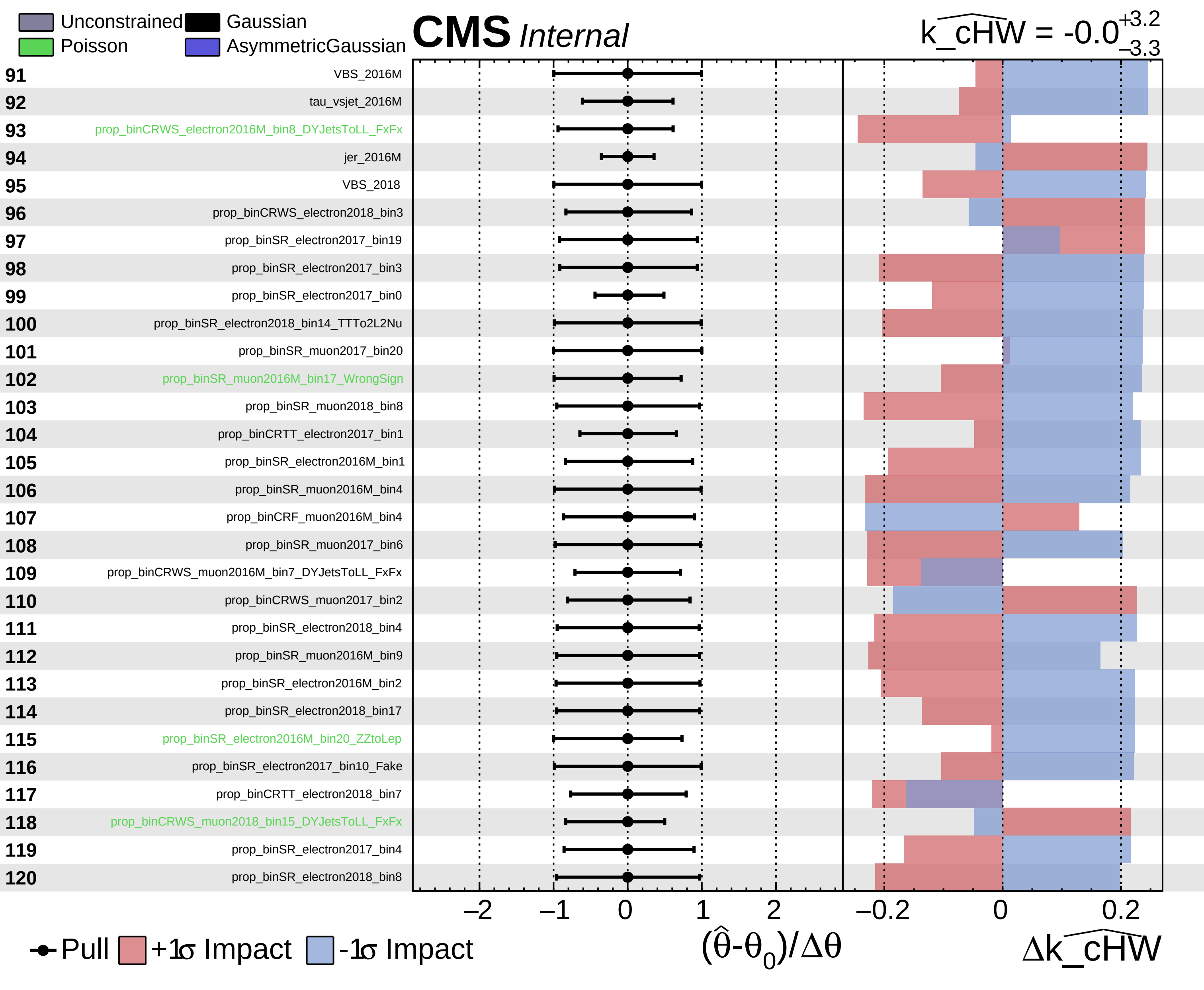


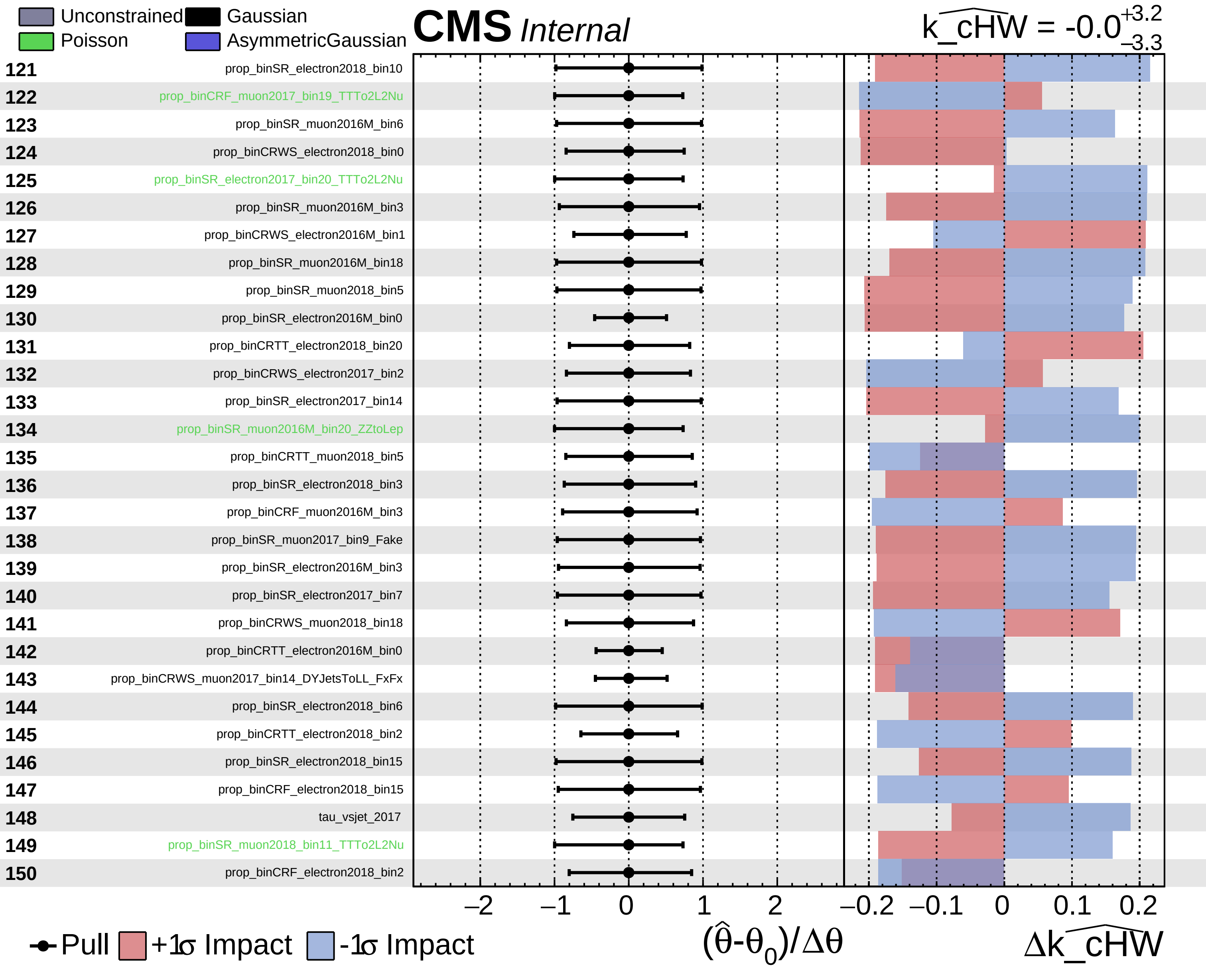
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
1 pdf_total
2 prop_binCRTT_muon2018_bin3
3 prop_binSR_muon2018_bin20
4 prop_binSR_electron2018_bin20
5 prop_binSR_electron2017_bin20_WZ
6 DYest_electron_2018
1.0 +0.7 −0.7
7 prop_binSR_electron2016M_bin20_Fake
8 DYest_muon_2017
1.0 +0.4 −0.4
9 prop_binCRF_muon2018_bin0
10 DYest_electron_2016M
1.0 +0.6 −0.6
11 TTest_electron_2016M
1.00 +0.07 −0.07
12 DYest_muon_2018
1.00 +0.31 −0.31
13 TTest_electron_2018
1.00 +0.05 −0.05
14 TTest_muon_2016M
1.00 +0.07 −0.07
15 DYest_muon_2016M
1.0 +0.6 −0.6
16 TTest_electron_2017
1.00 +0.06 −0.06
17 TTest_muon_2017
1.00 +0.06 −0.06
18 TTest_muon_2018
1.00 +0.05 −0.05
19 prop_binCRF_electron2018_bin0
20 prop_binCRWS_muon2018_bin1
21 PF
22 prop_binSR_muon2016M_bin7_Fake
23 prop_binCRWS_electron2018_bin1
24 prop_binCRWS_muon2017_bin17_DYJetsToLL_FxFx
25 prop_binCRTT_muon2018_bin2
26 FR_sys_muon_2017
27 prop_binCRF_muon2016M_bin0
28 prop_binSR_muon2018_bin4
29 prop_binCRTT_muon2018_bin0
30 prop_binSR_electron2018_bin0
−2 −1 0 1 2
−1 0 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
31 prop_binSR_electron2018_bin19
32 prop_binCRTT_electron2016M_bin17
33 FR_sys_electron_2018
34 prop_binCRF_muon2017_bin0
35 jes
36 prop_binCRF_electron2016M_bin0
37 prop_binCRWS_muon2018_bin2
38 FR_sys_electron_2017
39 mistag
40 prop_binSR_muon2017_bin1
41 prop_binSR_muon2017_bin19
42 FR_sys_muon_2016M
43 prop_binSR_electron2017_bin1
44 prop_binSR_muon2018_bin1
45 prop_binCRWS_electron2017_bin1
46 prop_binSR_muon2016M_bin20_TTTo2L2Nu
47 prop_binCRWS_muon2016M_bin1
48 prop_binCRWS_electron2018_bin4
49 prop_binCRWS_electron2016M_bin0
50 prop_binCRTT_muon2017_bin20
51 prop_binCRTT_muon2017_bin0
52 prop_binSR_electron2018_bin1
53 prop_binCRF_electron2017_bin0
54 prop_binSR_electron2018_bin18_VG
55 prop_binSR_muon2018_bin18
56 prop_binCRTT_electron2018_bin1
57 prop_binSR_electron2018_bin2
58 prop_binSR_muon2016M_bin0
59 prop_binSR_muon2016M_bin17_Fake
60 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

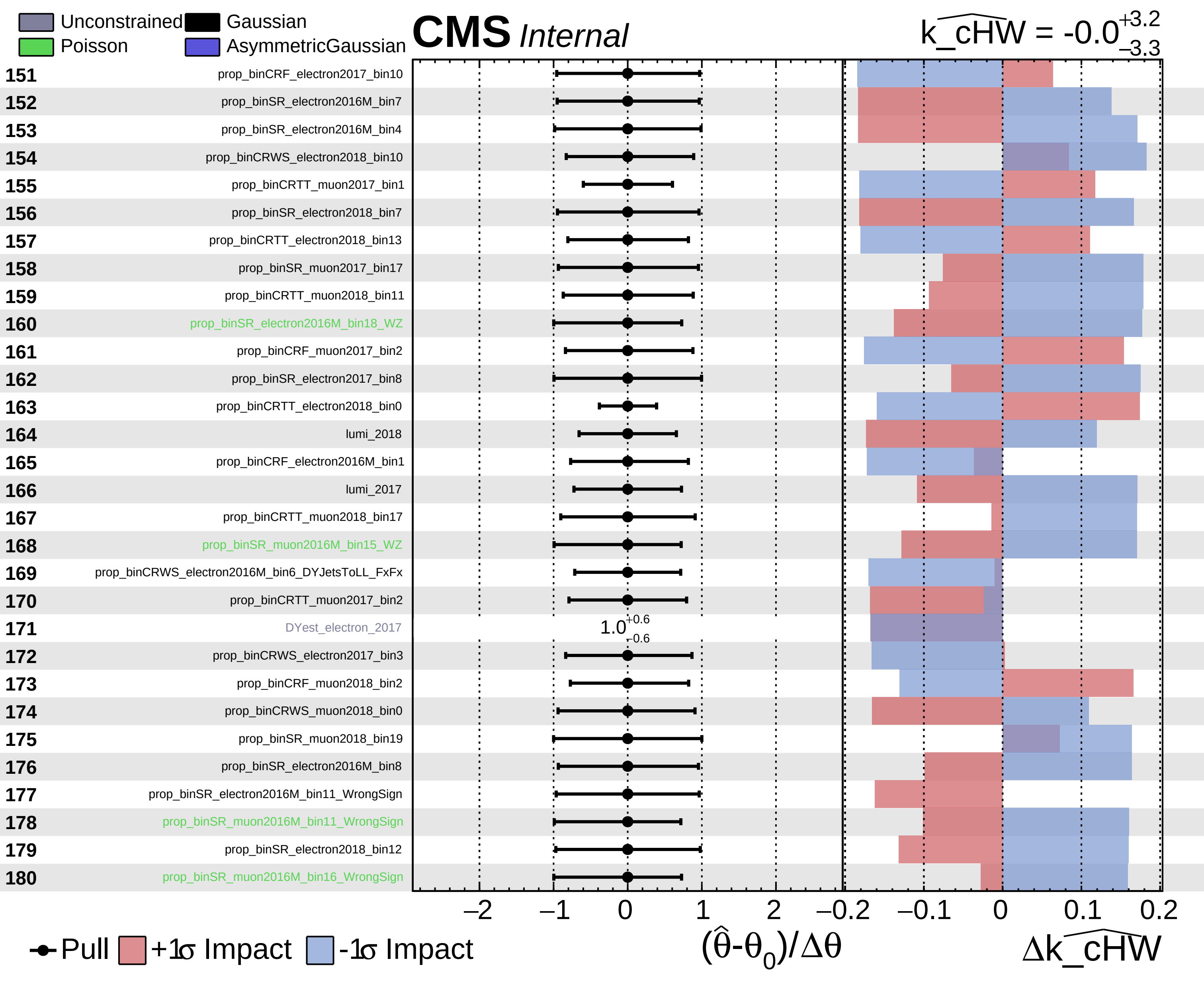

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
61 FR_sys_muon_2018
62 prop_binSR_muon2018_bin2
63 prop_binSR_electron2017_bin20_Triboson
64 prop_binCRWS_muon2018_bin6_Fake
65 prop_binSR_muon2016M_bin19
66 prop_binSR_muon2018_bin0
67 prop_binCRWS_muon2017_bin1
68 prop_binSR_muon2016M_bin1
69 prop_binSR_electron2017_bin18_WZ
70 prop_binCRTT_muon2016M_bin0
71 FR_sys_electron_2016M
72 prop_binCRF_electron2018_bin1
73 prop_binSR_muon2017_bin18_VG
74 VBS_2017
75 prop_binCRWS_muon2018_bin3
76 btag
77 prop_binCRF_electron2016M_bin20_TTTo2L2Nu
78 prop_binSR_electron2016M_bin19
79 pu
80 prop_binSR_muon2017_bin3
81 prop_binCRWS_electron2018_bin12_DYJetsToLL_FxFx
82 prop_binSR_electron2017_bin2
83 prop_binSR_muon2017_bin2
84 prop_binCRTT_muon2018_bin4
85 prop_binCRTT_muon2018_bin1
86 prop_binSR_muon2018_bin17
87 prop_binSR_muon2017_bin4
88 prop_binSR_muon2016M_bin2
89 prop_binSR_muon2018_bin3
90 TES
−2 −1 0 1 2
−0.2 0 0.2
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



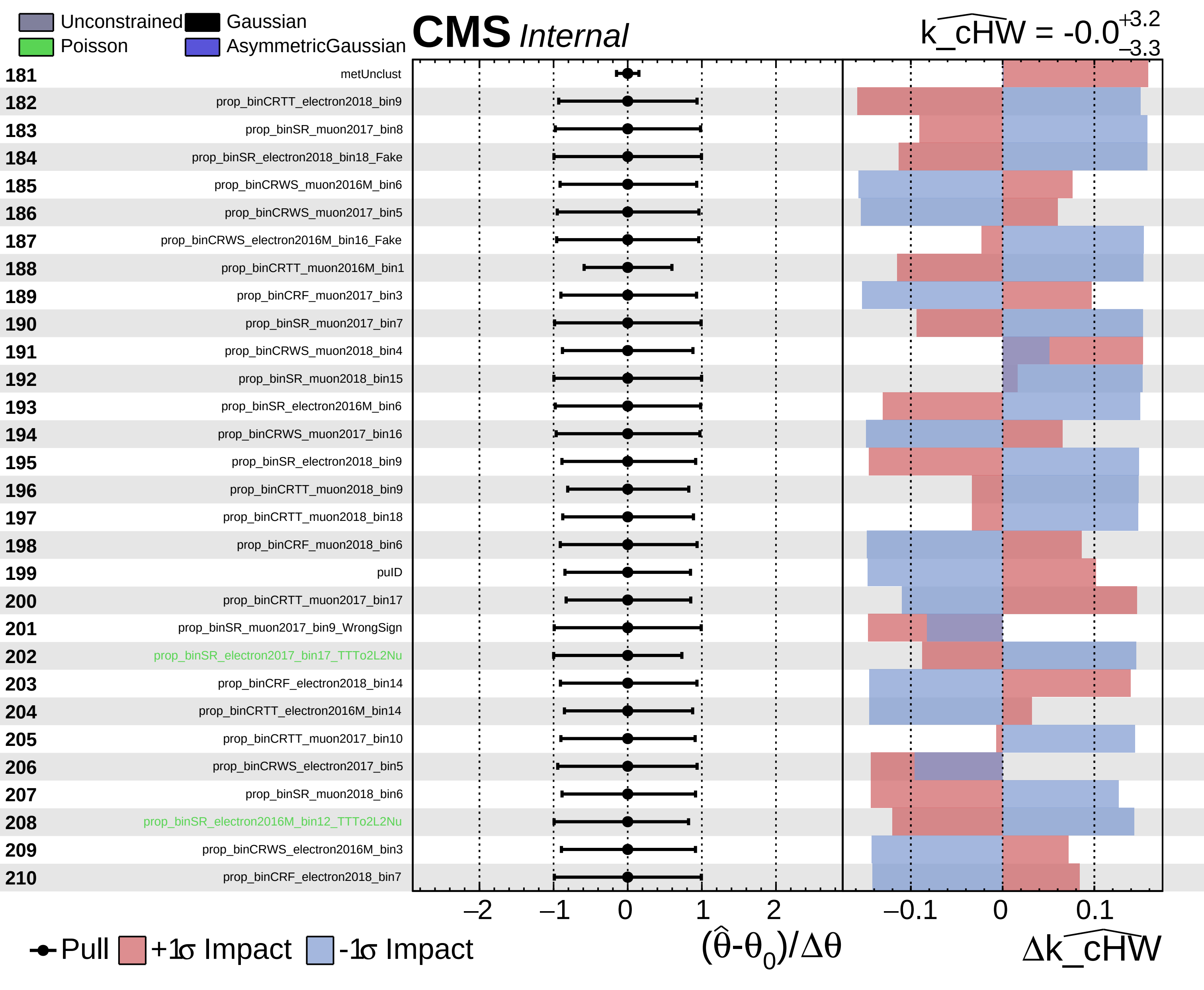
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
91 VBS_2016M
92 tau_vsjet_2016M
93 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
94 jer_2016M
95 VBS_2018
96 prop_binCRWS_electron2018_bin3
97 prop_binSR_electron2017_bin19
98 prop_binSR_electron2017_bin3
99 prop_binSR_electron2017_bin0
100 prop_binSR_electron2018_bin14_TTTo2L2Nu
101 prop_binSR_muon2017_bin20
102 prop_binSR_muon2016M_bin17_WrongSign
103 prop_binSR_muon2018_bin8
104 prop_binCRTT_electron2017_bin1
105 prop_binSR_electron2016M_bin1
106 prop_binSR_muon2016M_bin4
107 prop_binCRF_muon2016M_bin4
108 prop_binSR_muon2017_bin6
109 prop_binCRWS_muon2016M_bin7_DYJetsToLL_FxFx
110 prop_binCRWS_muon2017_bin2
111 prop_binSR_electron2018_bin4
112 prop_binSR_muon2016M_bin9
113 prop_binSR_electron2016M_bin2
114 prop_binSR_electron2018_bin17
115 prop_binSR_electron2016M_bin20_ZZtoLep
116 prop_binSR_electron2017_bin10_Fake
117 prop_binCRTT_electron2018_bin7
118 prop_binCRWS_muon2018_bin15_DYJetsToLL_FxFx
119 prop_binSR_electron2017_bin4
120 prop_binSR_electron2018_bin8
−2
−1
0
1
2
−0.2
0
0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



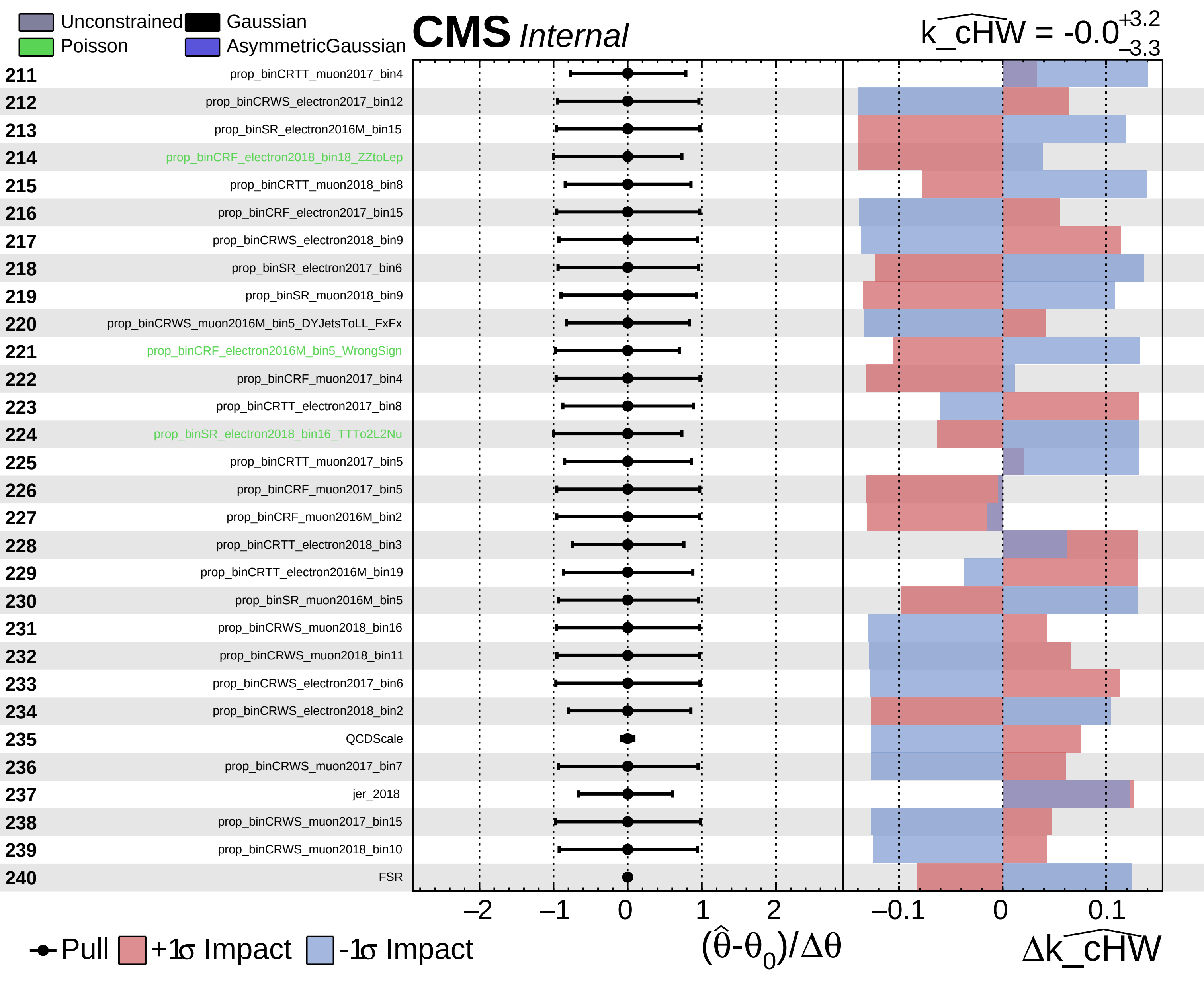
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
121 prop_binSR_electron2018_bin10
122 prop_binCRF_muon2017_bin19_TTTo2L2Nu
123 prop_binSR_muon2016M_bin6
124 prop_binCRWS_electron2018_bin0
125 prop_binSR_electron2017_bin20_TTTo2L2Nu
126 prop_binSR_muon2016M_bin3
127 prop_binCRWS_electron2016M_bin1
128 prop_binSR_muon2016M_bin18
129 prop_binSR_muon2018_bin5
130 prop_binSR_electron2016M_bin0
131 prop_binCRTT_electron2018_bin20
132 prop_binCRWS_electron2017_bin2
133 prop_binSR_electron2017_bin14
134 prop_binSR_muon2016M_bin20_ZZtoLep
135 prop_binCRTT_muon2018_bin5
136 prop_binSR_electron2018_bin3
137 prop_binCRF_muon2016M_bin3
138 prop_binSR_muon2017_bin9_Fake
139 prop_binSR_electron2016M_bin3
140 prop_binSR_electron2017_bin7
141 prop_binCRWS_muon2018_bin18
142 prop_binCRTT_electron2016M_bin0
143 prop_binCRWS_muon2017_bin14_DYJetsToLL_FxFx
144 prop_binSR_electron2018_bin6
145 prop_binCRTT_electron2018_bin2
146 prop_binSR_electron2018_bin15
147 prop_binCRF_electron2018_bin15
148 tau_vsjet_2017
149 prop_binSR_muon2018_bin11_TTTo2L2Nu
150 prop_binCRF_electron2018_bin2
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



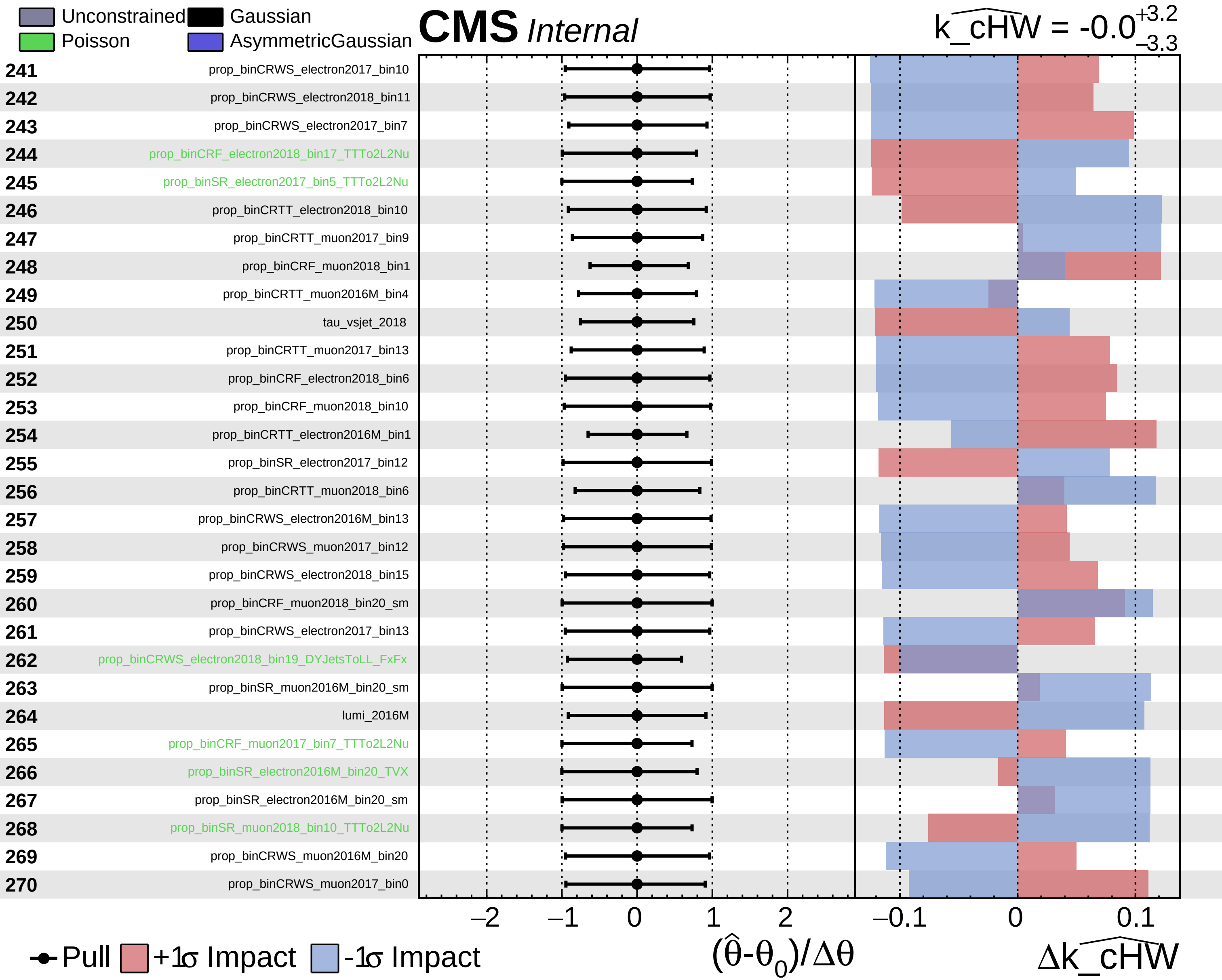
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
151 prop_binCRF_electron2017_bin10
152 prop_binSR_electron2016M_bin7
153 prop_binSR_electron2016M_bin4
154 prop_binCRWS_electron2018_bin10
155 prop_binCRTT_muon2017_bin1
156 prop_binSR_electron2018_bin7
157 prop_binCRTT_electron2018_bin13
158 prop_binSR_muon2017_bin17
159 prop_binCRTT_muon2018_bin11
160 prop_binSR_electron2016M_bin18_WZ
161 prop_binCRF_muon2017_bin2
162 prop_binSR_electron2017_bin8
163 prop_binCRTT_electron2018_bin0
164 lumi_2018
165 prop_binCRF_electron2016M_bin1
166 lumi_2017
167 prop_binCRTT_muon2018_bin17
168 prop_binSR_muon2016M_bin15_WZ
169 prop_binCRWS_electron2016M_bin6_DYJetsToLL_FxFx
170 prop_binCRTT_muon2017_bin2
171 DYest_electron_2017
1.0 +0.6 −0.6
172 prop_binCRWS_electron2017_bin3
173 prop_binCRF_muon2018_bin2
174 prop_binCRWS_muon2018_bin0
175 prop_binSR_muon2018_bin19
176 prop_binSR_electron2016M_bin8
177 prop_binSR_electron2016M_bin11_WrongSign
178 prop_binSR_muon2016M_bin11_WrongSign
179 prop_binSR_electron2018_bin12
180 prop_binSR_muon2016M_bin16_WrongSign
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

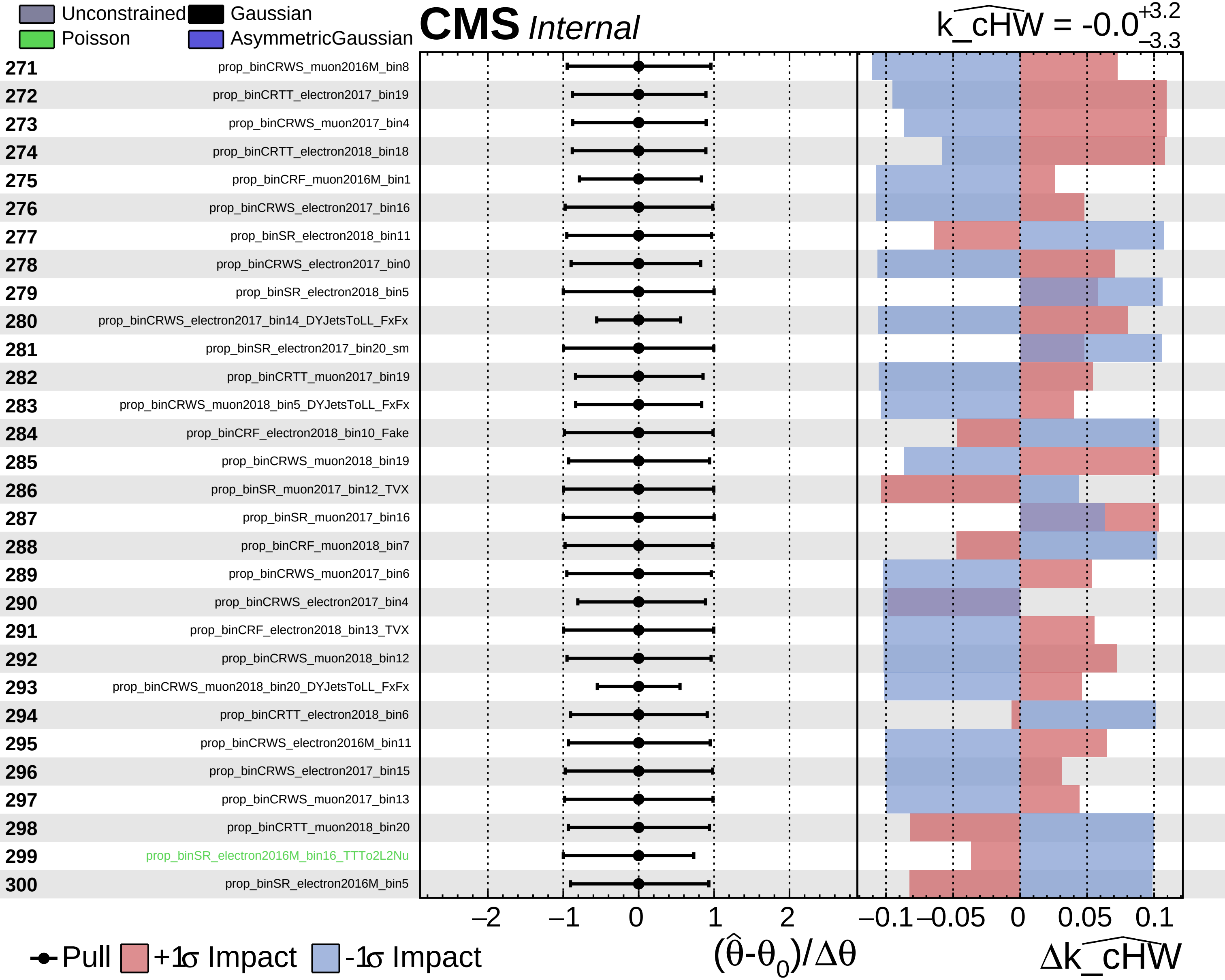


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
181 metUnclust
182 prop_binCRTT_electron2018_bin9
183 prop_binSR_muon2017_bin8
184 prop_binSR_electron2018_bin18_Fake
185 prop_binCRWS_muon2016M_bin6
186 prop_binCRWS_muon2017_bin5
187 prop_binCRWS_electron2016M_bin16_Fake
188 prop_binCRTT_muon2016M_bin1
189 prop_binCRF_muon2017_bin3
190 prop_binSR_muon2017_bin7
191 prop_binCRWS_muon2018_bin4
192 prop_binSR_muon2018_bin15
193 prop_binSR_electron2016M_bin6
194 prop_binCRWS_muon2017_bin16
195 prop_binSR_electron2018_bin9
196 prop_binCRTT_muon2018_bin9
197 prop_binCRTT_muon2018_bin18
198 prop_binCRF_muon2018_bin6
199 puID
200 prop_binCRTT_muon2017_bin17
201 prop_binSR_muon2017_bin9_WrongSign
202 prop_binSR_electron2017_bin17_TTTo2L2Nu
203 prop_binCRF_electron2018_bin14
204 prop_binCRTT_electron2016M_bin14
205 prop_binCRTT_muon2017_bin10
206 prop_binCRWS_electron2017_bin5
207 prop_binSR_muon2018_bin6
208 prop_binSR_electron2016M_bin12_TTTo2L2Nu
209 prop_binCRWS_electron2016M_bin3
210 prop_binCRF_electron2018_bin7
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

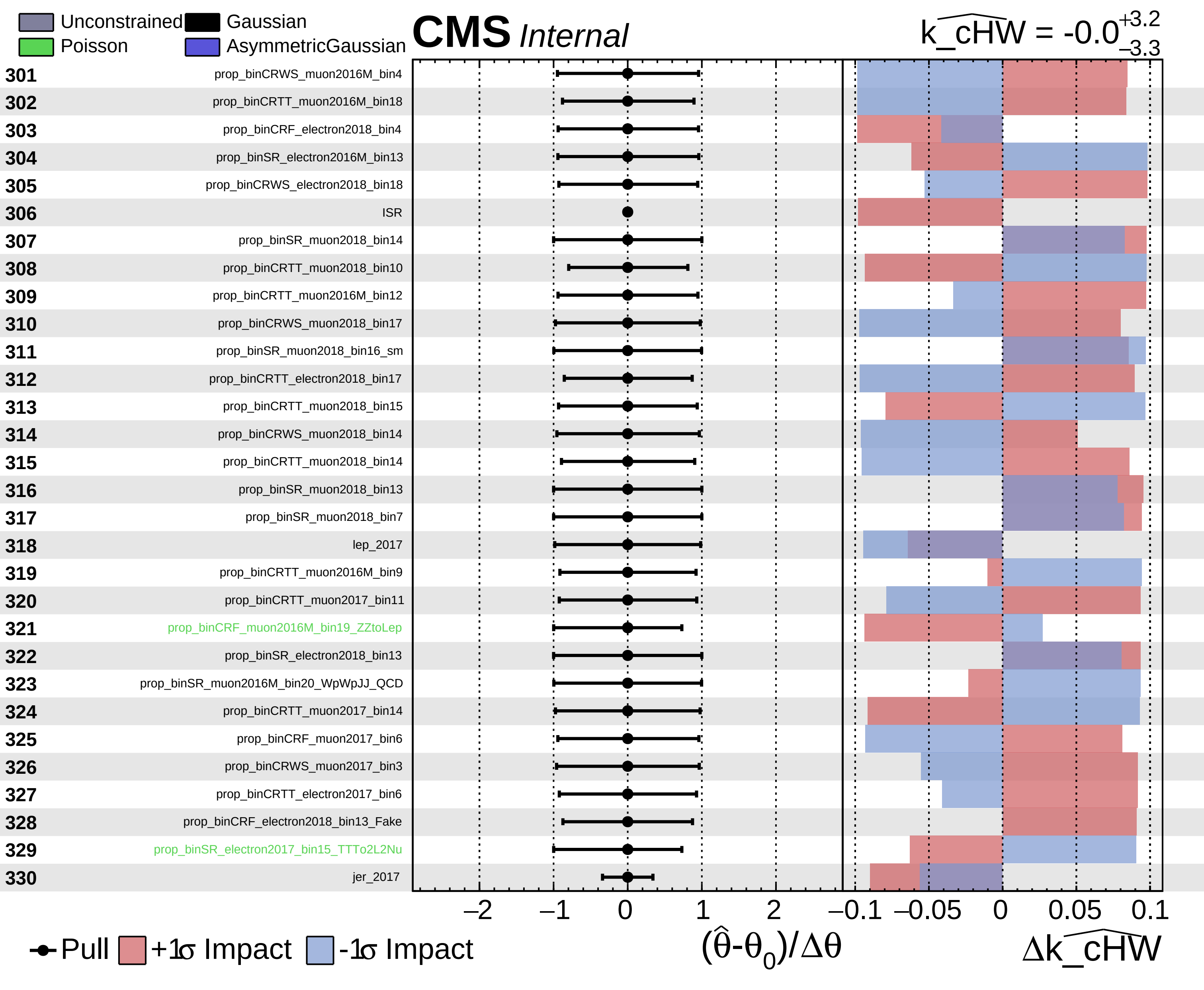


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
211 prop_binCRTT_muon2017_bin4
212 prop_binCRWS_electron2017_bin12
213 prop_binSR_electron2016M_bin15
214 prop_binCRF_electron2018_bin18_ZZtoLep
215 prop_binCRTT_muon2018_bin8
216 prop_binCRF_electron2017_bin15
217 prop_binCRWS_electron2018_bin9
218 prop_binSR_electron2017_bin6
219 prop_binSR_muon2018_bin9
220 prop_binCRWS_muon2016M_bin5_DYJetsToLL_FxFx
221 prop_binCRF_electron2016M_bin5_WrongSign
222 prop_binCRF_muon2017_bin4
223 prop_binCRTT_electron2017_bin8
224 prop_binSR_electron2018_bin16_TTTo2L2Nu
225 prop_binCRTT_muon2017_bin5
226 prop_binCRF_muon2017_bin5
227 prop_binCRF_muon2016M_bin2
228 prop_binCRTT_electron2018_bin3
229 prop_binCRTT_electron2016M_bin19
230 prop_binSR_muon2016M_bin5
231 prop_binCRWS_muon2018_bin16
232 prop_binCRWS_muon2018_bin11
233 prop_binCRWS_electron2017_bin6
234 prop_binCRWS_electron2018_bin2
235 QCDScale
236 prop_binCRWS_muon2017_bin7
237 jer_2018
238 prop_binCRWS_muon2017_bin15
239 prop_binCRWS_muon2018_bin10
240 FSR
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

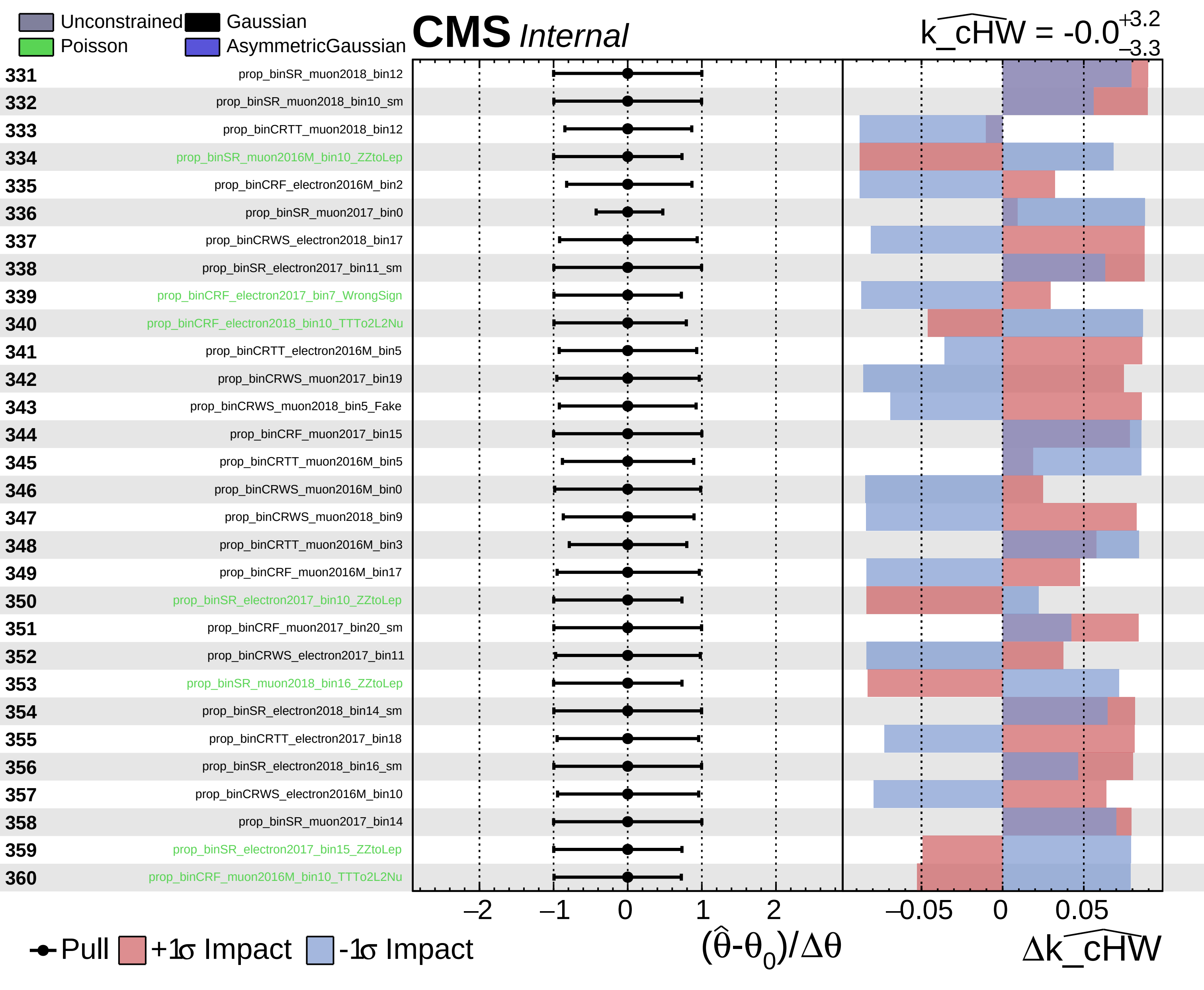

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
241 prop_binCRWS_electron2017_bin10
242 prop_binCRWS_electron2018_bin11
243 prop_binCRWS_electron2017_bin7
244 prop_binCRF_electron2018_bin17_TTTo2L2Nu
245 prop_binSR_electron2017_bin5_TTTo2L2Nu
246 prop_binCRTT_electron2018_bin10
247 prop_binCRTT_muon2017_bin9
248 prop_binCRF_muon2018_bin1
249 prop_binCRTT_muon2016M_bin4
250 tau_vsjet_2018
251 prop_binCRTT_muon2017_bin13
252 prop_binCRF_electron2018_bin6
253 prop_binCRF_muon2018_bin10
254 prop_binCRTT_electron2016M_bin1
255 prop_binSR_electron2017_bin12
256 prop_binCRTT_muon2018_bin6
257 prop_binCRWS_electron2016M_bin13
258 prop_binCRWS_muon2017_bin12
259 prop_binCRWS_electron2018_bin15
260 prop_binCRF_muon2018_bin20_sm
261 prop_binCRWS_electron2017_bin13
262 prop_binCRWS_electron2018_bin19_DYJetsToLL_FxFx
263 prop_binSR_muon2016M_bin20_sm
264 lumi_2016M
265 prop_binCRF_muon2017_bin7_TTTo2L2Nu
266 prop_binSR_electron2016M_bin20_TVX
267 prop_binSR_electron2016M_bin20_sm
268 prop_binSR_muon2018_bin10_TTTo2L2Nu
269 prop_binCRWS_muon2016M_bin20
270 prop_binCRWS_muon2017_bin0
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



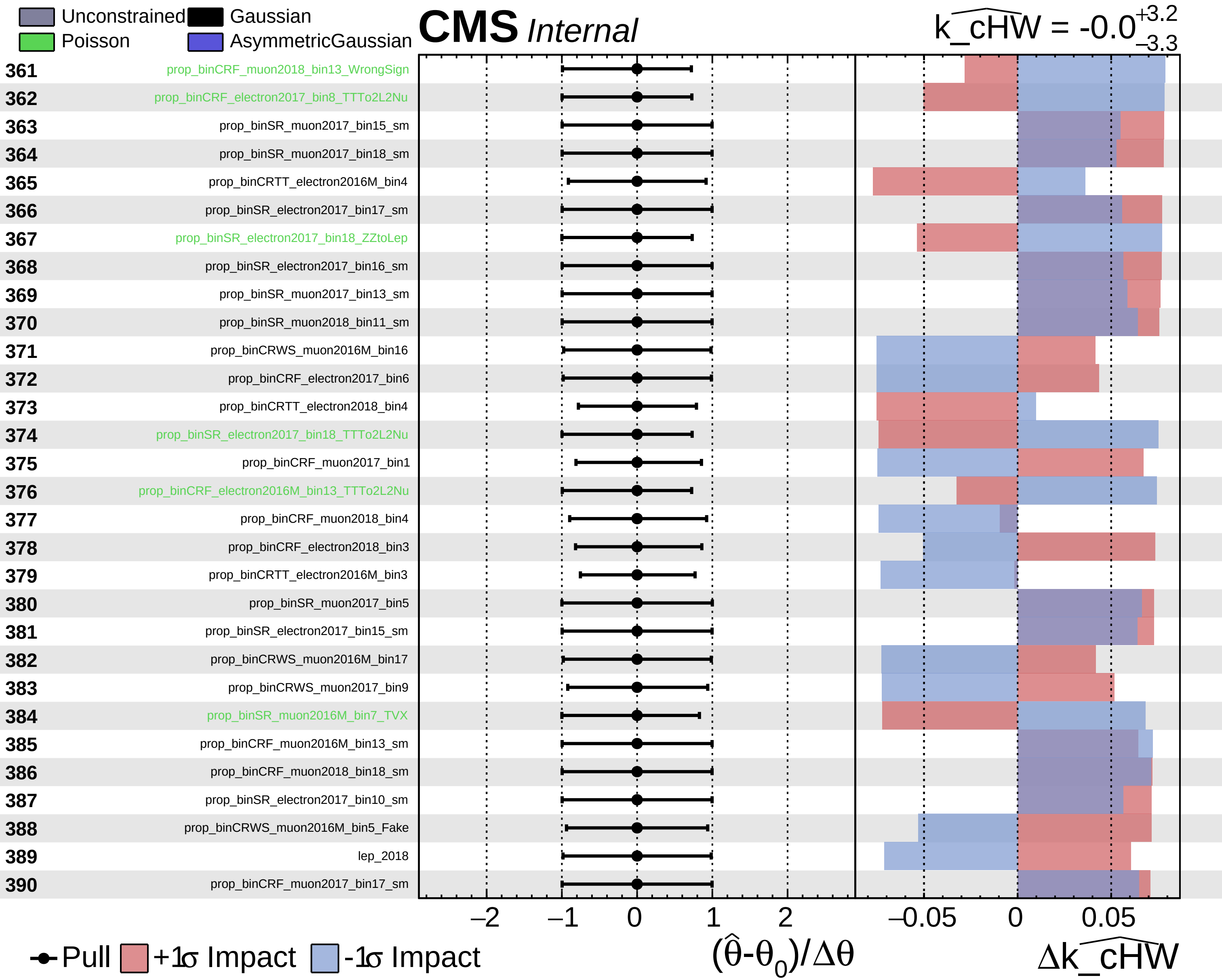
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0+3.2−3.3
271 prop_binCRWS_muon2016M_bin8
272 prop_binCRTT_electron2017_bin19
273 prop_binCRWS_muon2017_bin4
274 prop_binCRTT_electron2018_bin18
275 prop_binCRF_muon2016M_bin1
276 prop_binCRWS_electron2017_bin16
277 prop_binSR_electron2018_bin11
278 prop_binCRWS_electron2017_bin0
279 prop_binSR_electron2018_bin5
280 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
281 prop_binSR_electron2017_bin20_sm
282 prop_binCRTT_muon2017_bin19
283 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
284 prop_binCRF_electron2018_bin10_Fake
285 prop_binCRWS_muon2018_bin19
286 prop_binSR_muon2017_bin12_TVX
287 prop_binSR_muon2017_bin16
288 prop_binCRF_muon2018_bin7
289 prop_binCRWS_muon2017_bin6
290 prop_binCRWS_electron2017_bin4
291 prop_binCRF_electron2018_bin13_TVX
292 prop_binCRWS_muon2018_bin12
293 prop_binCRWS_muon2018_bin20_DYJetsToLL_FxFx
294 prop_binCRTT_electron2018_bin6
295 prop_binCRWS_electron2016M_bin11
296 prop_binCRWS_electron2017_bin15
297 prop_binCRWS_muon2017_bin13
298 prop_binCRTT_muon2018_bin20
299 prop_binSR_electron2016M_bin16_TTTo2L2Nu
300 prop_binSR_electron2016M_bin5
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



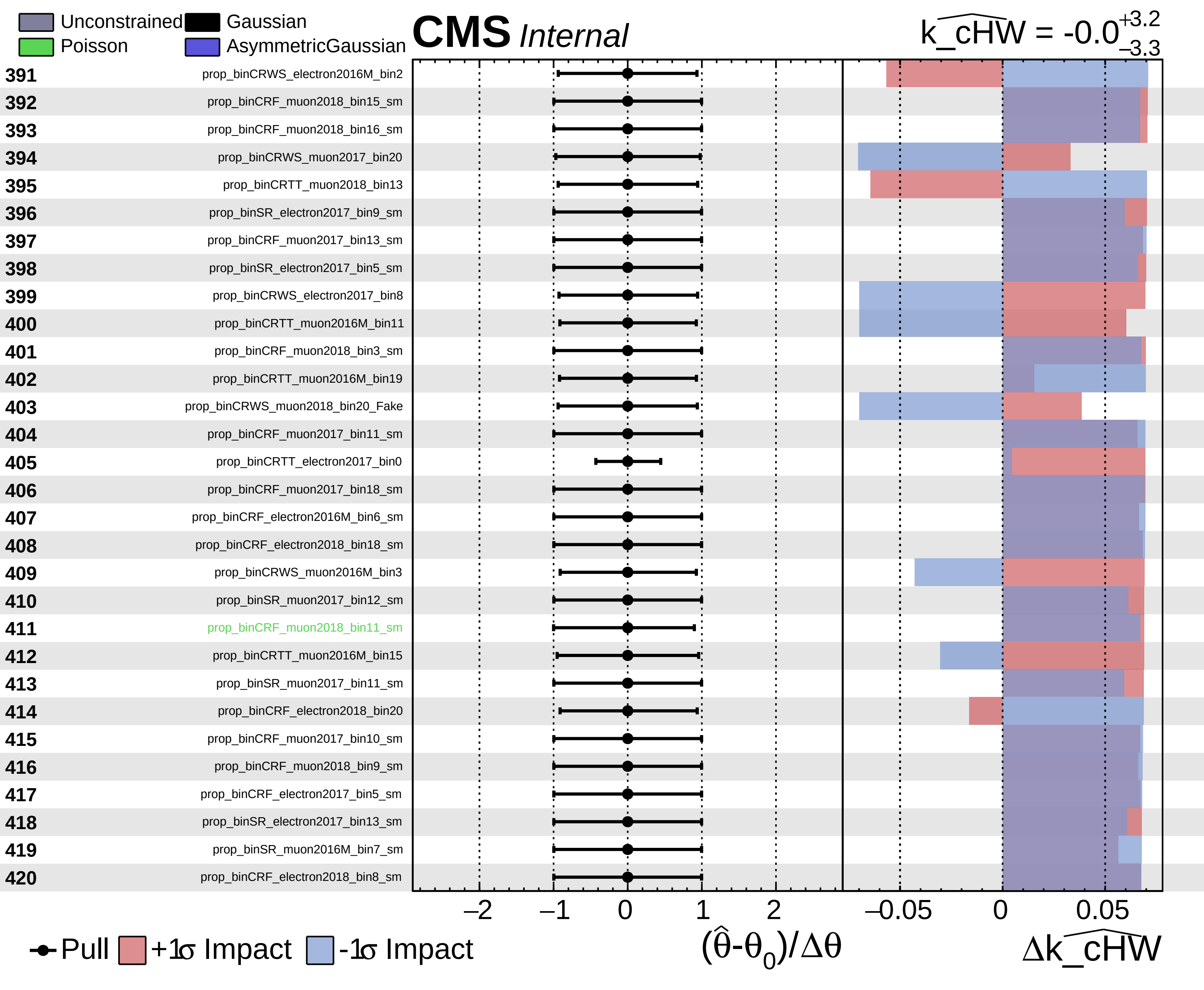
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
301 prop_binCRWS_muon2016M_bin4
302 prop_binCRTT_muon2016M_bin18
303 prop_binCRF_electron2018_bin4
304 prop_binSR_electron2016M_bin13
305 prop_binCRWS_electron2018_bin18
306 ISR
307 prop_binSR_muon2018_bin14
308 prop_binCRTT_muon2018_bin10
309 prop_binCRTT_muon2016M_bin12
310 prop_binCRWS_muon2018_bin17
311 prop_binSR_muon2018_bin16_sm
312 prop_binCRTT_electron2018_bin17
313 prop_binCRTT_muon2018_bin15
314 prop_binCRWS_muon2018_bin14
315 prop_binCRTT_muon2018_bin14
316 prop_binSR_muon2018_bin13
317 prop_binSR_muon2018_bin7
318 lep_2017
319 prop_binCRTT_muon2016M_bin9
320 prop_binCRTT_muon2017_bin11
321 prop_binCRF_muon2016M_bin19_ZZtoLep
322 prop_binSR_electron2018_bin13
323 prop_binSR_muon2016M_bin20_WpWpJJ_QCD
324 prop_binCRTT_muon2017_bin14
325 prop_binCRF_muon2017_bin6
326 prop_binCRWS_muon2017_bin3
327 prop_binCRTT_electron2017_bin6
328 prop_binCRF_electron2018_bin13_Fake
329 prop_binSR_electron2017_bin15_TTTo2L2Nu
330 jer_2017
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

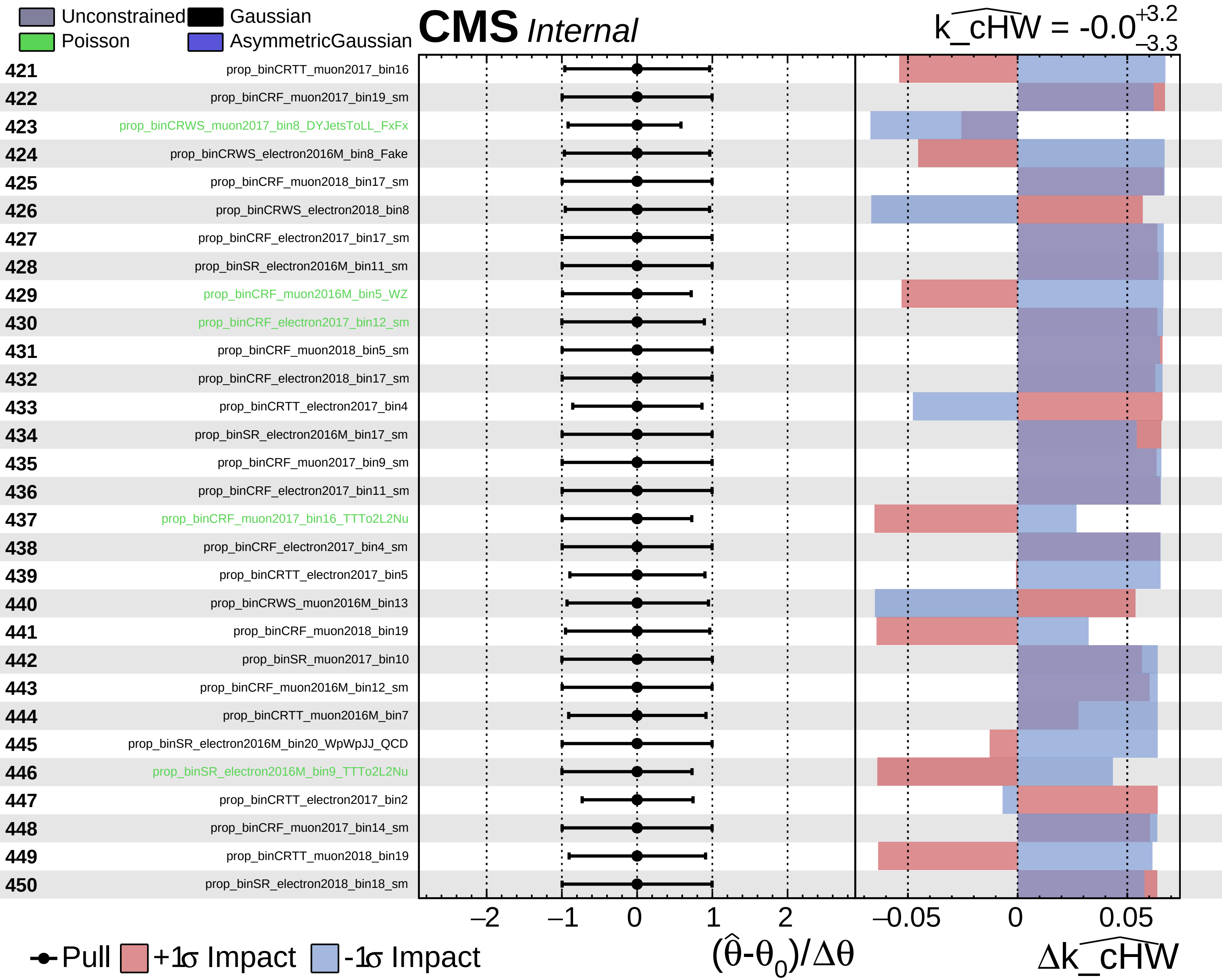


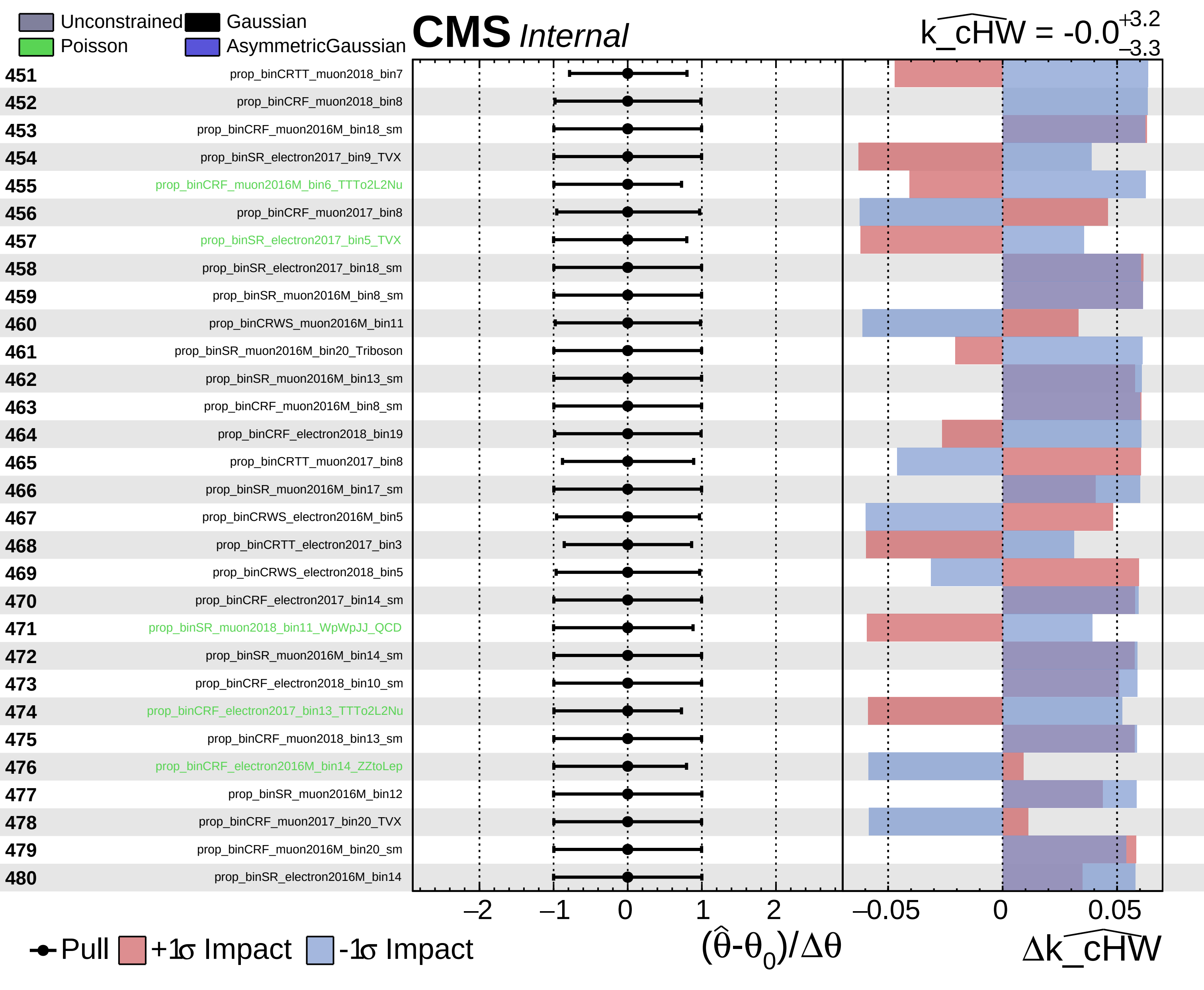
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
331 prop_binSR_muon2018_bin12
332 prop_binSR_muon2018_bin10_sm
333 prop_binCRTT_muon2018_bin12
334 prop_binSR_muon2016M_bin10_ZZtoLep
335 prop_binCRF_electron2016M_bin2
336 prop_binSR_muon2017_bin0
337 prop_binCRWS_electron2018_bin17
338 prop_binSR_electron2017_bin11_sm
339 prop_binCRF_electron2017_bin7_WrongSign
340 prop_binCRF_electron2018_bin10_TTTo2L2Nu
341 prop_binCRTT_electron2016M_bin5
342 prop_binCRWS_muon2017_bin19
343 prop_binCRWS_muon2018_bin5_Fake
344 prop_binCRF_muon2017_bin15
345 prop_binCRTT_muon2016M_bin5
346 prop_binCRWS_muon2016M_bin0
347 prop_binCRWS_muon2018_bin9
348 prop_binCRTT_muon2016M_bin3
349 prop_binCRF_muon2016M_bin17
350 prop_binSR_electron2017_bin10_ZZtoLep
351 prop_binCRF_muon2017_bin20_sm
352 prop_binCRWS_electron2017_bin11
353 prop_binSR_muon2018_bin16_ZZtoLep
354 prop_binSR_electron2018_bin14_sm
355 prop_binCRTT_electron2017_bin18
356 prop_binSR_electron2018_bin16_sm
357 prop_binCRWS_electron2016M_bin10
358 prop_binSR_muon2017_bin14
359 prop_binSR_electron2017_bin15_ZZtoLep
360 prop_binCRF_muon2016M_bin10_TTTo2L2Nu
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



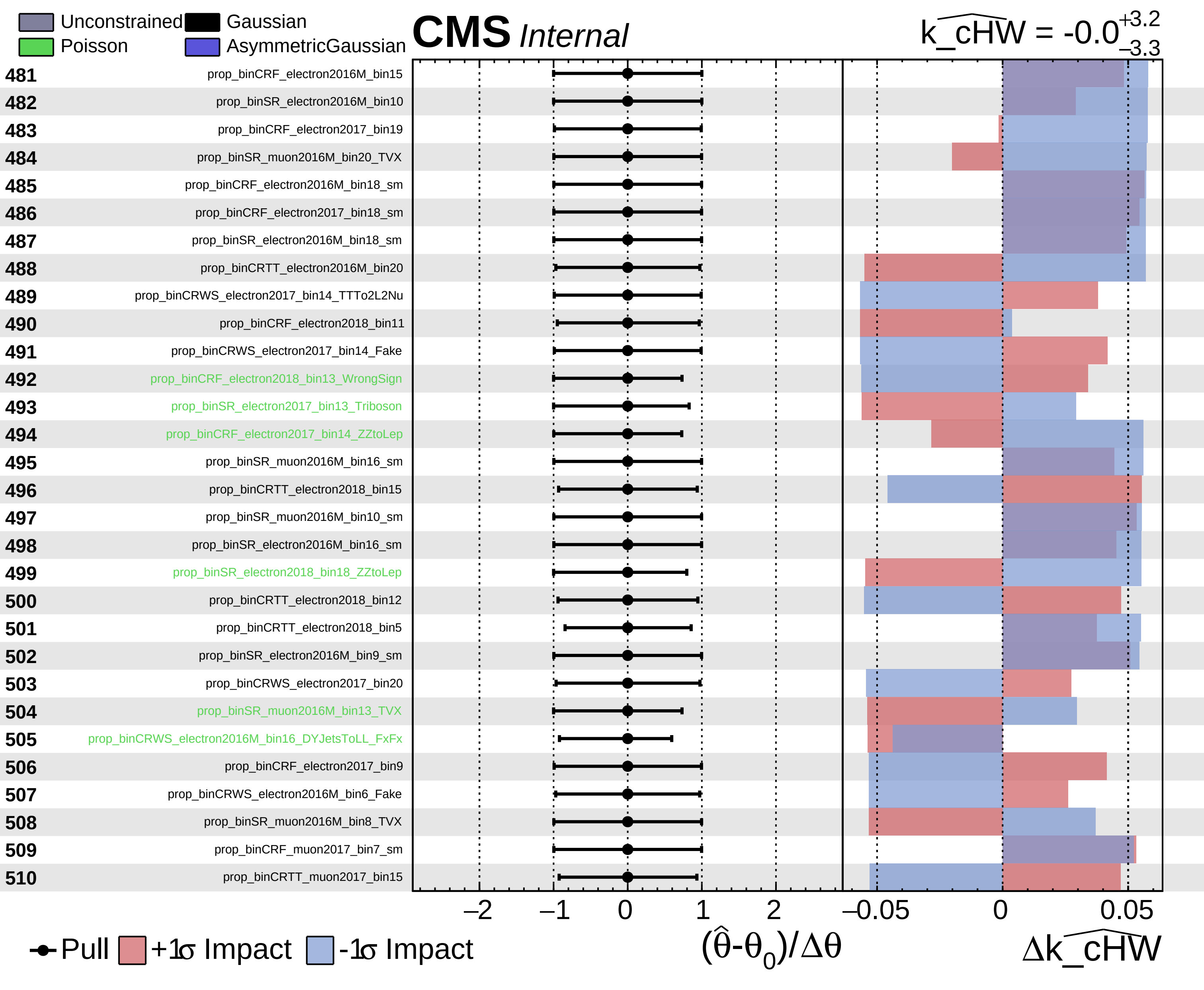
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
361 prop_binCRF_muon2018_bin13_WrongSign
362 prop_binCRF_electron2017_bin8_TTTo2L2Nu
363 prop_binSR_muon2017_bin15_sm
364 prop_binSR_muon2017_bin18_sm
365 prop_binCRTT_electron2016M_bin4
366 prop_binSR_electron2017_bin17_sm
367 prop_binSR_electron2017_bin18_ZZtoLep
368 prop_binSR_electron2017_bin16_sm
369 prop_binSR_muon2017_bin13_sm
370 prop_binSR_muon2018_bin11_sm
371 prop_binCRWS_muon2016M_bin16
372 prop_binCRF_electron2017_bin6
373 prop_binCRTT_electron2018_bin4
374 prop_binSR_electron2017_bin18_TTTo2L2Nu
375 prop_binCRF_muon2017_bin1
376 prop_binCRF_electron2016M_bin13_TTTo2L2Nu
377 prop_binCRF_muon2018_bin4
378 prop_binCRF_electron2018_bin3
379 prop_binCRTT_electron2016M_bin3
380 prop_binSR_muon2017_bin5
381 prop_binSR_electron2017_bin15_sm
382 prop_binCRWS_muon2016M_bin17
383 prop_binCRWS_muon2017_bin9
384 prop_binSR_muon2016M_bin7_TVX
385 prop_binCRF_muon2016M_bin13_sm
386 prop_binCRF_muon2018_bin18_sm
387 prop_binSR_electron2017_bin10_sm
388 prop_binCRWS_muon2016M_bin5_Fake
389 lep_2018
390 prop_binCRF_muon2017_bin17_sm
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0+3.2 −3.3
391 prop_binCRWS_electron2016M_bin2
392 prop_binCRF_muon2018_bin15_sm
393 prop_binCRF_muon2018_bin16_sm
394 prop_binCRWS_muon2017_bin20
395 prop_binCRTT_muon2018_bin13
396 prop_binSR_electron2017_bin9_sm
397 prop_binCRF_muon2017_bin13_sm
398 prop_binSR_electron2017_bin5_sm
399 prop_binCRWS_electron2017_bin8
400 prop_binCRTT_muon2016M_bin11
401 prop_binCRF_muon2018_bin3_sm
402 prop_binCRTT_muon2016M_bin19
403 prop_binCRWS_muon2018_bin20_Fake
404 prop_binCRF_muon2017_bin11_sm
405 prop_binCRTT_electron2017_bin0
406 prop_binCRF_muon2017_bin18_sm
407 prop_binCRF_electron2016M_bin6_sm
408 prop_binCRF_electron2018_bin18_sm
409 prop_binCRWS_muon2016M_bin3
410 prop_binSR_muon2017_bin12_sm
411 prop_binCRF_muon2018_bin11_sm
412 prop_binCRTT_muon2016M_bin15
413 prop_binSR_muon2017_bin11_sm
414 prop_binCRF_electron2018_bin20
415 prop_binCRF_muon2017_bin10_sm
416 prop_binCRF_muon2018_bin9_sm
417 prop_binCRF_electron2017_bin5_sm
418 prop_binSR_electron2017_bin13_sm
419 prop_binSR_muon2016M_bin7_sm
420 prop_binCRF_electron2018_bin8_sm
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

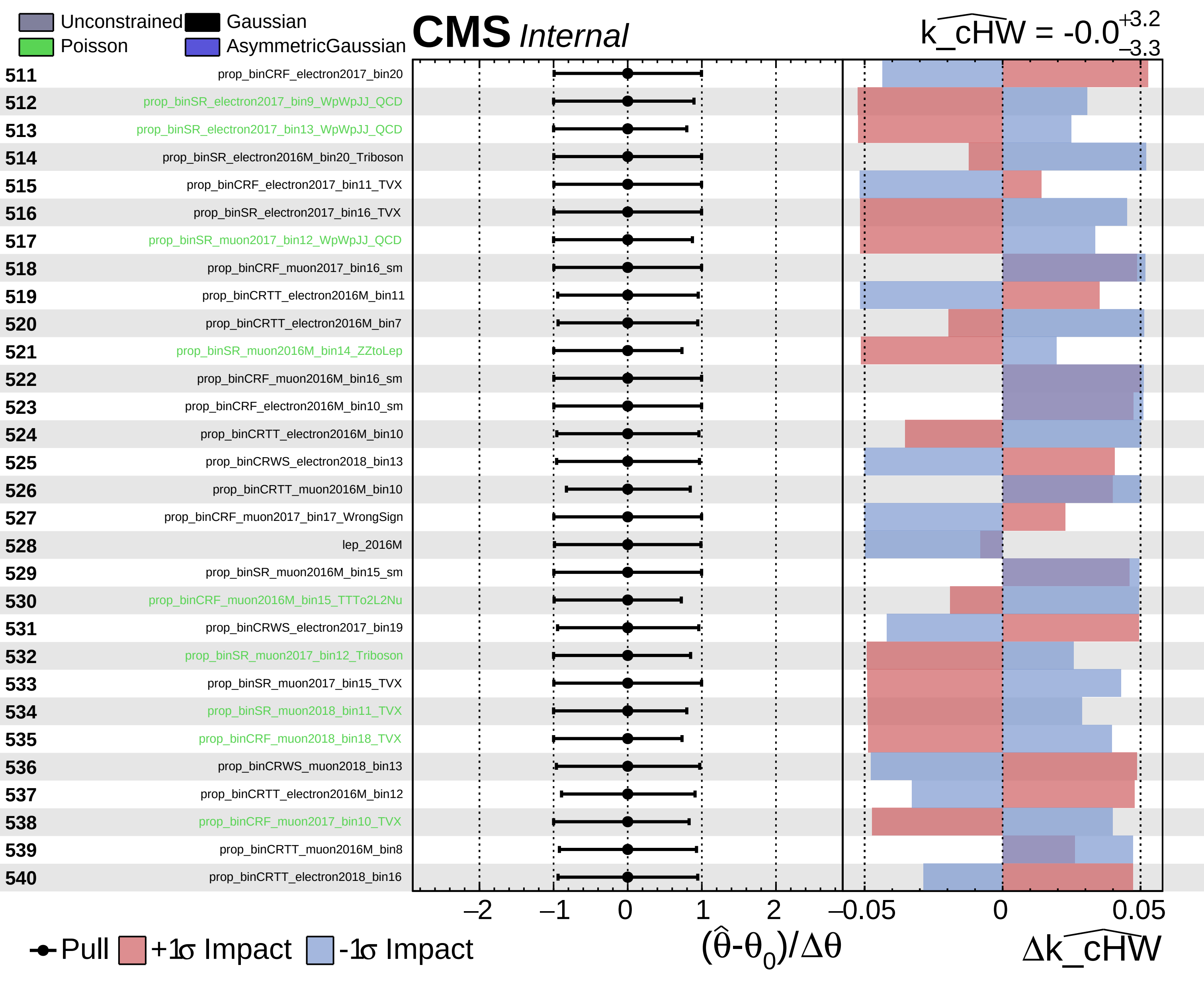

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
421 prop_binCRTT_muon2017_bin16
422 prop_binCRF_muon2017_bin19_sm
423 prop_binCRWS_muon2017_bin8_DYJetsToLL_FxFx
424 prop_binCRWS_electron2016M_bin8_Fake
425 prop_binCRF_muon2018_bin17_sm
426 prop_binCRWS_electron2018_bin8
427 prop_binCRF_electron2017_bin17_sm
428 prop_binSR_electron2016M_bin11_sm
429 prop_binCRF_muon2016M_bin5_WZ
430 prop_binCRF_electron2017_bin12_sm
431 prop_binCRF_muon2018_bin5_sm
432 prop_binCRF_electron2018_bin17_sm
433 prop_binCRTT_electron2017_bin4
434 prop_binSR_electron2016M_bin17_sm
435 prop_binCRF_muon2017_bin9_sm
436 prop_binCRF_electron2017_bin11_sm
437 prop_binCRF_muon2017_bin16_TTTo2L2Nu
438 prop_binCRF_electron2017_bin4_sm
439 prop_binCRTT_electron2017_bin5
440 prop_binCRWS_muon2016M_bin13
441 prop_binCRF_muon2018_bin19
442 prop_binSR_muon2017_bin10
443 prop_binCRF_muon2016M_bin12_sm
444 prop_binCRTT_muon2016M_bin7
445 prop_binSR_electron2016M_bin20_WpWpJJ_QCD
446 prop_binSR_electron2016M_bin9_TTTo2L2Nu
447 prop_binCRTT_electron2017_bin2
448 prop_binCRF_muon2017_bin14_sm
449 prop_binCRTT_muon2018_bin19
450 prop_binSR_electron2018_bin18_sm
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



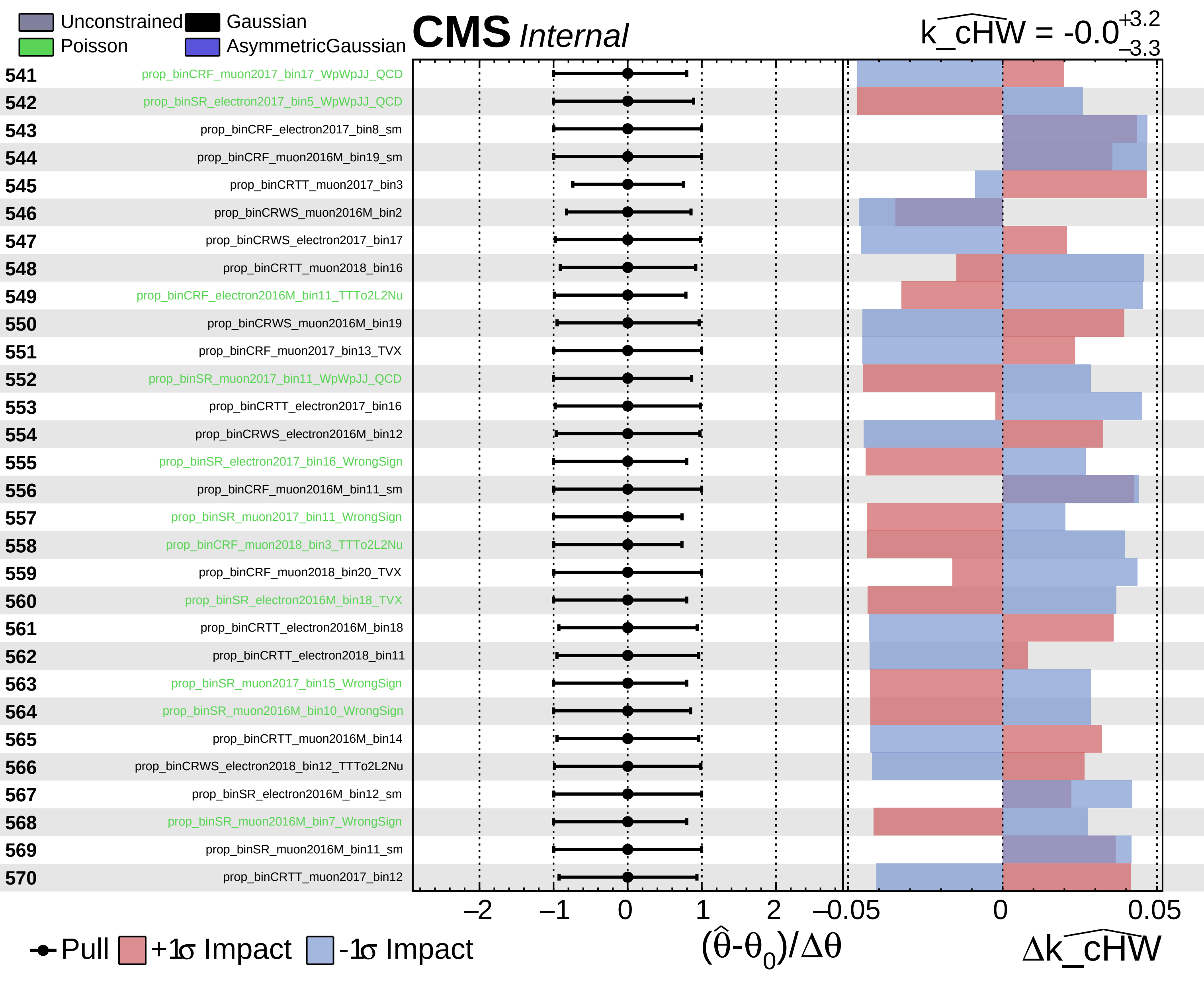
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
451 prop_binCRTT_muon2018_bin7
452 prop_binCRF_muon2018_bin8
453 prop_binCRF_muon2016M_bin18_sm
454 prop_binSR_electron2017_bin9_TVX
455 prop_binCRF_muon2016M_bin6_TTTo2L2Nu
456 prop_binCRF_muon2017_bin8
457 prop_binSR_electron2017_bin5_TVX
458 prop_binSR_electron2017_bin18_sm
459 prop_binSR_muon2016M_bin8_sm
460 prop_binCRWS_muon2016M_bin11
461 prop_binSR_muon2016M_bin20_Triboson
462 prop_binSR_muon2016M_bin13_sm
463 prop_binCRF_muon2016M_bin8_sm
464 prop_binCRF_electron2018_bin19
465 prop_binCRTT_muon2017_bin8
466 prop_binSR_muon2016M_bin17_sm
467 prop_binCRWS_electron2016M_bin5
468 prop_binCRTT_electron2017_bin3
469 prop_binCRWS_electron2018_bin5
470 prop_binCRF_electron2017_bin14_sm
471 prop_binSR_muon2018_bin11_WpWpJJ_QCD
472 prop_binSR_muon2016M_bin14_sm
473 prop_binCRF_electron2018_bin10_sm
474 prop_binCRF_electron2017_bin13_TTTo2L2Nu
475 prop_binCRF_muon2018_bin13_sm
476 prop_binCRF_electron2016M_bin14_ZZtoLep
477 prop_binSR_muon2016M_bin12
478 prop_binCRF_muon2017_bin20_TVX
479 prop_binCRF_muon2016M_bin20_sm
480 prop_binSR_electron2016M_bin14
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW



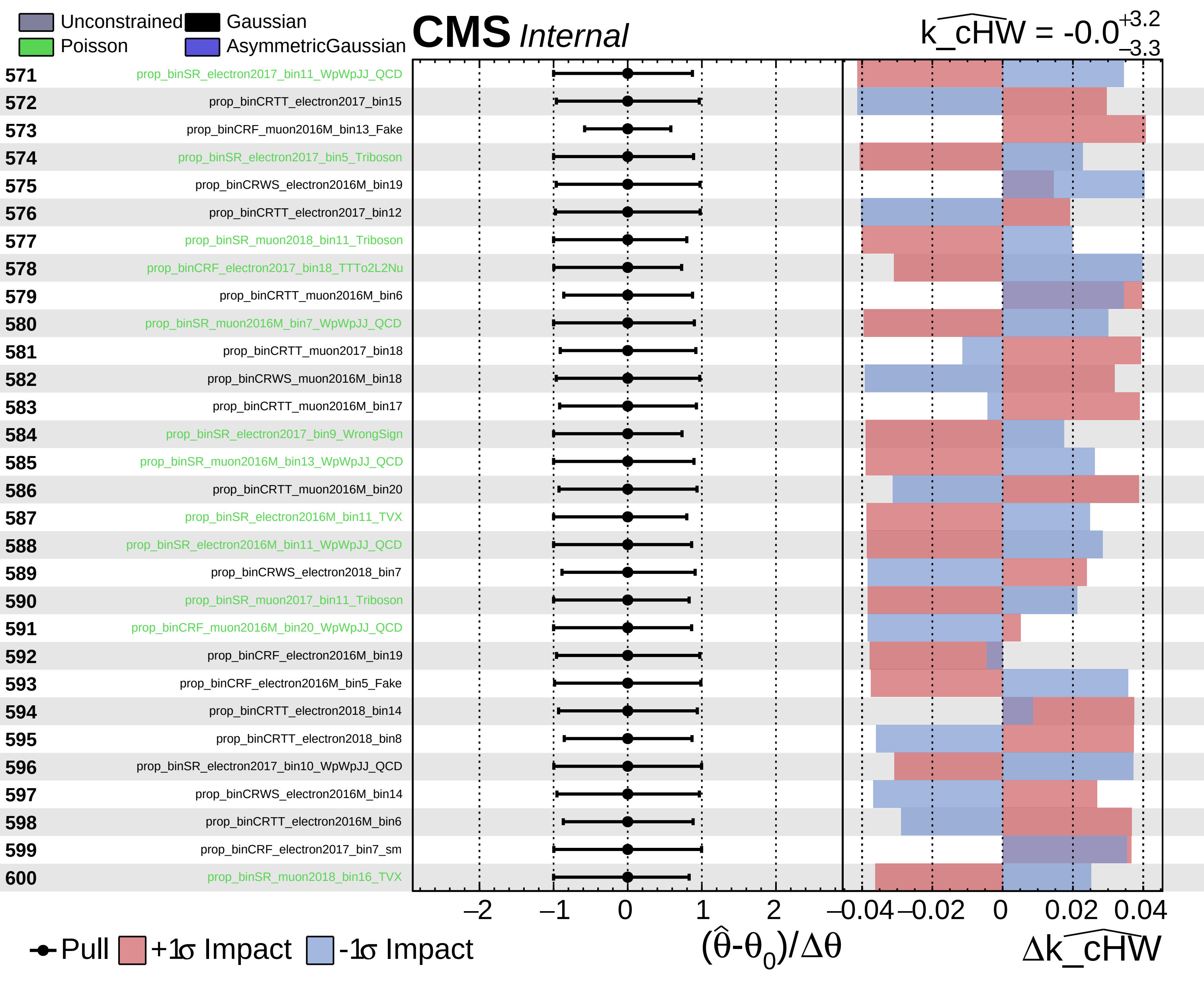
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
481 prop_binCRF_electron2016M_bin15
482 prop_binSR_electron2016M_bin10
483 prop_binCRF_electron2017_bin19
484 prop_binSR_muon2016M_bin20_TVX
485 prop_binCRF_electron2016M_bin18_sm
486 prop_binCRF_electron2017_bin18_sm
487 prop_binSR_electron2016M_bin18_sm
488 prop_binCRTT_electron2016M_bin20
489 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
490 prop_binCRF_electron2018_bin11
491 prop_binCRWS_electron2017_bin14_Fake
492 prop_binCRF_electron2018_bin13_WrongSign
493 prop_binSR_electron2017_bin13_Triboson
494 prop_binCRF_electron2017_bin14_ZZtoLep
495 prop_binSR_muon2016M_bin16_sm
496 prop_binCRTT_electron2018_bin15
497 prop_binSR_muon2016M_bin10_sm
498 prop_binSR_electron2016M_bin16_sm
499 prop_binSR_electron2018_bin18_ZZtoLep
500 prop_binCRTT_electron2018_bin12
501 prop_binCRTT_electron2018_bin5
502 prop_binSR_electron2016M_bin9_sm
503 prop_binCRWS_electron2017_bin20
504 prop_binSR_muon2016M_bin13_TVX
505 prop_binCRWS_electron2016M_bin16_DYJetsToLL_FxFx
506 prop_binCRF_electron2017_bin9
507 prop_binCRWS_electron2016M_bin6_Fake
508 prop_binSR_muon2016M_bin8_TVX
509 prop_binCRF_muon2017_bin7_sm
510 prop_binCRTT_muon2017_bin15
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

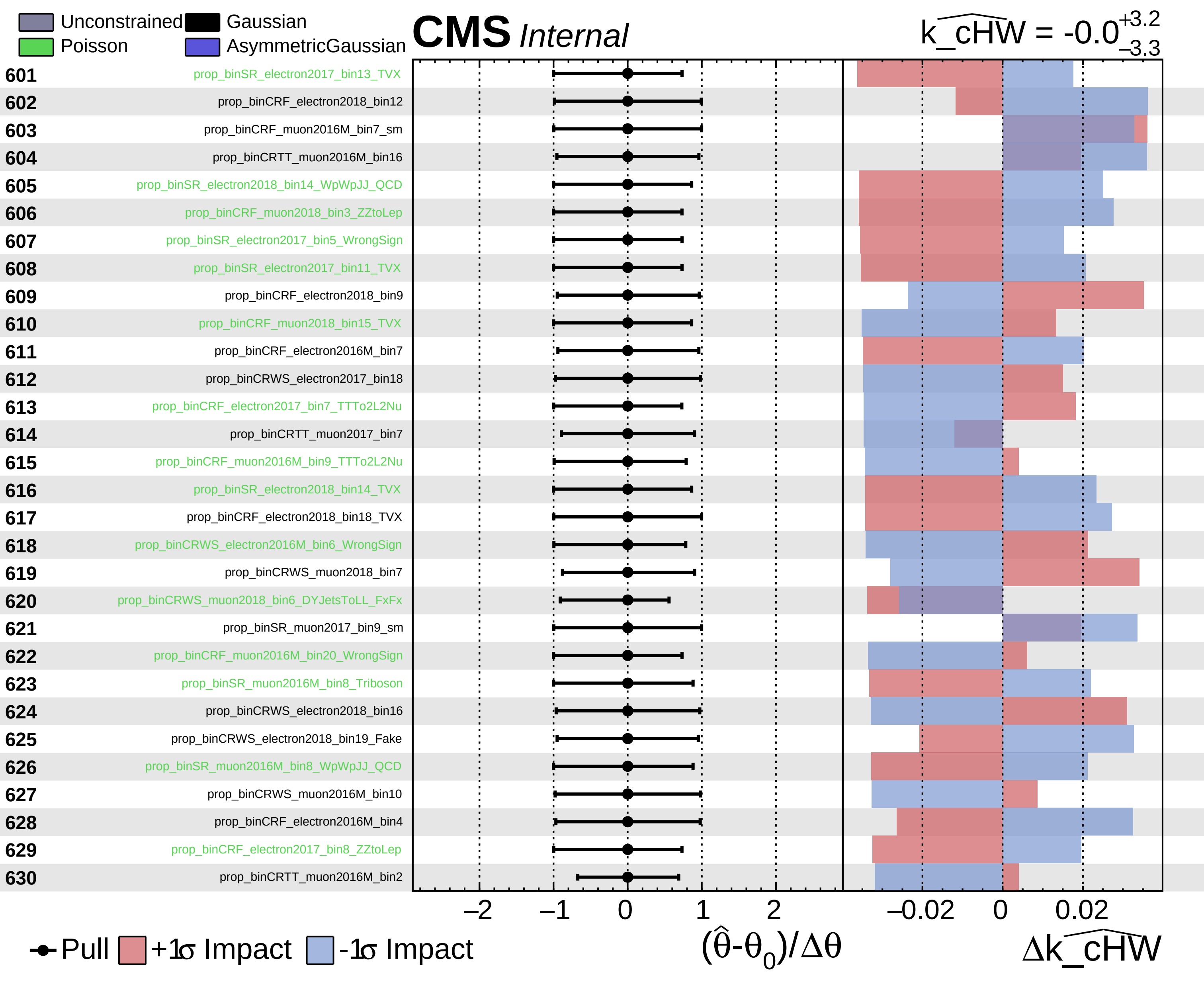


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
511 prop_binCRF_electron2017_bin20
512 prop_binSR_electron2017_bin9_WpWpJJ_QCD
513 prop_binSR_electron2017_bin13_WpWpJJ_QCD
514 prop_binSR_electron2016M_bin20_Triboson
515 prop_binCRF_electron2017_bin11_TVX
516 prop_binSR_electron2017_bin16_TVX
517 prop_binSR_muon2017_bin12_WpWpJJ_QCD
518 prop_binCRF_muon2017_bin16_sm
519 prop_binCRTT_electron2016M_bin11
520 prop_binCRTT_electron2016M_bin7
521 prop_binSR_muon2016M_bin14_ZZtoLep
522 prop_binCRF_muon2016M_bin16_sm
523 prop_binCRF_electron2016M_bin10_sm
524 prop_binCRTT_electron2016M_bin10
525 prop_binCRWS_electron2018_bin13
526 prop_binCRTT_muon2016M_bin10
527 prop_binCRF_muon2017_bin17_WrongSign
528 lep_2016M
529 prop_binSR_muon2016M_bin15_sm
530 prop_binCRF_muon2016M_bin15_TTTo2L2Nu
531 prop_binCRWS_electron2017_bin19
532 prop_binSR_muon2017_bin12_Triboson
533 prop_binSR_muon2017_bin15_TVX
534 prop_binSR_muon2018_bin11_TVX
535 prop_binCRF_muon2018_bin18_TVX
536 prop_binCRWS_muon2018_bin13
537 prop_binCRTT_electron2016M_bin12
538 prop_binCRF_muon2017_bin10_TVX
539 prop_binCRTT_muon2016M_bin8
540 prop_binCRTT_electron2018_bin16
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

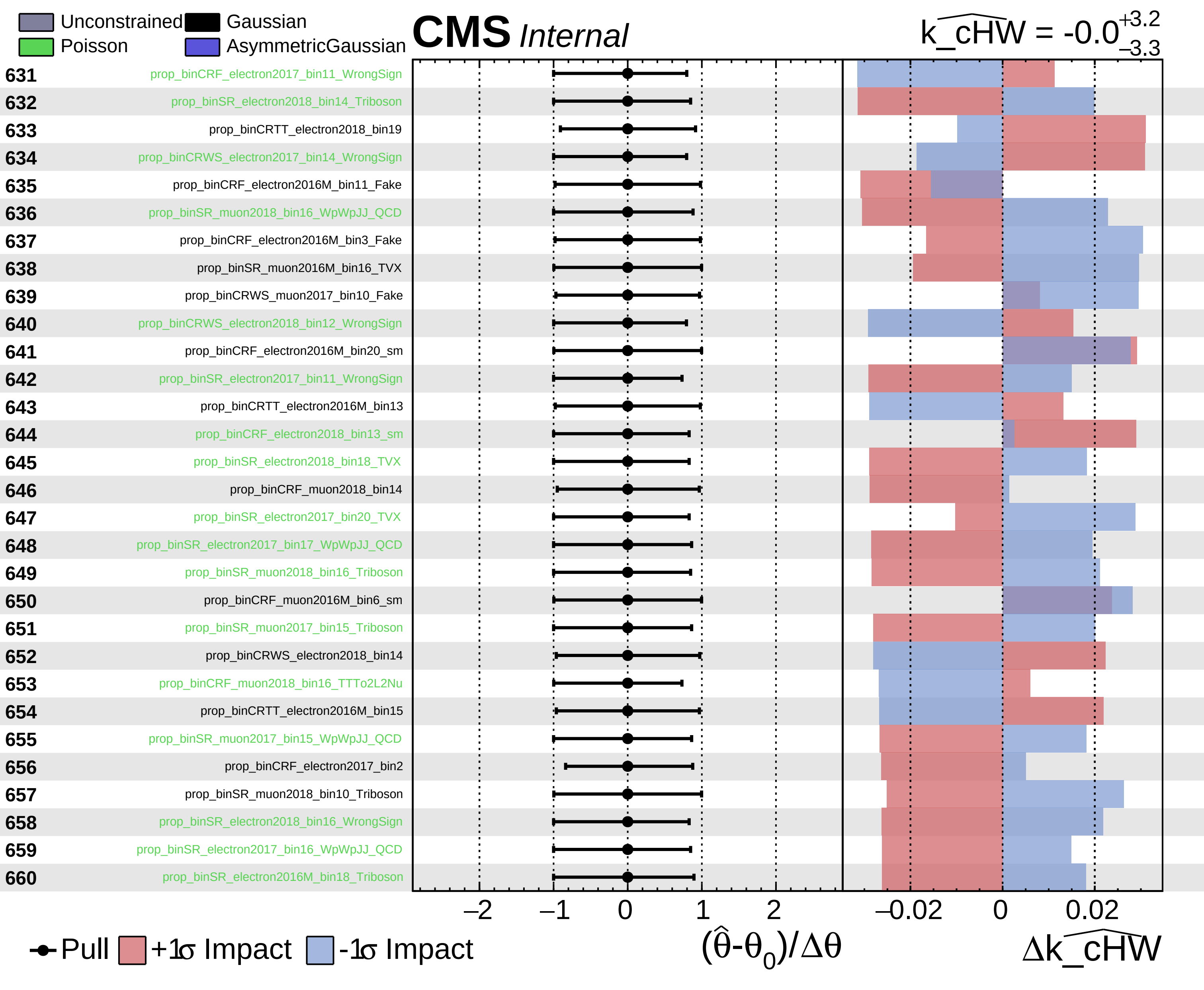


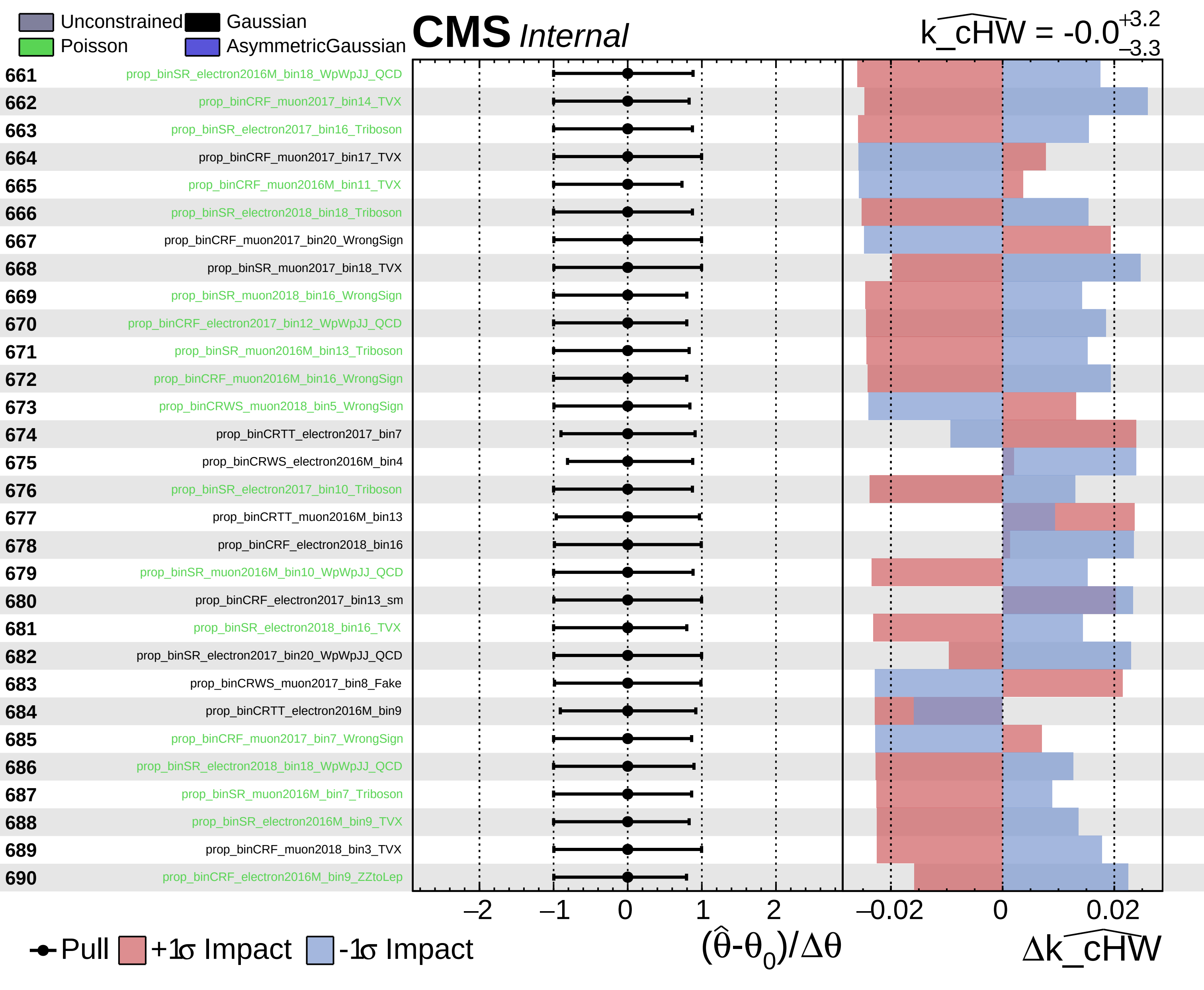
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
541 prop_binCRF_muon2017_bin17_WpWpJJ_QCD
542 prop_binSR_electron2017_bin5_WpWpJJ_QCD
543 prop_binCRF_electron2017_bin8_sm
544 prop_binCRF_muon2016M_bin19_sm
545 prop_binCRTT_muon2017_bin3
546 prop_binCRWS_muon2016M_bin2
547 prop_binCRWS_electron2017_bin17
548 prop_binCRTT_muon2018_bin16
549 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
550 prop_binCRWS_muon2016M_bin19
551 prop_binCRF_muon2017_bin13_TVX
552 prop_binSR_muon2017_bin11_WpWpJJ_QCD
553 prop_binCRTT_electron2017_bin16
554 prop_binCRWS_electron2016M_bin12
555 prop_binSR_electron2017_bin16_WrongSign
556 prop_binCRF_muon2016M_bin11_sm
557 prop_binSR_muon2017_bin11_WrongSign
558 prop_binCRF_muon2018_bin3_TTTo2L2Nu
559 prop_binCRF_muon2018_bin20_TVX
560 prop_binSR_electron2016M_bin18_TVX
561 prop_binCRTT_electron2016M_bin18
562 prop_binCRTT_electron2018_bin11
563 prop_binSR_muon2017_bin15_WrongSign
564 prop_binSR_muon2016M_bin10_WrongSign
565 prop_binCRTT_muon2016M_bin14
566 prop_binCRWS_electron2018_bin12_TTTo2L2Nu
567 prop_binSR_electron2016M_bin12_sm
568 prop_binSR_muon2016M_bin7_WrongSign
569 prop_binSR_muon2016M_bin11_sm
570 prop_binCRTT_muon2017_bin12
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW

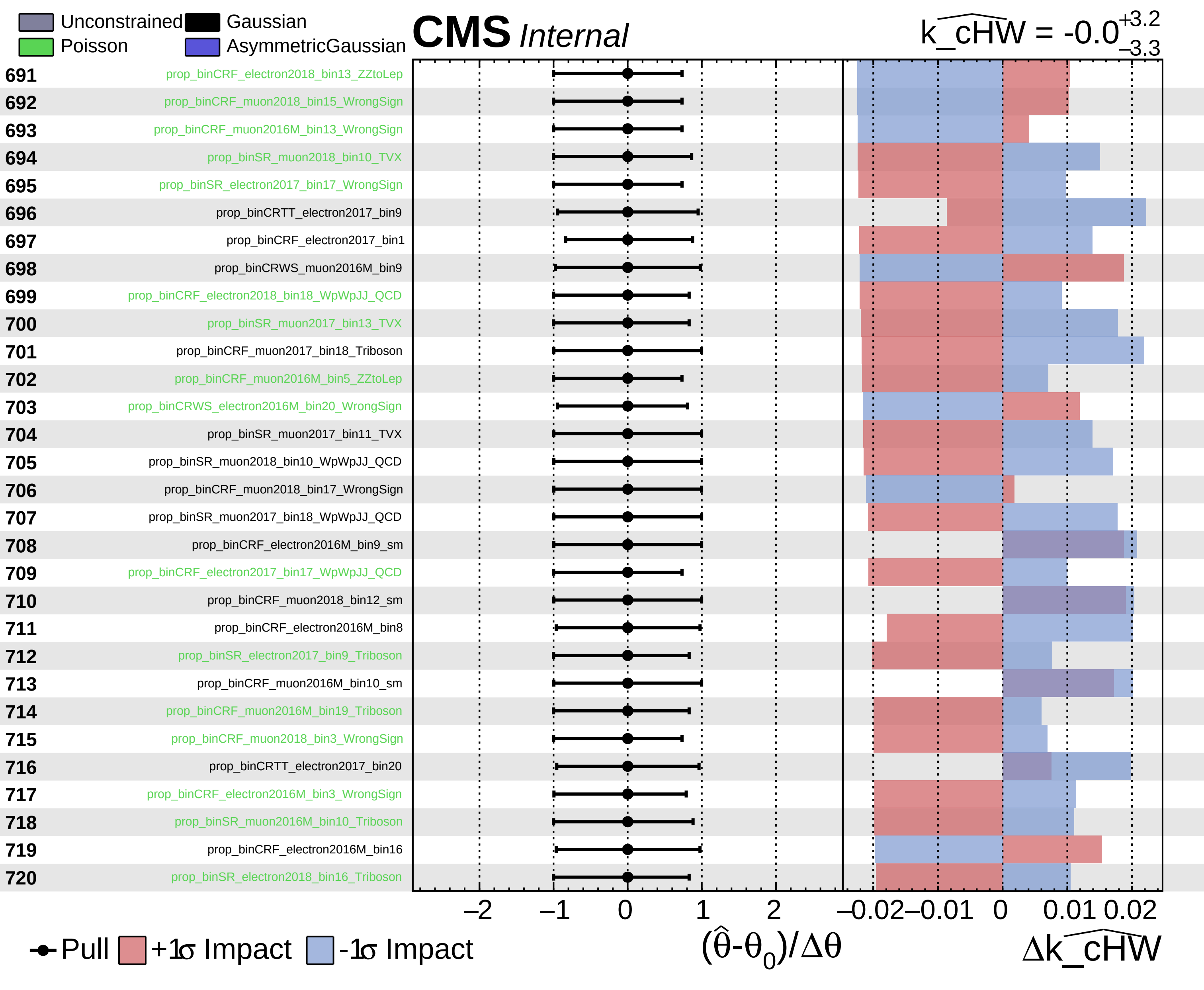

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
571 prop_binSR_electron2017_bin11_WpWpJJ_QCD
572 prop_binCRTT_electron2017_bin15
573 prop_binCRF_muon2016M_bin13_Fake
574 prop_binSR_electron2017_bin5_Triboson
575 prop_binCRWS_electron2016M_bin19
576 prop_binCRTT_electron2017_bin12
577 prop_binSR_muon2018_bin11_Triboson
578 prop_binCRF_electron2017_bin18_TTTo2L2Nu
579 prop_binCRTT_muon2016M_bin6
580 prop_binSR_muon2016M_bin7_WpWpJJ_QCD
581 prop_binCRTT_muon2017_bin18
582 prop_binCRWS_muon2016M_bin18
583 prop_binCRTT_muon2016M_bin17
584 prop_binSR_electron2017_bin9_WrongSign
585 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
586 prop_binCRTT_muon2016M_bin20
587 prop_binSR_electron2016M_bin11_TVX
588 prop_binSR_electron2016M_bin11_WpWpJJ_QCD
589 prop_binCRWS_electron2018_bin7
590 prop_binSR_muon2017_bin11_Triboson
591 prop_binCRF_muon2016M_bin20_WpWpJJ_QCD
592 prop_binCRF_electron2016M_bin19
593 prop_binCRF_electron2016M_bin5_Fake
594 prop_binCRTT_electron2018_bin14
595 prop_binCRTT_electron2018_bin8
596 prop_binSR_electron2017_bin10_WpWpJJ_QCD
597 prop_binCRWS_electron2016M_bin14
598 prop_binCRTT_electron2016M_bin6
599 prop_binCRF_electron2017_bin7_sm
600 prop_binSR_muon2018_bin16_TVX
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW



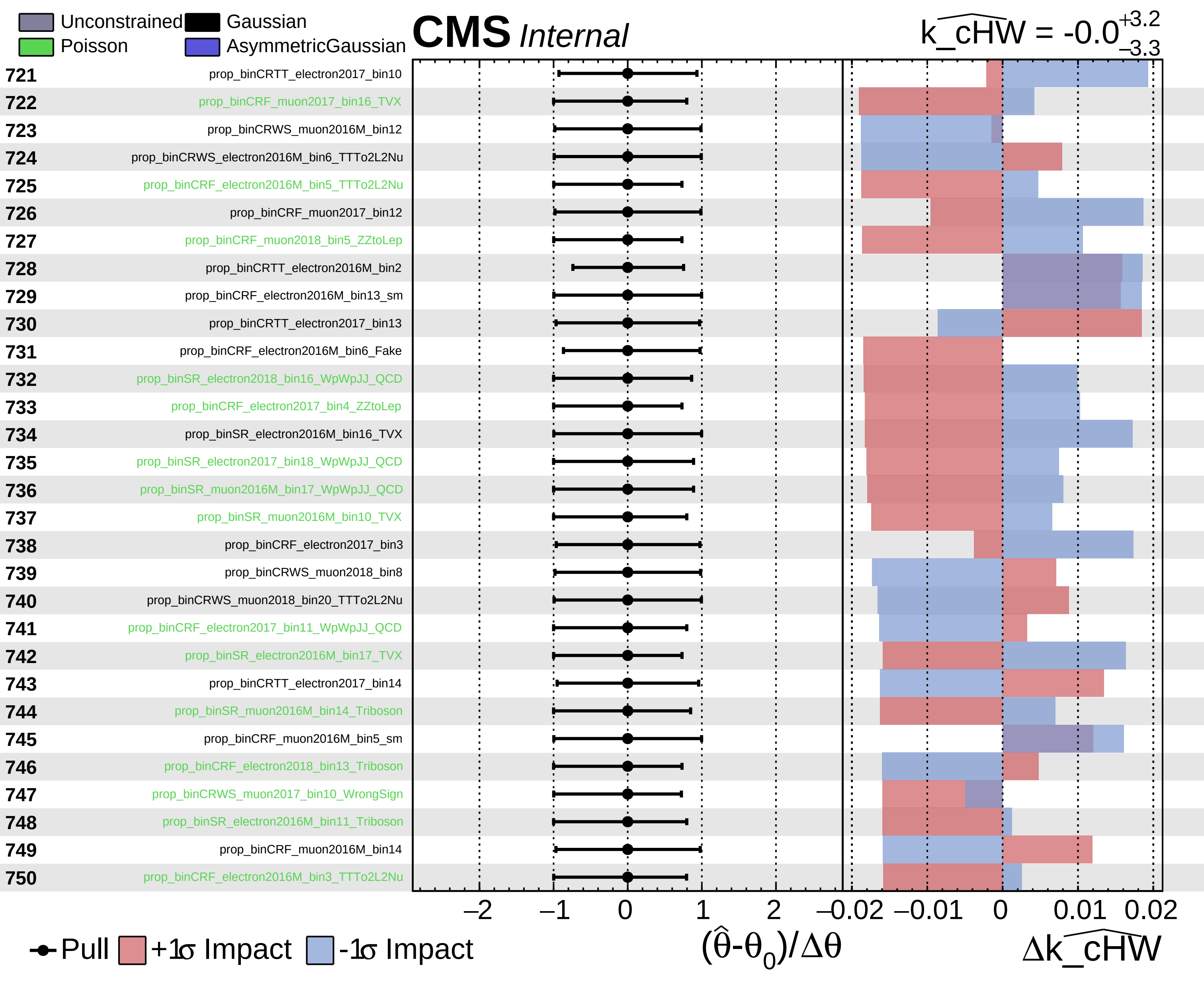
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
601 prop_binSR_electron2017_bin13_TVX
602 prop_binCRF_electron2018_bin12
603 prop_binCRF_muon2016M_bin7_sm
604 prop_binCRTT_muon2016M_bin16
605 prop_binSR_electron2018_bin14_WpWpJJ_QCD
606 prop_binCRF_muon2018_bin3_ZZtoLep
607 prop_binSR_electron2017_bin5_WrongSign
608 prop_binSR_electron2017_bin11_TVX
609 prop_binCRF_electron2018_bin9
610 prop_binCRF_muon2018_bin15_TVX
611 prop_binCRF_electron2016M_bin7
612 prop_binCRWS_electron2017_bin18
613 prop_binCRF_electron2017_bin7_TTTo2L2Nu
614 prop_binCRTT_muon2017_bin7
615 prop_binCRF_muon2016M_bin9_TTTo2L2Nu
616 prop_binSR_electron2018_bin14_TVX
617 prop_binCRF_electron2018_bin18_TVX
618 prop_binCRWS_electron2016M_bin6_WrongSign
619 prop_binCRWS_muon2018_bin7
620 prop_binCRWS_muon2018_bin6_DYJetsToLL_FxFx
621 prop_binSR_muon2017_bin9_sm
622 prop_binCRF_muon2016M_bin20_WrongSign
623 prop_binSR_muon2016M_bin8_Triboson
624 prop_binCRWS_electron2018_bin16
625 prop_binCRWS_electron2018_bin19_Fake
626 prop_binSR_muon2016M_bin8_WpWpJJ_QCD
627 prop_binCRWS_muon2016M_bin10
628 prop_binCRF_electron2016M_bin4
629 prop_binCRF_electron2017_bin8_ZZtoLep
630 prop_binCRTT_muon2016M_bin2
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
631 prop_binCRF_electron2017_bin11_WrongSign
632 prop_binSR_electron2018_bin14_Triboson
633 prop_binCRTT_electron2018_bin19
634 prop_binCRWS_electron2017_bin14_WrongSign
635 prop_binCRF_electron2016M_bin11_Fake
636 prop_binSR_muon2018_bin16_WpWpJJ_QCD
637 prop_binCRF_electron2016M_bin3_Fake
638 prop_binSR_muon2016M_bin16_TVX
639 prop_binCRWS_muon2017_bin10_Fake
640 prop_binCRWS_electron2018_bin12_WrongSign
641 prop_binCRF_electron2016M_bin20_sm
642 prop_binSR_electron2017_bin11_WrongSign
643 prop_binCRTT_electron2016M_bin13
644 prop_binCRF_electron2018_bin13_sm
645 prop_binSR_electron2018_bin18_TVX
646 prop_binCRF_muon2018_bin14
647 prop_binSR_electron2017_bin20_TVX
648 prop_binSR_electron2017_bin17_WpWpJJ_QCD
649 prop_binSR_muon2018_bin16_Triboson
650 prop_binCRF_muon2016M_bin6_sm
651 prop_binSR_muon2017_bin15_Triboson
652 prop_binCRWS_electron2018_bin14
653 prop_binCRF_muon2018_bin16_TTTo2L2Nu
654 prop_binCRTT_electron2016M_bin15
655 prop_binSR_muon2017_bin15_WpWpJJ_QCD
656 prop_binCRF_electron2017_bin2
657 prop_binSR_muon2018_bin10_Triboson
658 prop_binSR_electron2018_bin16_WrongSign
659 prop_binSR_electron2017_bin16_WpWpJJ_QCD
660 prop_binSR_electron2016M_bin18_Triboson
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

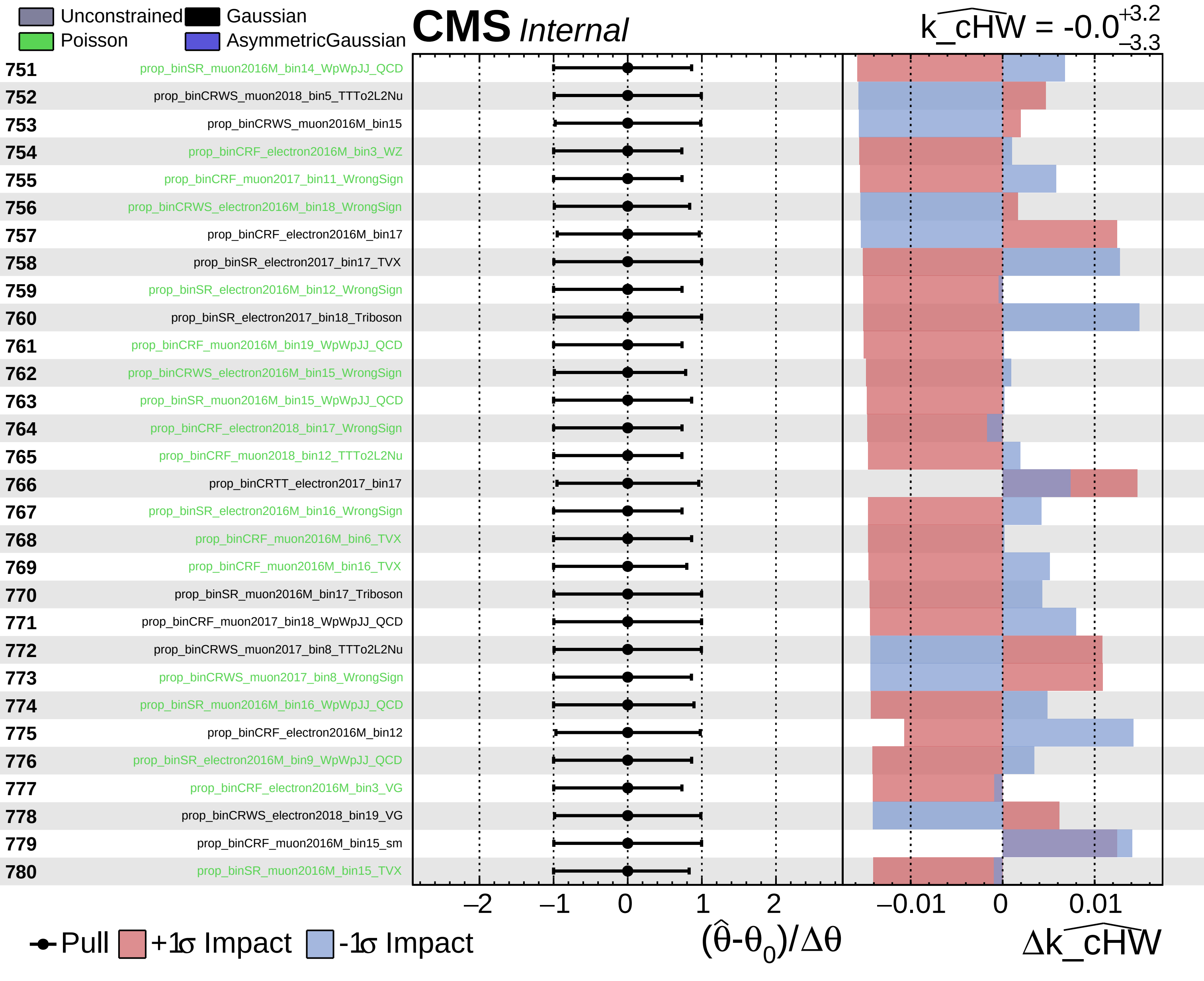

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
661 prop_binSR_electron2016M_bin18_WpWpJJ_QCD
662 prop_binCRF_muon2017_bin14_TVX
663 prop_binSR_electron2017_bin16_Triboson
664 prop_binCRF_muon2017_bin17_TVX
665 prop_binCRF_muon2016M_bin11_TVX
666 prop_binSR_electron2018_bin18_Triboson
667 prop_binCRF_muon2017_bin20_WrongSign
668 prop_binSR_muon2017_bin18_TVX
669 prop_binSR_muon2018_bin16_WrongSign
670 prop_binCRF_electron2017_bin12_WpWpJJ_QCD
671 prop_binSR_muon2016M_bin13_Triboson
672 prop_binCRF_muon2016M_bin16_WrongSign
673 prop_binCRWS_muon2018_bin5_WrongSign
674 prop_binCRTT_electron2017_bin7
675 prop_binCRWS_electron2016M_bin4
676 prop_binSR_electron2017_bin10_Triboson
677 prop_binCRTT_muon2016M_bin13
678 prop_binCRF_electron2018_bin16
679 prop_binSR_muon2016M_bin10_WpWpJJ_QCD
680 prop_binCRF_electron2017_bin13_sm
681 prop_binSR_electron2018_bin16_TVX
682 prop_binSR_electron2017_bin20_WpWpJJ_QCD
683 prop_binCRWS_muon2017_bin8_Fake
684 prop_binCRTT_electron2016M_bin9
685 prop_binCRF_muon2017_bin7_WrongSign
686 prop_binSR_electron2018_bin18_WpWpJJ_QCD
687 prop_binSR_muon2016M_bin7_Triboson
688 prop_binSR_electron2016M_bin9_TVX
689 prop_binCRF_muon2018_bin3_TVX
690 prop_binCRF_electron2016M_bin9_ZZtoLep
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

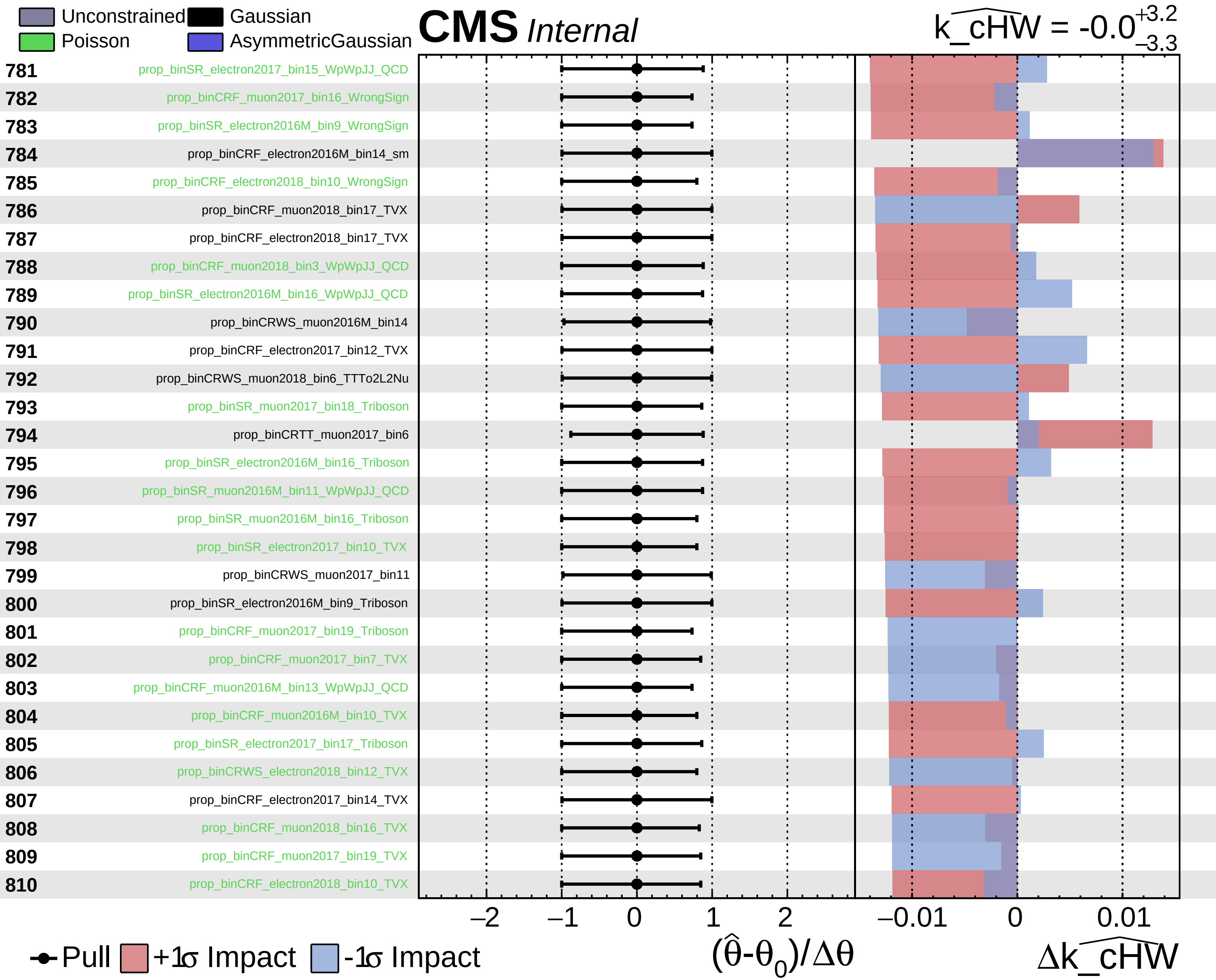


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
691 prop_binCRF_electron2018_bin13_ZZtoLep
692 prop_binCRF_muon2018_bin15_WrongSign
693 prop_binCRF_muon2016M_bin13_WrongSign
694 prop_binSR_muon2018_bin10_TVX
695 prop_binSR_electron2017_bin17_WrongSign
696 prop_binCRTT_electron2017_bin9
697 prop_binCRF_electron2017_bin1
698 prop_binCRWS_muon2016M_bin9
699 prop_binCRF_electron2018_bin18_WpWpJJ_QCD
700 prop_binSR_muon2017_bin13_TVX
701 prop_binCRF_muon2017_bin18_Triboson
702 prop_binCRF_muon2016M_bin5_ZZtoLep
703 prop_binCRWS_electron2016M_bin20_WrongSign
704 prop_binSR_muon2017_bin11_TVX
705 prop_binSR_muon2018_bin10_WpWpJJ_QCD
706 prop_binCRF_muon2018_bin17_WrongSign
707 prop_binSR_muon2017_bin18_WpWpJJ_QCD
708 prop_binCRF_electron2016M_bin9_sm
709 prop_binCRF_electron2017_bin17_WpWpJJ_QCD
710 prop_binCRF_muon2018_bin12_sm
711 prop_binCRF_electron2016M_bin8
712 prop_binSR_electron2017_bin9_Triboson
713 prop_binCRF_muon2016M_bin10_sm
714 prop_binCRF_muon2016M_bin19_Triboson
715 prop_binCRF_muon2018_bin3_WrongSign
716 prop_binCRTT_electron2017_bin20
717 prop_binCRF_electron2016M_bin3_WrongSign
718 prop_binSR_muon2016M_bin10_Triboson
719 prop_binCRF_electron2016M_bin16
720 prop_binSR_electron2018_bin16_Triboson
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

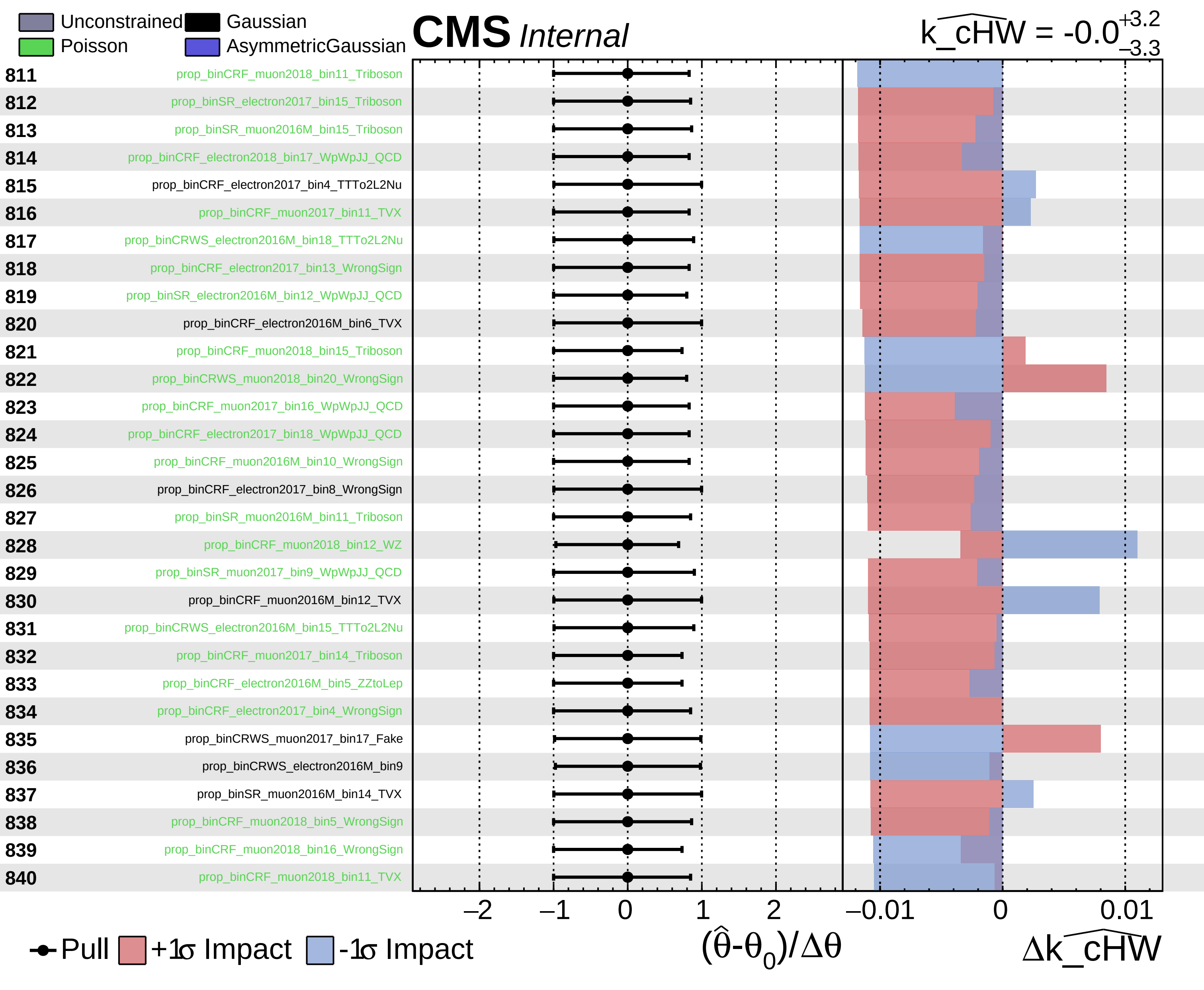


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0+3.2 −3.3
721 prop_binCRTT_electron2017_bin10
722 prop_binCRF_muon2017_bin16_TVX
723 prop_binCRWS_muon2016M_bin12
724 prop_binCRWS_electron2016M_bin6_TTTo2L2Nu
725 prop_binCRF_electron2016M_bin5_TTTo2L2Nu
726 prop_binCRF_muon2017_bin12
727 prop_binCRF_muon2018_bin5_ZZtoLep
728 prop_binCRTT_electron2016M_bin2
729 prop_binCRF_electron2016M_bin13_sm
730 prop_binCRTT_electron2017_bin13
731 prop_binCRF_electron2016M_bin6_Fake
732 prop_binSR_electron2018_bin16_WpWpJJ_QCD
733 prop_binCRF_electron2017_bin4_ZZtoLep
734 prop_binSR_electron2016M_bin16_TVX
735 prop_binSR_electron2017_bin18_WpWpJJ_QCD
736 prop_binSR_muon2016M_bin17_WpWpJJ_QCD
737 prop_binSR_muon2016M_bin10_TVX
738 prop_binCRF_electron2017_bin3
739 prop_binCRWS_muon2018_bin8
740 prop_binCRWS_muon2018_bin20_TTTo2L2Nu
741 prop_binCRF_electron2017_bin11_WpWpJJ_QCD
742 prop_binSR_electron2016M_bin17_TVX
743 prop_binCRTT_electron2017_bin14
744 prop_binSR_muon2016M_bin14_Triboson
745 prop_binCRF_muon2016M_bin5_sm
746 prop_binCRF_electron2018_bin13_Triboson
747 prop_binCRWS_muon2017_bin10_WrongSign
748 prop_binSR_electron2016M_bin11_Triboson
749 prop_binCRF_muon2016M_bin14
750 prop_binCRF_electron2016M_bin3_TTTo2L2Nu
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

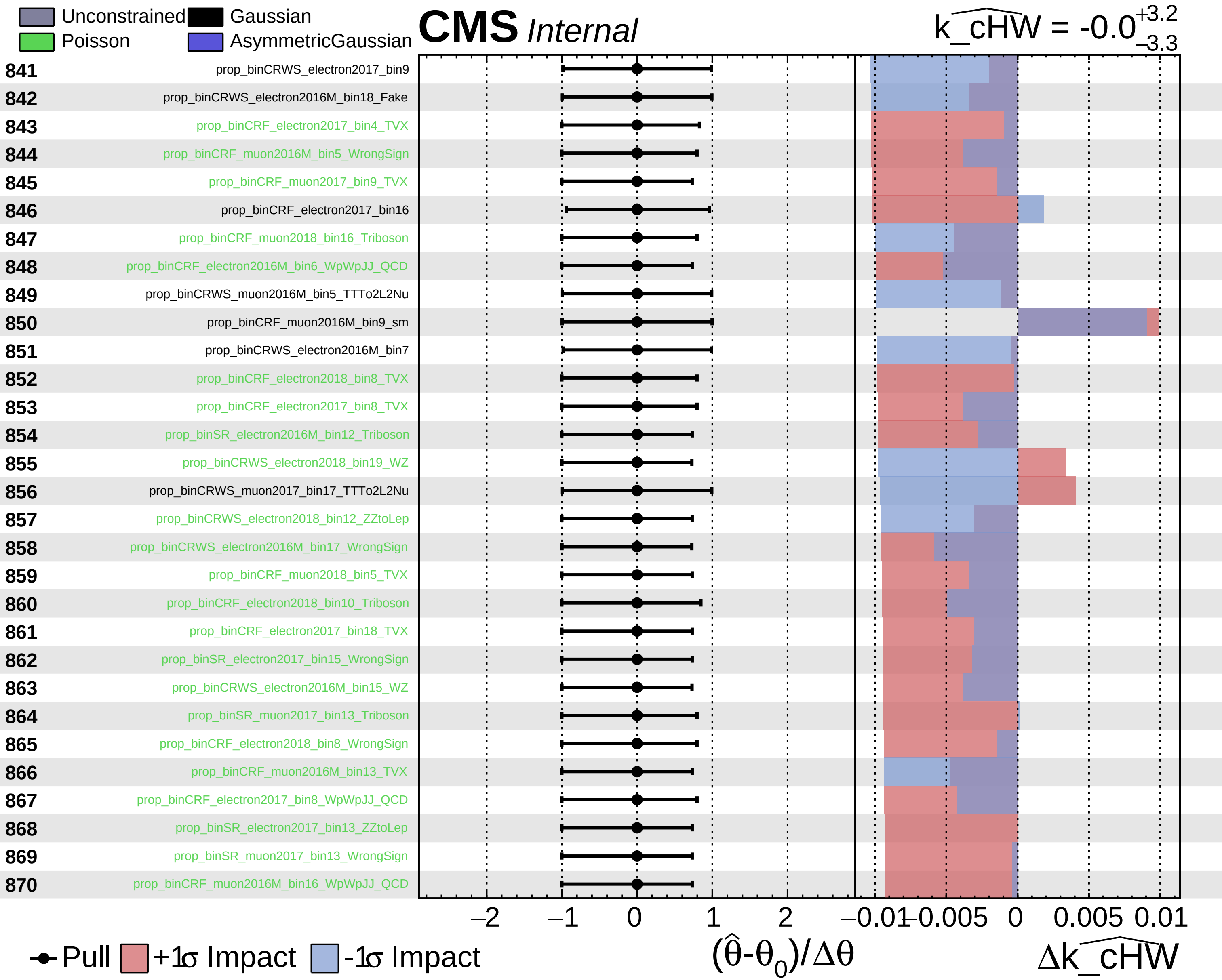

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
751 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
752 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
753 prop_binCRWS_muon2016M_bin15
754 prop_binCRF_electron2016M_bin3_WZ
755 prop_binCRF_muon2017_bin11_WrongSign
756 prop_binCRWS_electron2016M_bin18_WrongSign
757 prop_binCRF_electron2016M_bin17
758 prop_binSR_electron2017_bin17_TVX
759 prop_binSR_electron2016M_bin12_WrongSign
760 prop_binSR_electron2017_bin18_Triboson
761 prop_binCRF_muon2016M_bin19_WpWpJJ_QCD
762 prop_binCRWS_electron2016M_bin15_WrongSign
763 prop_binSR_muon2016M_bin15_WpWpJJ_QCD
764 prop_binCRF_electron2018_bin17_WrongSign
765 prop_binCRF_muon2018_bin12_TTTo2L2Nu
766 prop_binCRTT_electron2017_bin17
767 prop_binSR_electron2016M_bin16_WrongSign
768 prop_binCRF_muon2016M_bin6_TVX
769 prop_binCRF_muon2016M_bin16_TVX
770 prop_binSR_muon2016M_bin17_Triboson
771 prop_binCRF_muon2017_bin18_WpWpJJ_QCD
772 prop_binCRWS_muon2017_bin8_TTTo2L2Nu
773 prop_binCRWS_muon2017_bin8_WrongSign
774 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
775 prop_binCRF_electron2016M_bin12
776 prop_binSR_electron2016M_bin9_WpWpJJ_QCD
777 prop_binCRF_electron2016M_bin3_VG
778 prop_binCRWS_electron2018_bin19_VG
779 prop_binCRF_muon2016M_bin15_sm
780 prop_binSR_muon2016M_bin15_TVX
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW



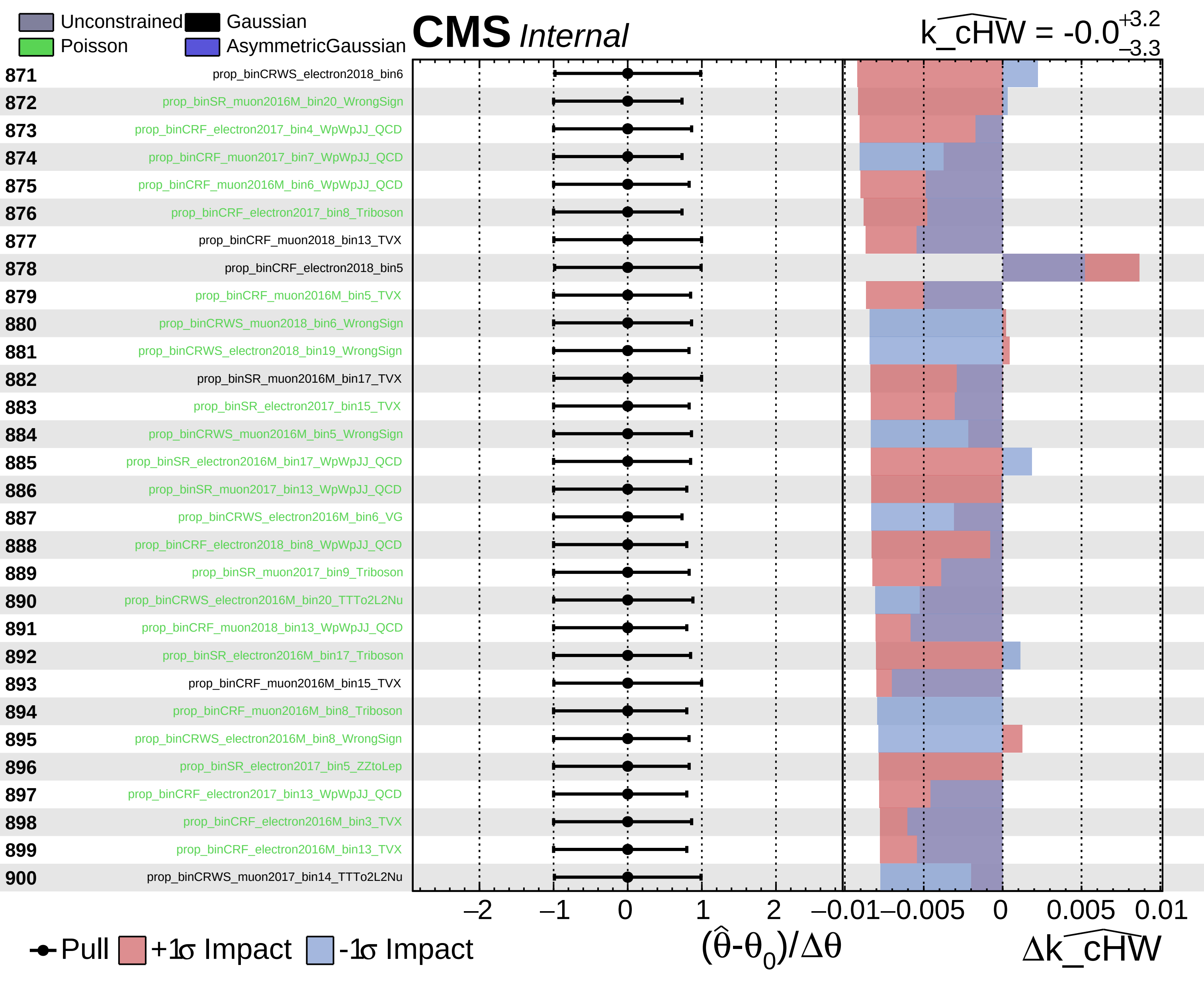
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
781 prop_binSR_electron2017_bin15_WpWpJJ_QCD
782 prop_binCRF_muon2017_bin16_WrongSign
783 prop_binSR_electron2016M_bin9_WrongSign
784 prop_binCRF_electron2016M_bin14_sm
785 prop_binCRF_electron2018_bin10_WrongSign
786 prop_binCRF_muon2018_bin17_TVX
787 prop_binCRF_electron2018_bin17_TVX
788 prop_binCRF_muon2018_bin3_WpWpJJ_QCD
789 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
790 prop_binCRWS_muon2016M_bin14
791 prop_binCRF_electron2017_bin12_TVX
792 prop_binCRWS_muon2018_bin6_TTTo2L2Nu
793 prop_binSR_muon2017_bin18_Triboson
794 prop_binCRTT_muon2017_bin6
795 prop_binSR_electron2016M_bin16_Triboson
796 prop_binSR_muon2016M_bin11_WpWpJJ_QCD
797 prop_binSR_muon2016M_bin16_Triboson
798 prop_binSR_electron2017_bin10_TVX
799 prop_binCRWS_muon2017_bin11
800 prop_binSR_electron2016M_bin9_Triboson
801 prop_binCRF_muon2017_bin19_Triboson
802 prop_binCRF_muon2017_bin7_TVX
803 prop_binCRF_muon2016M_bin13_WpWpJJ_QCD
804 prop_binCRF_muon2016M_bin10_TVX
805 prop_binSR_electron2017_bin17_Triboson
806 prop_binCRWS_electron2018_bin12_TVX
807 prop_binCRF_electron2017_bin14_TVX
808 prop_binCRF_muon2018_bin16_TVX
809 prop_binCRF_muon2017_bin19_TVX
810 prop_binCRF_electron2018_bin10_TVX
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



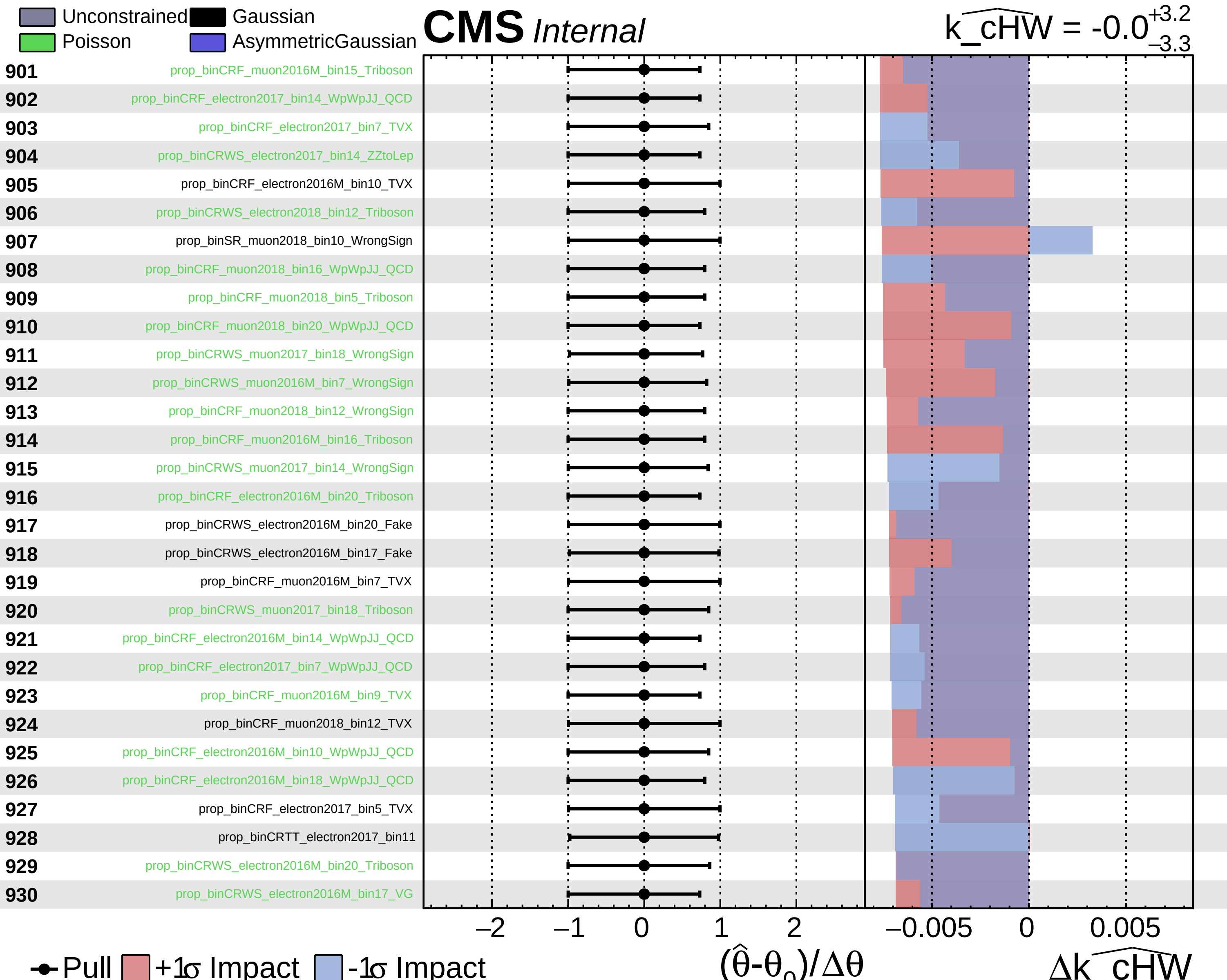
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
811 prop_binCRF_muon2018_bin11_Triboson
812 prop_binSR_electron2017_bin15_Triboson
813 prop_binSR_muon2016M_bin15_Triboson
814 prop_binCRF_electron2018_bin17_WpWpJJ_QCD
815 prop_binCRF_electron2017_bin4_TTTo2L2Nu
816 prop_binCRF_muon2017_bin11_TVX
817 prop_binCRWS_electron2016M_bin18_TTTo2L2Nu
818 prop_binCRF_electron2017_bin13_WrongSign
819 prop_binSR_electron2016M_bin12_WpWpJJ_QCD
820 prop_binCRF_electron2016M_bin6_TVX
821 prop_binCRF_muon2018_bin15_Triboson
822 prop_binCRWS_muon2018_bin20_WrongSign
823 prop_binCRF_muon2017_bin16_WpWpJJ_QCD
824 prop_binCRF_electron2017_bin18_WpWpJJ_QCD
825 prop_binCRF_muon2016M_bin10_WrongSign
826 prop_binCRF_electron2017_bin8_WrongSign
827 prop_binSR_muon2016M_bin11_Triboson
828 prop_binCRF_muon2018_bin12_WZ
829 prop_binSR_muon2017_bin9_WpWpJJ_QCD
830 prop_binCRF_muon2016M_bin12_TVX
831 prop_binCRWS_electron2016M_bin15_TTTo2L2Nu
832 prop_binCRF_muon2017_bin14_Triboson
833 prop_binCRF_electron2016M_bin5_ZZtoLep
834 prop_binCRF_electron2017_bin4_WrongSign
835 prop_binCRWS_muon2017_bin17_Fake
836 prop_binCRWS_electron2016M_bin9
837 prop_binSR_muon2016M_bin14_TVX
838 prop_binCRF_muon2018_bin5_WrongSign
839 prop_binCRF_muon2018_bin16_WrongSign
840 prop_binCRF_muon2018_bin11_TVX
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

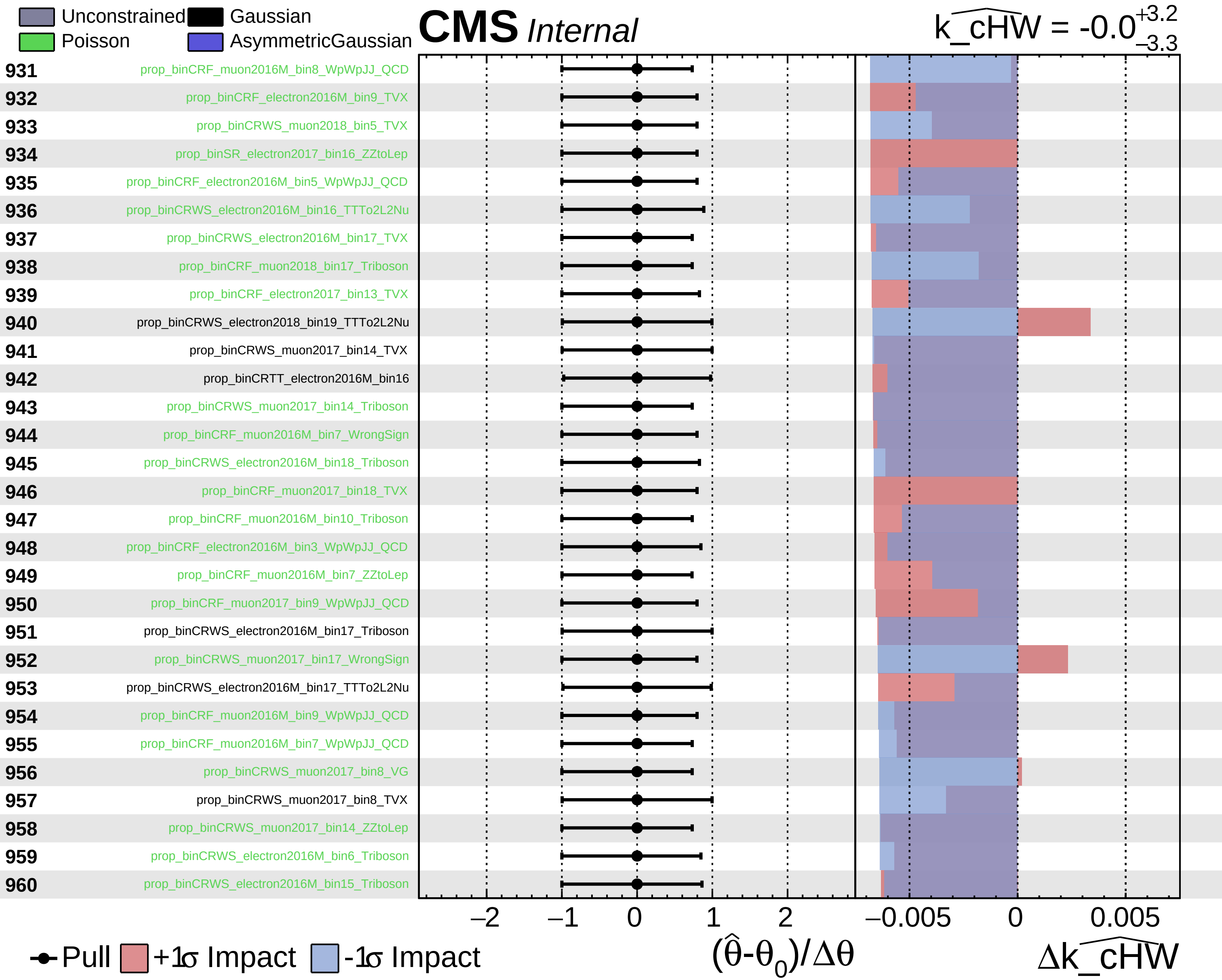


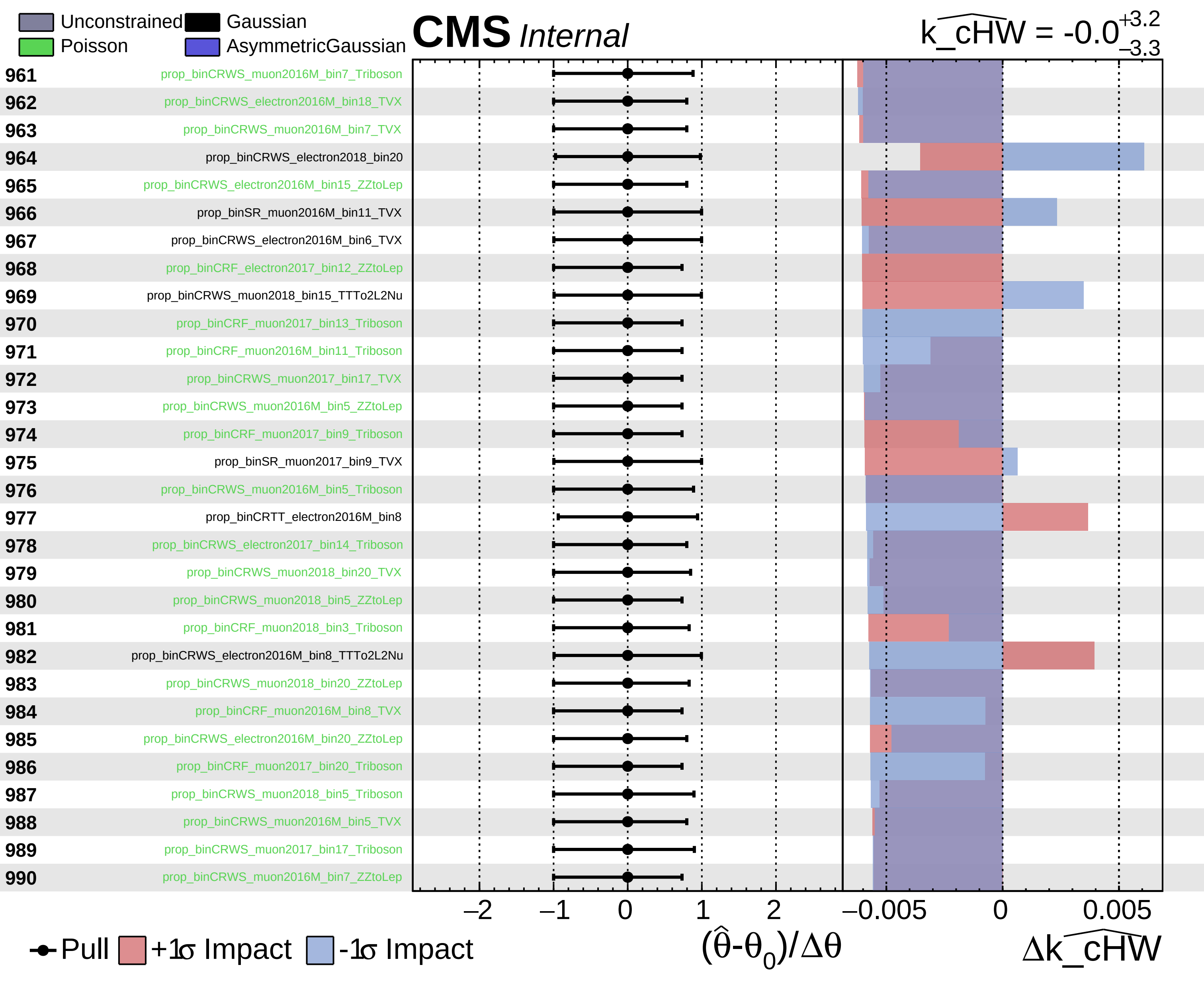
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
841 prop_binCRWS_electron2017_bin9
842 prop_binCRWS_electron2016M_bin18_Fake
843 prop_binCRF_electron2017_bin4_TVX
844 prop_binCRF_muon2016M_bin5_WrongSign
845 prop_binCRF_muon2017_bin9_TVX
846 prop_binCRF_electron2017_bin16
847 prop_binCRF_muon2018_bin16_Triboson
848 prop_binCRF_electron2016M_bin6_WpWpJJ_QCD
849 prop_binCRWS_muon2016M_bin5_TTTo2L2Nu
850 prop_binCRF_muon2016M_bin9_sm
851 prop_binCRWS_electron2016M_bin7
852 prop_binCRF_electron2018_bin8_TVX
853 prop_binCRF_electron2017_bin8_TVX
854 prop_binSR_electron2016M_bin12_Triboson
855 prop_binCRWS_electron2018_bin19_WZ
856 prop_binCRWS_muon2017_bin17_TTTo2L2Nu
857 prop_binCRWS_electron2018_bin12_ZZtoLep
858 prop_binCRWS_electron2016M_bin17_WrongSign
859 prop_binCRF_muon2018_bin5_TVX
860 prop_binCRF_electron2018_bin10_Triboson
861 prop_binCRF_electron2017_bin18_TVX
862 prop_binSR_electron2017_bin15_WrongSign
863 prop_binCRWS_electron2016M_bin15_WZ
864 prop_binSR_muon2017_bin13_Triboson
865 prop_binCRF_electron2018_bin8_WrongSign
866 prop_binCRF_muon2016M_bin13_TVX
867 prop_binCRF_electron2017_bin8_WpWpJJ_QCD
868 prop_binSR_electron2017_bin13_ZZtoLep
869 prop_binSR_muon2017_bin13_WrongSign
870 prop_binCRF_muon2016M_bin16_WpWpJJ_QCD
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

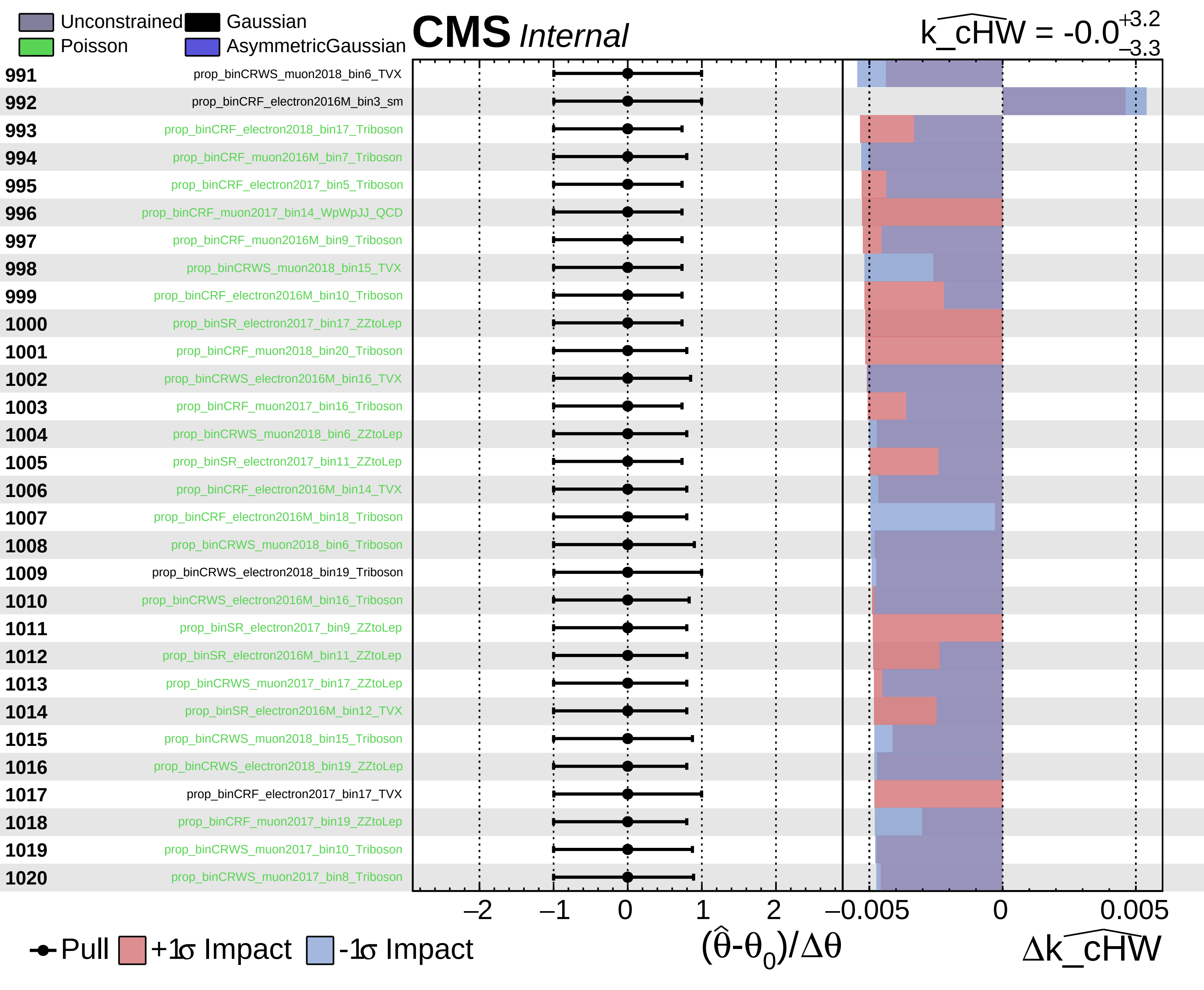


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
871 prop_binCRWS_electron2018_bin6
872 prop_binSR_muon2016M_bin20_WrongSign
873 prop_binCRF_electron2017_bin4_WpWpJJ_QCD
874 prop_binCRF_muon2017_bin7_WpWpJJ_QCD
875 prop_binCRF_muon2016M_bin6_WpWpJJ_QCD
876 prop_binCRF_electron2017_bin8_Triboson
877 prop_binCRF_muon2018_bin13_TVX
878 prop_binCRF_electron2018_bin5
879 prop_binCRF_muon2016M_bin5_TVX
880 prop_binCRWS_muon2018_bin6_WrongSign
881 prop_binCRWS_electron2018_bin19_WrongSign
882 prop_binSR_muon2016M_bin17_TVX
883 prop_binSR_electron2017_bin15_TVX
884 prop_binCRWS_muon2016M_bin5_WrongSign
885 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
886 prop_binSR_muon2017_bin13_WpWpJJ_QCD
887 prop_binCRWS_electron2016M_bin6_VG
888 prop_binCRF_electron2018_bin8_WpWpJJ_QCD
889 prop_binSR_muon2017_bin9_Triboson
890 prop_binCRWS_electron2016M_bin20_TTTo2L2Nu
891 prop_binCRF_muon2018_bin13_WpWpJJ_QCD
892 prop_binSR_electron2016M_bin17_Triboson
893 prop_binCRF_muon2016M_bin15_TVX
894 prop_binCRF_muon2016M_bin8_Triboson
895 prop_binCRWS_electron2016M_bin8_WrongSign
896 prop_binSR_electron2017_bin5_ZZtoLep
897 prop_binCRF_electron2017_bin13_WpWpJJ_QCD
898 prop_binCRF_electron2016M_bin3_TVX
899 prop_binCRF_electron2016M_bin13_TVX
900 prop_binCRWS_muon2017_bin14_TTTo2L2Nu
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW

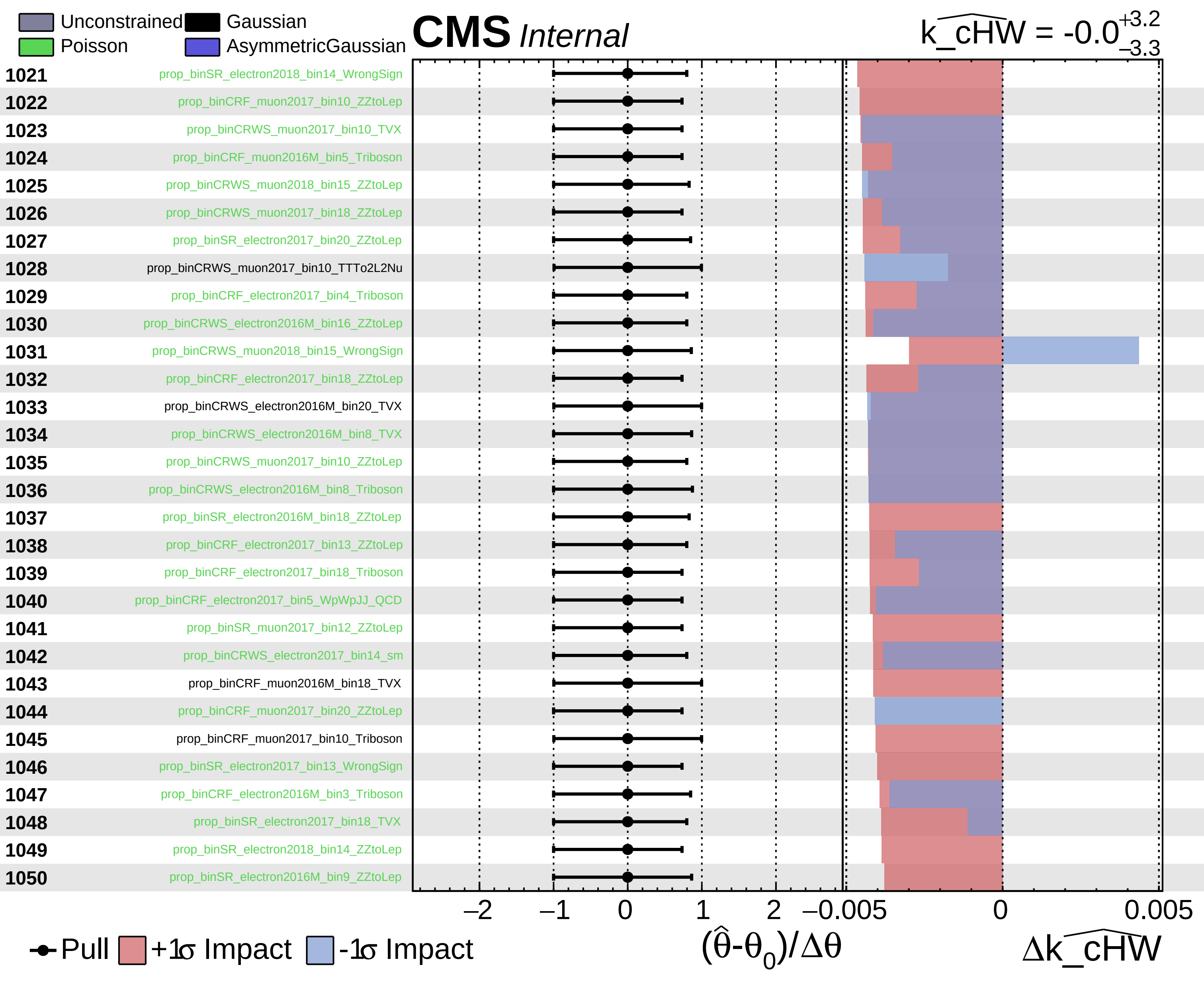

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
901 prop_binCRF_muon2016M_bin15_Triboson
902 prop_binCRF_electron2017_bin14_WpWpJJ_QCD
903 prop_binCRF_electron2017_bin7_TVX
904 prop_binCRWS_electron2017_bin14_ZZtoLep
905 prop_binCRF_electron2016M_bin10_TVX
906 prop_binCRWS_electron2018_bin12_Triboson
907 prop_binSR_muon2018_bin10_WrongSign
908 prop_binCRF_muon2018_bin16_WpWpJJ_QCD
909 prop_binCRF_muon2018_bin5_Triboson
910 prop_binCRF_muon2018_bin20_WpWpJJ_QCD
911 prop_binCRWS_muon2017_bin18_WrongSign
912 prop_binCRWS_muon2016M_bin7_WrongSign
913 prop_binCRF_muon2018_bin12_WrongSign
914 prop_binCRF_muon2016M_bin16_Triboson
915 prop_binCRWS_muon2017_bin14_WrongSign
916 prop_binCRF_electron2016M_bin20_Triboson
917 prop_binCRWS_electron2016M_bin20_Fake
918 prop_binCRWS_electron2016M_bin17_Fake
919 prop_binCRF_muon2016M_bin7_TVX
920 prop_binCRWS_muon2017_bin18_Triboson
921 prop_binCRF_electron2016M_bin14_WpWpJJ_QCD
922 prop_binCRF_electron2017_bin7_WpWpJJ_QCD
923 prop_binCRF_muon2016M_bin9_TVX
924 prop_binCRF_muon2018_bin12_TVX
925 prop_binCRF_electron2016M_bin10_WpWpJJ_QCD
926 prop_binCRF_electron2016M_bin18_WpWpJJ_QCD
927 prop_binCRF_electron2017_bin5_TVX
928 prop_binCRTT_electron2017_bin11
929 prop_binCRWS_electron2016M_bin20_Triboson
930 prop_binCRWS_electron2016M_bin17_VG
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
931 prop_binCRF_muon2016M_bin8_WpWpJJ_QCD
932 prop_binCRF_electron2016M_bin9_TVX
933 prop_binCRWS_muon2018_bin5_TVX
934 prop_binSR_electron2017_bin16_ZZtoLep
935 prop_binCRF_electron2016M_bin5_WpWpJJ_QCD
936 prop_binCRWS_electron2016M_bin16_TTTo2L2Nu
937 prop_binCRWS_electron2016M_bin17_TVX
938 prop_binCRF_muon2018_bin17_Triboson
939 prop_binCRF_electron2017_bin13_TVX
940 prop_binCRWS_electron2018_bin19_TTTo2L2Nu
941 prop_binCRWS_muon2017_bin14_TVX
942 prop_binCRTT_electron2016M_bin16
943 prop_binCRWS_muon2017_bin14_Triboson
944 prop_binCRF_muon2016M_bin7_WrongSign
945 prop_binCRWS_electron2016M_bin18_Triboson
946 prop_binCRF_muon2017_bin18_TVX
947 prop_binCRF_muon2016M_bin10_Triboson
948 prop_binCRF_electron2016M_bin3_WpWpJJ_QCD
949 prop_binCRF_muon2016M_bin7_ZZtoLep
950 prop_binCRF_muon2017_bin9_WpWpJJ_QCD
951 prop_binCRWS_electron2016M_bin17_Triboson
952 prop_binCRWS_muon2017_bin17_WrongSign
953 prop_binCRWS_electron2016M_bin17_TTTo2L2Nu
954 prop_binCRF_muon2016M_bin9_WpWpJJ_QCD
955 prop_binCRF_muon2016M_bin7_WpWpJJ_QCD
956 prop_binCRWS_muon2017_bin8_VG
957 prop_binCRWS_muon2017_bin8_TVX
958 prop_binCRWS_muon2017_bin14_ZZtoLep
959 prop_binCRWS_electron2016M_bin6_Triboson
960 prop_binCRWS_electron2016M_bin15_Triboson
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

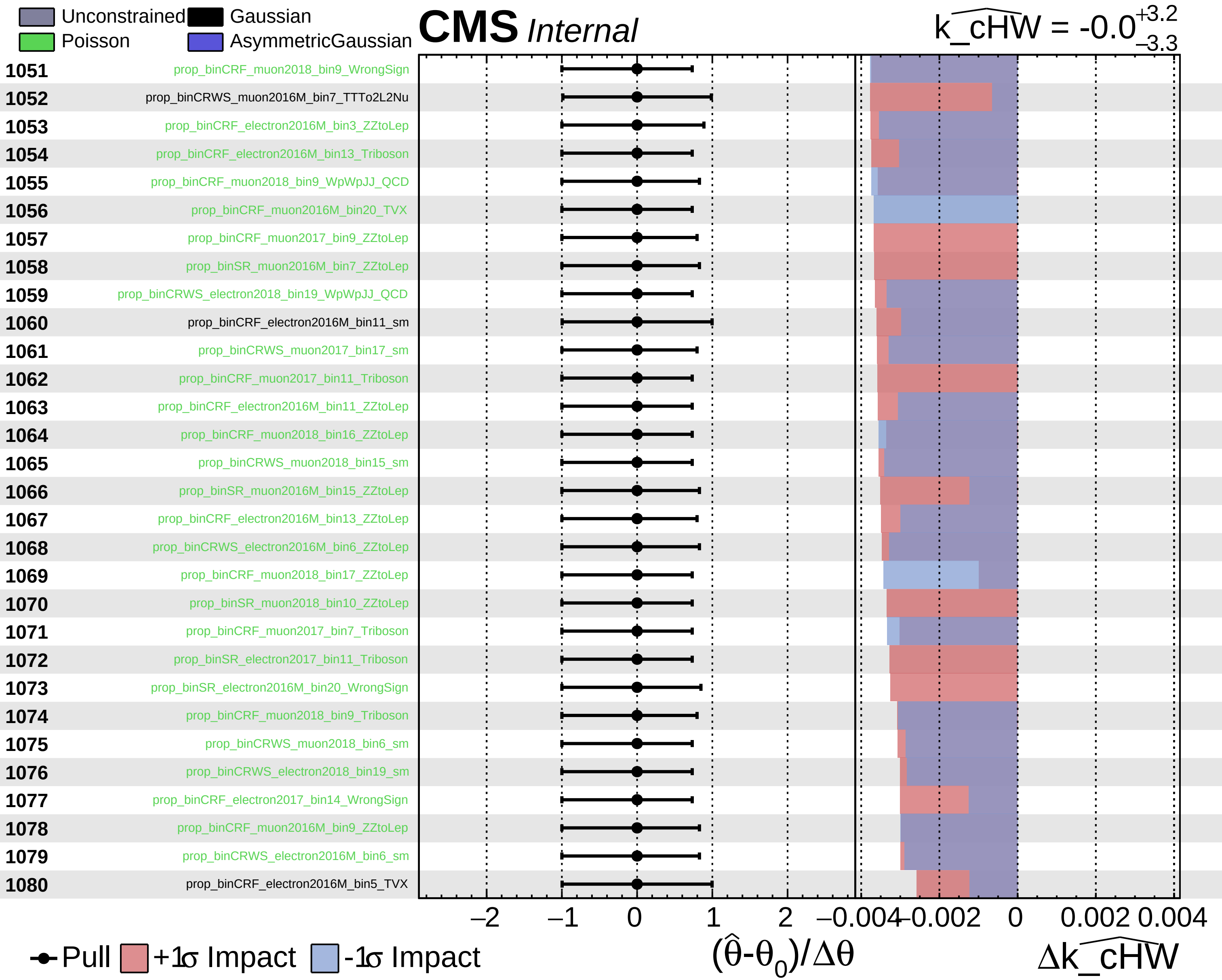

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
961 prop_binCRWS_muon2016M_bin7_Triboson
962 prop_binCRWS_electron2016M_bin18_TVX
963 prop_binCRWS_muon2016M_bin7_TVX
964 prop_binCRWS_electron2018_bin20
965 prop_binCRWS_electron2016M_bin15_ZZtoLep
966 prop_binSR_muon2016M_bin11_TVX
967 prop_binCRWS_electron2016M_bin6_TVX
968 prop_binCRF_electron2017_bin12_ZZtoLep
969 prop_binCRWS_muon2018_bin15_TTTo2L2Nu
970 prop_binCRF_muon2017_bin13_Triboson
971 prop_binCRF_muon2016M_bin11_Triboson
972 prop_binCRWS_muon2017_bin17_TVX
973 prop_binCRWS_muon2016M_bin5_ZZtoLep
974 prop_binCRF_muon2017_bin9_Triboson
975 prop_binSR_muon2017_bin9_TVX
976 prop_binCRWS_muon2016M_bin5_Triboson
977 prop_binCRTT_electron2016M_bin8
978 prop_binCRWS_electron2017_bin14_Triboson
979 prop_binCRWS_muon2018_bin20_TVX
980 prop_binCRWS_muon2018_bin5_ZZtoLep
981 prop_binCRF_muon2018_bin3_Triboson
982 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
983 prop_binCRWS_muon2018_bin20_ZZtoLep
984 prop_binCRF_muon2016M_bin8_TVX
985 prop_binCRWS_electron2016M_bin20_ZZtoLep
986 prop_binCRF_muon2017_bin20_Triboson
987 prop_binCRWS_muon2018_bin5_Triboson
988 prop_binCRWS_muon2016M_bin5_TVX
989 prop_binCRWS_muon2017_bin17_Triboson
990 prop_binCRWS_muon2016M_bin7_ZZtoLep
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



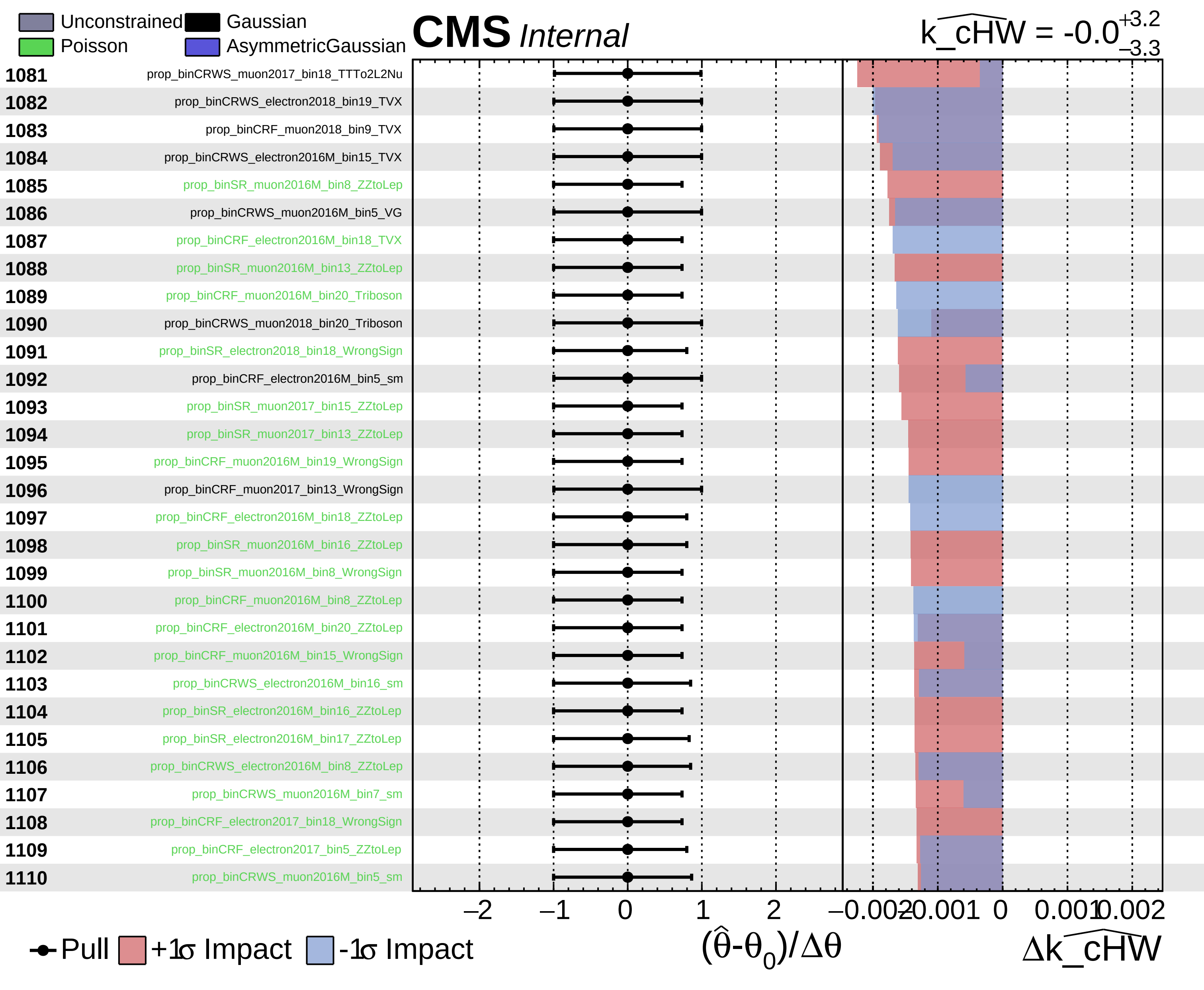
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
991 prop_binCRWS_muon2018_bin6_TVX
992 prop_binCRF_electron2016M_bin3_sm
993 prop_binCRF_electron2018_bin17_Triboson
994 prop_binCRF_muon2016M_bin7_Triboson
995 prop_binCRF_electron2017_bin5_Triboson
996 prop_binCRF_muon2017_bin14_WpWpJJ_QCD
997 prop_binCRF_muon2016M_bin9_Triboson
998 prop_binCRWS_muon2018_bin15_TVX
999 prop_binCRF_electron2016M_bin10_Triboson
1000 prop_binSR_electron2017_bin17_ZZtoLep
1001 prop_binCRF_muon2018_bin20_Triboson
1002 prop_binCRWS_electron2016M_bin16_TVX
1003 prop_binCRF_muon2017_bin16_Triboson
1004 prop_binCRWS_muon2018_bin6_ZZtoLep
1005 prop_binSR_electron2017_bin11_ZZtoLep
1006 prop_binCRF_electron2016M_bin14_TVX
1007 prop_binCRF_electron2016M_bin18_Triboson
1008 prop_binCRWS_muon2018_bin6_Triboson
1009 prop_binCRWS_electron2018_bin19_Triboson
1010 prop_binCRWS_electron2016M_bin16_Triboson
1011 prop_binSR_electron2017_bin9_ZZtoLep
1012 prop_binSR_electron2016M_bin11_ZZtoLep
1013 prop_binCRWS_muon2017_bin17_ZZtoLep
1014 prop_binSR_electron2016M_bin12_TVX
1015 prop_binCRWS_muon2018_bin15_Triboson
1016 prop_binCRWS_electron2018_bin19_ZZtoLep
1017 prop_binCRF_electron2017_bin17_TVX
1018 prop_binCRF_muon2017_bin19_ZZtoLep
1019 prop_binCRWS_muon2017_bin10_Triboson
1020 prop_binCRWS_muon2017_bin8_Triboson
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



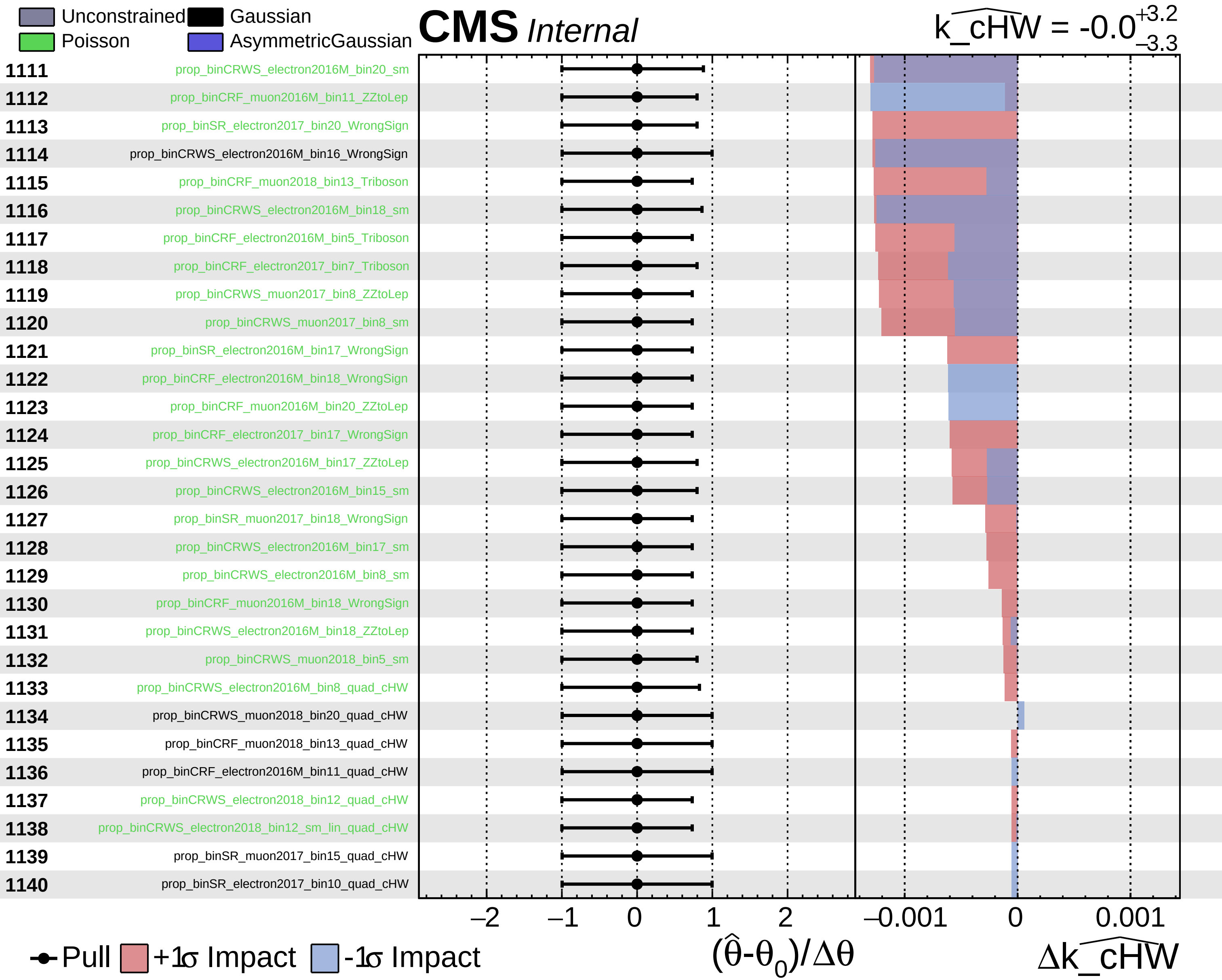
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
1021 prop_binSR_electron2018_bin14_WrongSign
1022 prop_binCRF_muon2017_bin10_ZZtoLep
1023 prop_binCRWS_muon2017_bin10_TVX
1024 prop_binCRF_muon2016M_bin5_Triboson
1025 prop_binCRWS_muon2018_bin15_ZZtoLep
1026 prop_binCRWS_muon2017_bin18_ZZtoLep
1027 prop_binSR_electron2017_bin20_ZZtoLep
1028 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
1029 prop_binCRF_electron2017_bin4_Triboson
1030 prop_binCRWS_electron2016M_bin16_ZZtoLep
1031 prop_binCRWS_muon2018_bin15_WrongSign
1032 prop_binCRF_electron2017_bin18_ZZtoLep
1033 prop_binCRWS_electron2016M_bin20_TVX
1034 prop_binCRWS_electron2016M_bin8_TVX
1035 prop_binCRWS_muon2017_bin10_ZZtoLep
1036 prop_binCRWS_electron2016M_bin8_Triboson
1037 prop_binSR_electron2016M_bin18_ZZtoLep
1038 prop_binCRF_electron2017_bin13_ZZtoLep
1039 prop_binCRF_electron2017_bin18_Triboson
1040 prop_binCRF_electron2017_bin5_WpWpJJ_QCD
1041 prop_binSR_muon2017_bin12_ZZtoLep
1042 prop_binCRWS_electron2017_bin14_sm
1043 prop_binCRF_muon2016M_bin18_TVX
1044 prop_binCRF_muon2017_bin20_ZZtoLep
1045 prop_binCRF_muon2017_bin10_Triboson
1046 prop_binSR_electron2017_bin13_WrongSign
1047 prop_binCRF_electron2016M_bin3_Triboson
1048 prop_binSR_electron2017_bin18_TVX
1049 prop_binSR_electron2018_bin14_ZZtoLep
1050 prop_binSR_electron2016M_bin9_ZZtoLep
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

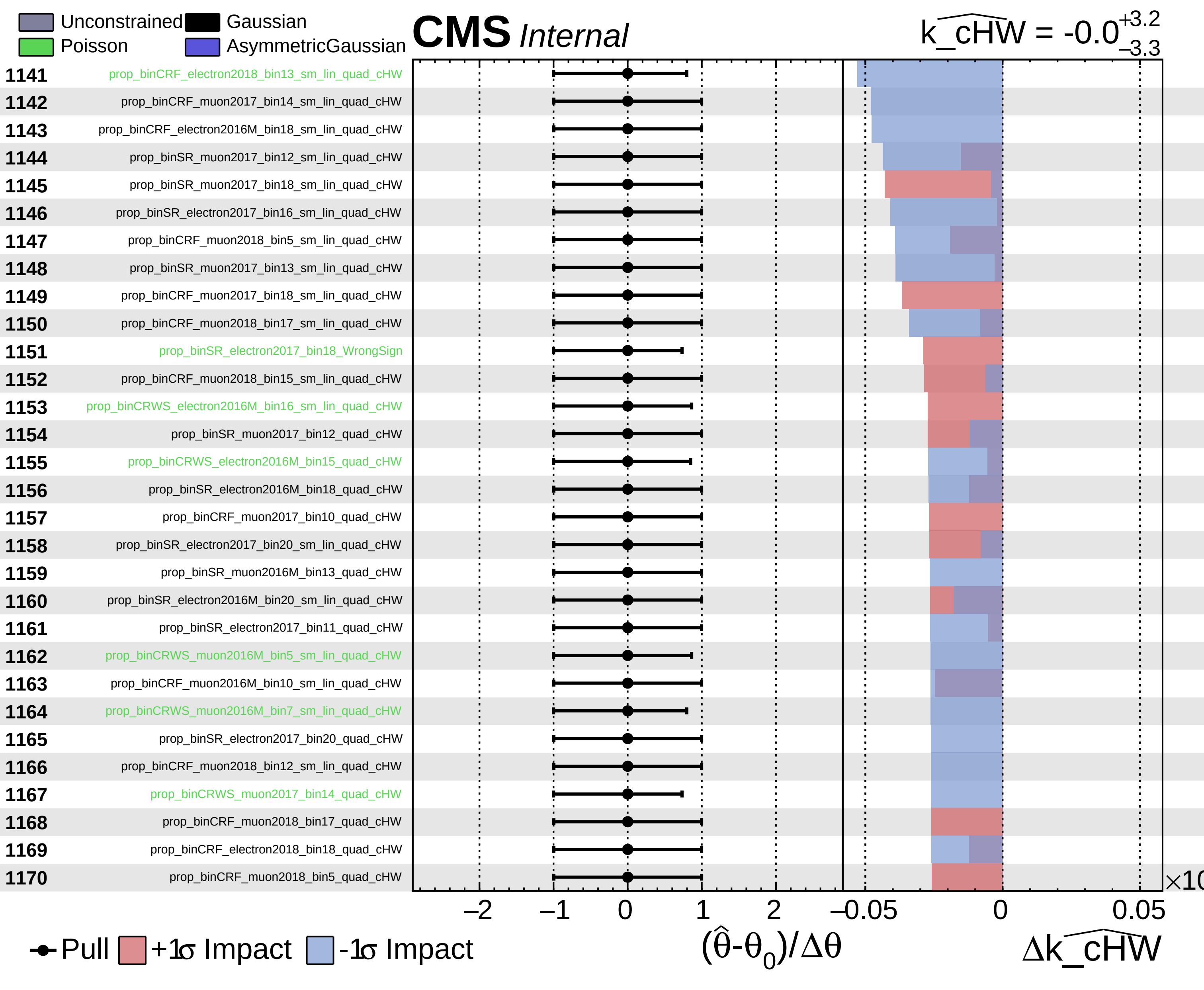


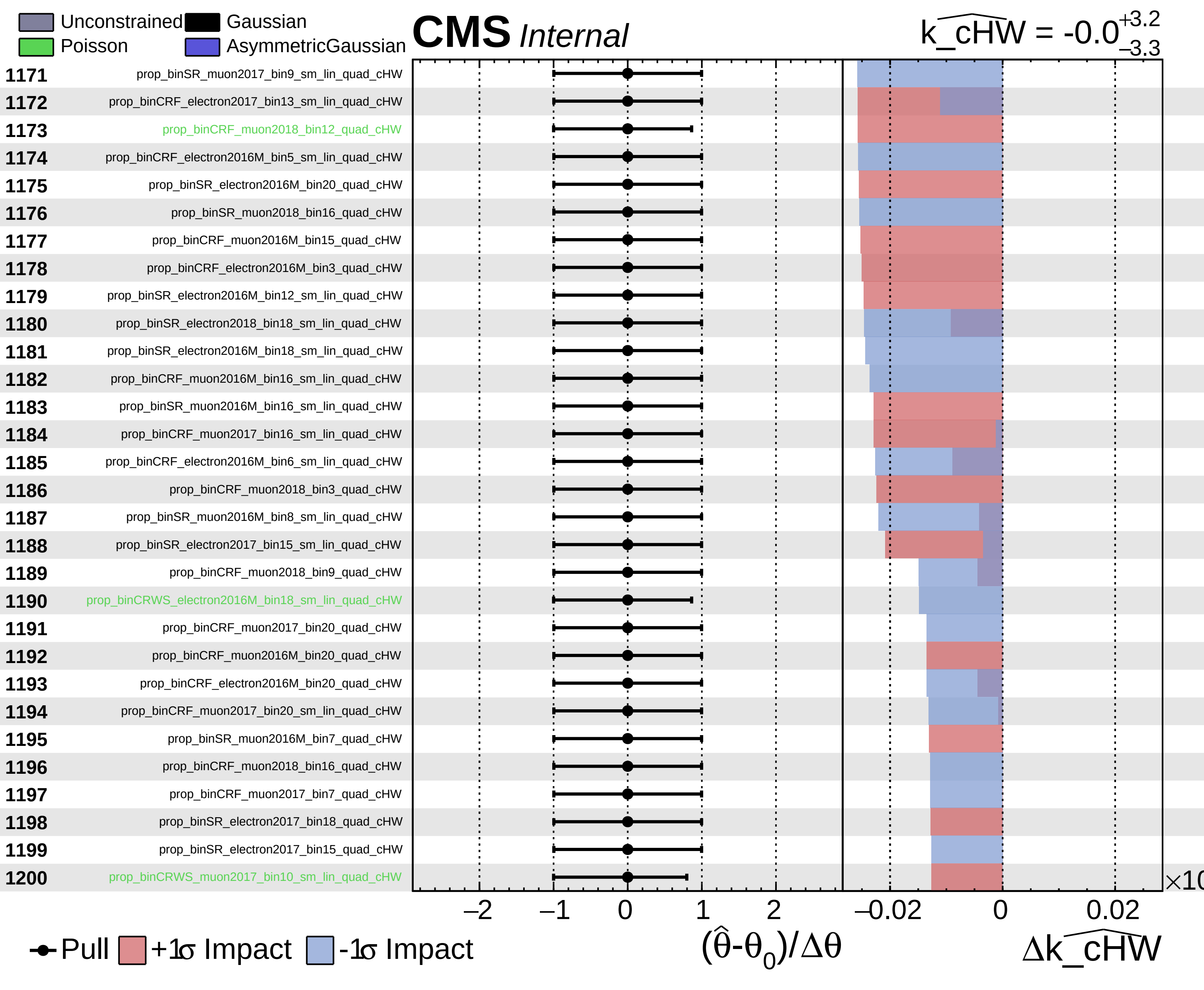
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0$^{+3.2}_{-3.3}$
1051 prop_binCRF_muon2018_bin9_WrongSign
1052 prop_binCRWS_muon2016M_bin7_TTTo2L2Nu
1053 prop_binCRF_electron2016M_bin3_ZZtoLep
1054 prop_binCRF_electron2016M_bin13_Triboson
1055 prop_binCRF_muon2018_bin9_WpWpJJ_QCD
1056 prop_binCRF_muon2016M_bin20_TVX
1057 prop_binCRF_muon2017_bin9_ZZtoLep
1058 prop_binSR_muon2016M_bin7_ZZtoLep
1059 prop_binCRWS_electron2018_bin19_WpWpJJ_QCD
1060 prop_binCRF_electron2016M_bin11_sm
1061 prop_binCRWS_muon2017_bin17_sm
1062 prop_binCRF_muon2017_bin11_Triboson
1063 prop_binCRF_electron2016M_bin11_ZZtoLep
1064 prop_binCRF_muon2018_bin16_ZZtoLep
1065 prop_binCRWS_muon2018_bin15_sm
1066 prop_binSR_muon2016M_bin15_ZZtoLep
1067 prop_binCRF_electron2016M_bin13_ZZtoLep
1068 prop_binCRWS_electron2016M_bin6_ZZtoLep
1069 prop_binCRF_muon2018_bin17_ZZtoLep
1070 prop_binSR_muon2018_bin10_ZZtoLep
1071 prop_binCRF_muon2017_bin7_Triboson
1072 prop_binSR_electron2017_bin11_Triboson
1073 prop_binSR_electron2016M_bin20_WrongSign
1074 prop_binCRF_muon2018_bin9_Triboson
1075 prop_binCRWS_muon2018_bin6_sm
1076 prop_binCRWS_electron2018_bin19_sm
1077 prop_binCRF_electron2017_bin14_WrongSign
1078 prop_binCRF_muon2016M_bin9_ZZtoLep
1079 prop_binCRWS_electron2016M_bin6_sm
1080 prop_binCRF_electron2016M_bin5_TVX
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW

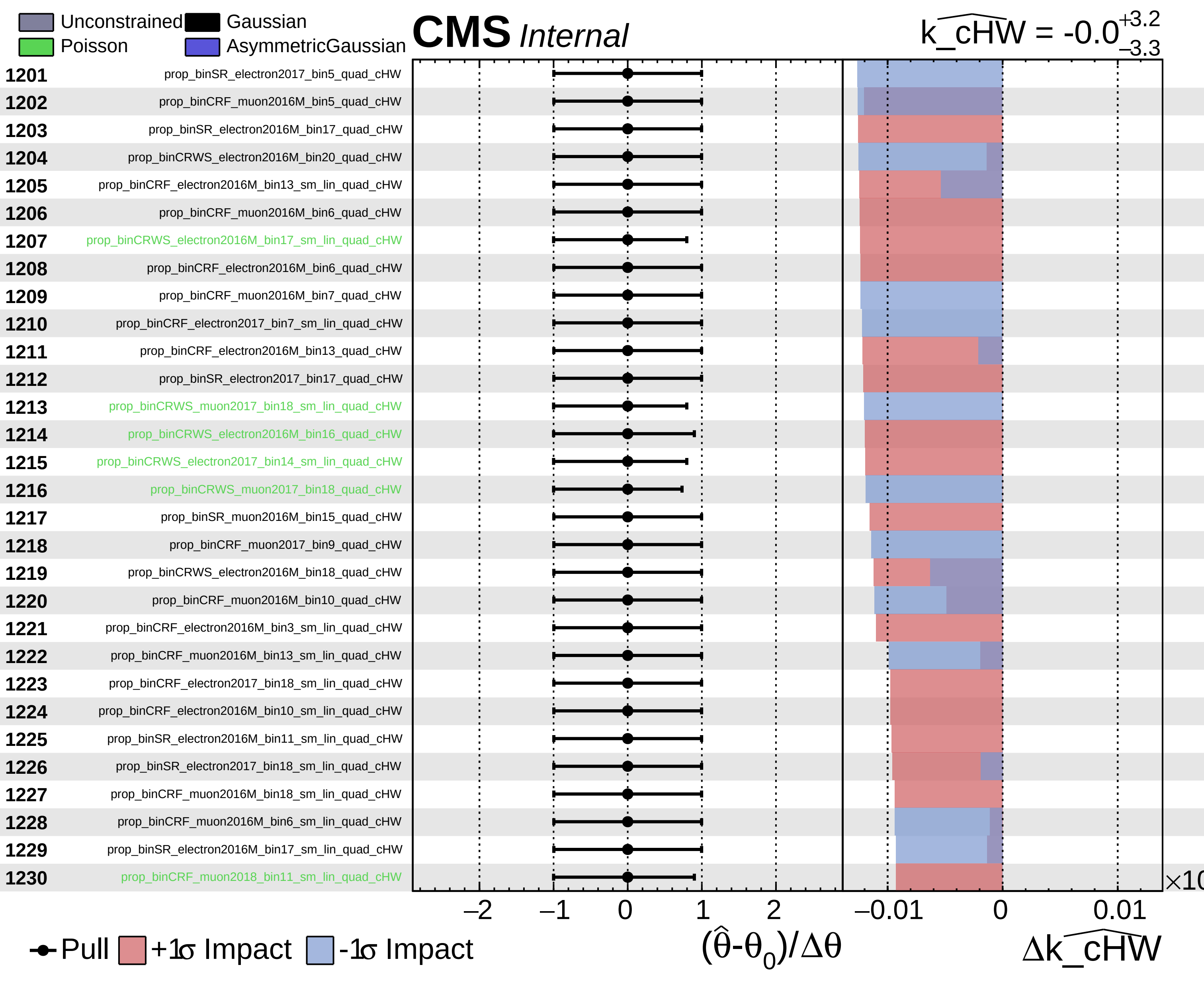


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
1081 prop_binCRWS_muon2017_bin18_TTTo2L2Nu
1082 prop_binCRWS_electron2018_bin19_TVX
1083 prop_binCRF_muon2018_bin9_TVX
1084 prop_binCRWS_electron2016M_bin15_TVX
1085 prop_binSR_muon2016M_bin8_ZZtoLep
1086 prop_binCRWS_muon2016M_bin5_VG
1087 prop_binCRF_electron2016M_bin18_TVX
1088 prop_binSR_muon2016M_bin13_ZZtoLep
1089 prop_binCRF_muon2016M_bin20_Triboson
1090 prop_binCRWS_muon2018_bin20_Triboson
1091 prop_binSR_electron2018_bin18_WrongSign
1092 prop_binCRF_electron2016M_bin5_sm
1093 prop_binSR_muon2017_bin15_ZZtoLep
1094 prop_binSR_muon2017_bin13_ZZtoLep
1095 prop_binCRF_muon2016M_bin19_WrongSign
1096 prop_binCRF_muon2017_bin13_WrongSign
1097 prop_binCRF_electron2016M_bin18_ZZtoLep
1098 prop_binSR_muon2016M_bin16_ZZtoLep
1099 prop_binSR_muon2016M_bin8_WrongSign
1100 prop_binCRF_muon2016M_bin8_ZZtoLep
1101 prop_binCRF_electron2016M_bin20_ZZtoLep
1102 prop_binCRF_muon2016M_bin15_WrongSign
1103 prop_binCRWS_electron2016M_bin16_sm
1104 prop_binSR_electron2016M_bin16_ZZtoLep
1105 prop_binSR_electron2016M_bin17_ZZtoLep
1106 prop_binCRWS_electron2016M_bin8_ZZtoLep
1107 prop_binCRWS_muon2016M_bin7_sm
1108 prop_binCRF_electron2017_bin18_WrongSign
1109 prop_binCRF_electron2017_bin5_ZZtoLep
1110 prop_binCRWS_muon2016M_bin5_sm
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
1111 prop_binCRWS_electron2016M_bin20_sm
1112 prop_binCRF_muon2016M_bin11_ZZtoLep
1113 prop_binSR_electron2017_bin20_WrongSign
1114 prop_binCRWS_electron2016M_bin16_WrongSign
1115 prop_binCRF_muon2018_bin13_Triboson
1116 prop_binCRWS_electron2016M_bin18_sm
1117 prop_binCRF_electron2016M_bin5_Triboson
1118 prop_binCRF_electron2017_bin7_Triboson
1119 prop_binCRWS_muon2017_bin8_ZZtoLep
1120 prop_binCRWS_muon2017_bin8_sm
1121 prop_binSR_electron2016M_bin17_WrongSign
1122 prop_binCRF_electron2016M_bin18_WrongSign
1123 prop_binCRF_muon2016M_bin20_ZZtoLep
1124 prop_binCRF_electron2017_bin17_WrongSign
1125 prop_binCRWS_electron2016M_bin17_ZZtoLep
1126 prop_binCRWS_electron2016M_bin15_sm
1127 prop_binSR_muon2017_bin18_WrongSign
1128 prop_binCRWS_electron2016M_bin17_sm
1129 prop_binCRWS_electron2016M_bin8_sm
1130 prop_binCRF_muon2016M_bin18_WrongSign
1131 prop_binCRWS_electron2016M_bin18_ZZtoLep
1132 prop_binCRWS_muon2018_bin5_sm
1133 prop_binCRWS_electron2016M_bin8_quad_cHW
1134 prop_binCRWS_muon2018_bin20_quad_cHW
1135 prop_binCRF_muon2018_bin13_quad_cHW
1136 prop_binCRF_electron2016M_bin11_quad_cHW
1137 prop_binCRWS_electron2018_bin12_quad_cHW
1138 prop_binCRWS_electron2018_bin12_sm_lin_quad_cHW
1139 prop_binSR_muon2017_bin15_quad_cHW
1140 prop_binSR_electron2017_bin10_quad_cHW
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
1141 prop_binCRF_electron2018_bin13_sm_lin_quad_cHW
1142 prop_binCRF_muon2017_bin14_sm_lin_quad_cHW
1143 prop_binCRF_electron2016M_bin18_sm_lin_quad_cHW
1144 prop_binSR_muon2017_bin12_sm_lin_quad_cHW
1145 prop_binSR_muon2017_bin18_sm_lin_quad_cHW
1146 prop_binSR_electron2017_bin16_sm_lin_quad_cHW
1147 prop_binCRF_muon2018_bin5_sm_lin_quad_cHW
1148 prop_binSR_muon2017_bin13_sm_lin_quad_cHW
1149 prop_binCRF_muon2017_bin18_sm_lin_quad_cHW
1150 prop_binCRF_muon2018_bin17_sm_lin_quad_cHW
1151 prop_binSR_electron2017_bin18_WrongSign
1152 prop_binCRF_muon2018_bin15_sm_lin_quad_cHW
1153 prop_binCRWS_electron2016M_bin16_sm_lin_quad_cHW
1154 prop_binSR_muon2017_bin12_quad_cHW
1155 prop_binCRWS_electron2016M_bin15_quad_cHW
1156 prop_binSR_electron2016M_bin18_quad_cHW
1157 prop_binCRF_muon2017_bin10_quad_cHW
1158 prop_binSR_electron2017_bin20_sm_lin_quad_cHW
1159 prop_binSR_muon2016M_bin13_quad_cHW
1160 prop_binSR_electron2016M_bin20_sm_lin_quad_cHW
1161 prop_binSR_electron2017_bin11_quad_cHW
1162 prop_binCRWS_muon2016M_bin5_sm_lin_quad_cHW
1163 prop_binCRF_muon2016M_bin10_sm_lin_quad_cHW
1164 prop_binCRWS_muon2016M_bin7_sm_lin_quad_cHW
1165 prop_binSR_electron2017_bin20_quad_cHW
1166 prop_binCRF_muon2018_bin12_sm_lin_quad_cHW
1167 prop_binCRWS_muon2017_bin14_quad_cHW
1168 prop_binCRF_muon2018_bin17_quad_cHW
1169 prop_binCRF_electron2018_bin18_quad_cHW
1170 prop_binCRF_muon2018_bin5_quad_cHW
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
1171 prop_binSR_muon2017_bin9_sm_lin_quad_cHW
1172 prop_binCRF_electron2017_bin13_sm_lin_quad_cHW
1173 prop_binCRF_muon2018_bin12_quad_cHW
1174 prop_binCRF_electron2016M_bin5_sm_lin_quad_cHW
1175 prop_binSR_electron2016M_bin20_quad_cHW
1176 prop_binSR_muon2018_bin16_quad_cHW
1177 prop_binCRF_muon2016M_bin15_quad_cHW
1178 prop_binCRF_electron2016M_bin3_quad_cHW
1179 prop_binSR_electron2016M_bin12_sm_lin_quad_cHW
1180 prop_binSR_electron2018_bin18_sm_lin_quad_cHW
1181 prop_binSR_electron2016M_bin18_sm_lin_quad_cHW
1182 prop_binCRF_muon2016M_bin16_sm_lin_quad_cHW
1183 prop_binSR_muon2016M_bin16_sm_lin_quad_cHW
1184 prop_binCRF_muon2017_bin16_sm_lin_quad_cHW
1185 prop_binCRF_electron2016M_bin6_sm_lin_quad_cHW
1186 prop_binCRF_muon2018_bin3_quad_cHW
1187 prop_binSR_muon2016M_bin8_sm_lin_quad_cHW
1188 prop_binSR_electron2017_bin15_sm_lin_quad_cHW
1189 prop_binCRF_muon2018_bin9_quad_cHW
1190 prop_binCRWS_electron2016M_bin18_sm_lin_quad_cHW
1191 prop_binCRF_muon2017_bin20_quad_cHW
1192 prop_binCRF_muon2016M_bin20_quad_cHW
1193 prop_binCRF_electron2016M_bin20_quad_cHW
1194 prop_binCRF_muon2017_bin20_sm_lin_quad_cHW
1195 prop_binSR_muon2016M_bin7_quad_cHW
1196 prop_binCRF_muon2018_bin16_quad_cHW
1197 prop_binCRF_muon2017_bin7_quad_cHW
1198 prop_binSR_electron2017_bin18_quad_cHW
1199 prop_binSR_electron2017_bin15_quad_cHW
1200 prop_binCRWS_muon2017_bin10_sm_lin_quad_cHW
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
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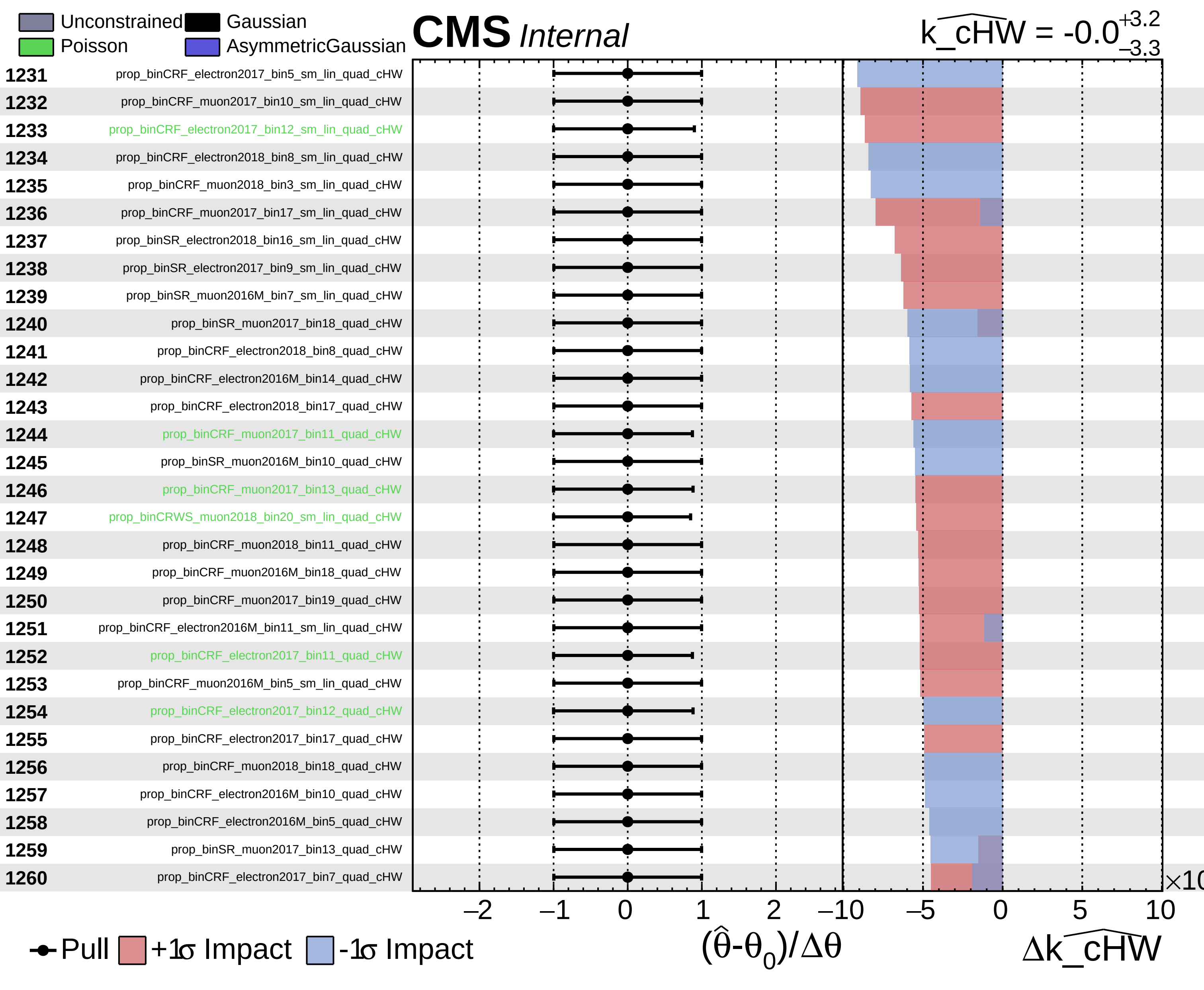
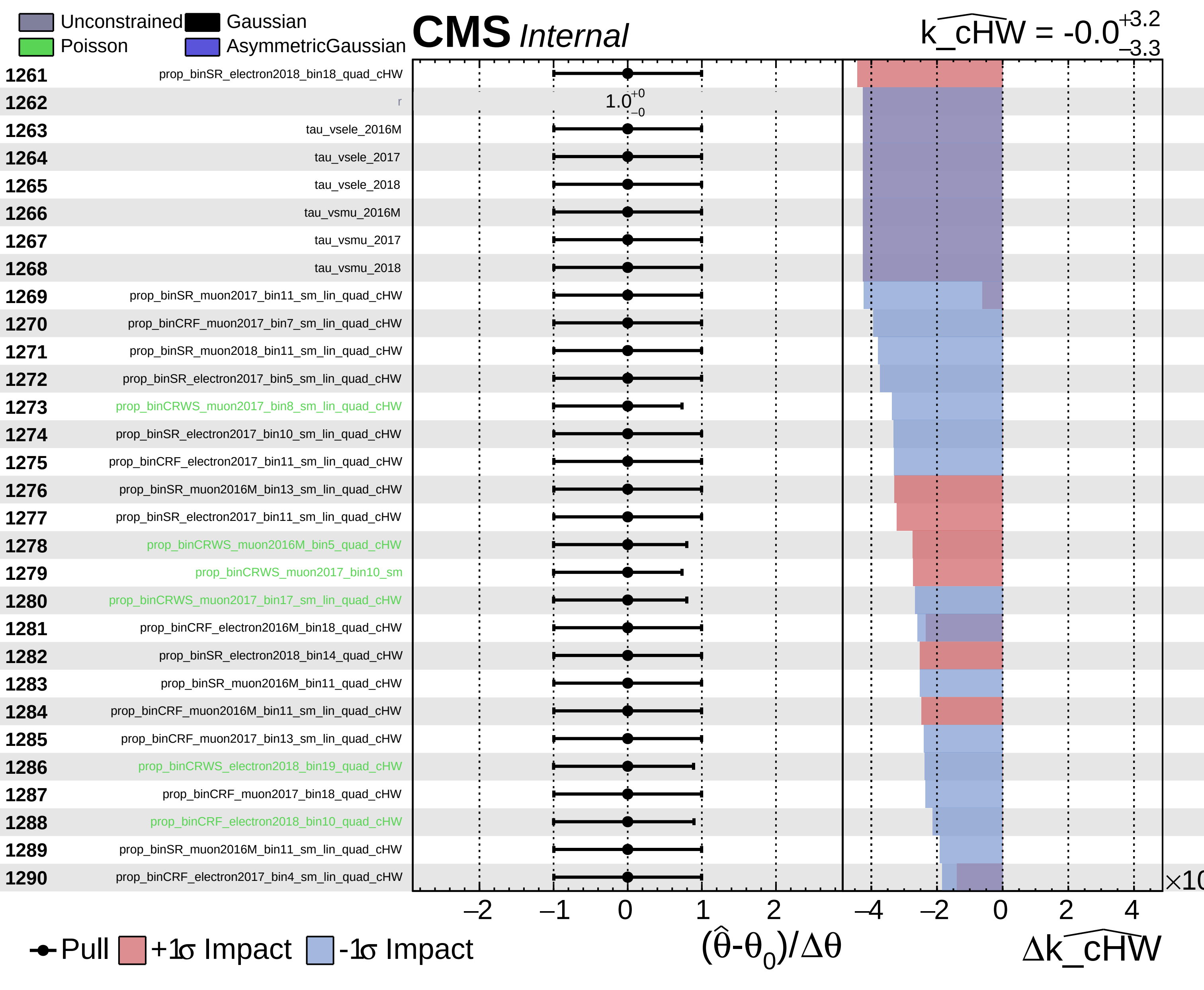
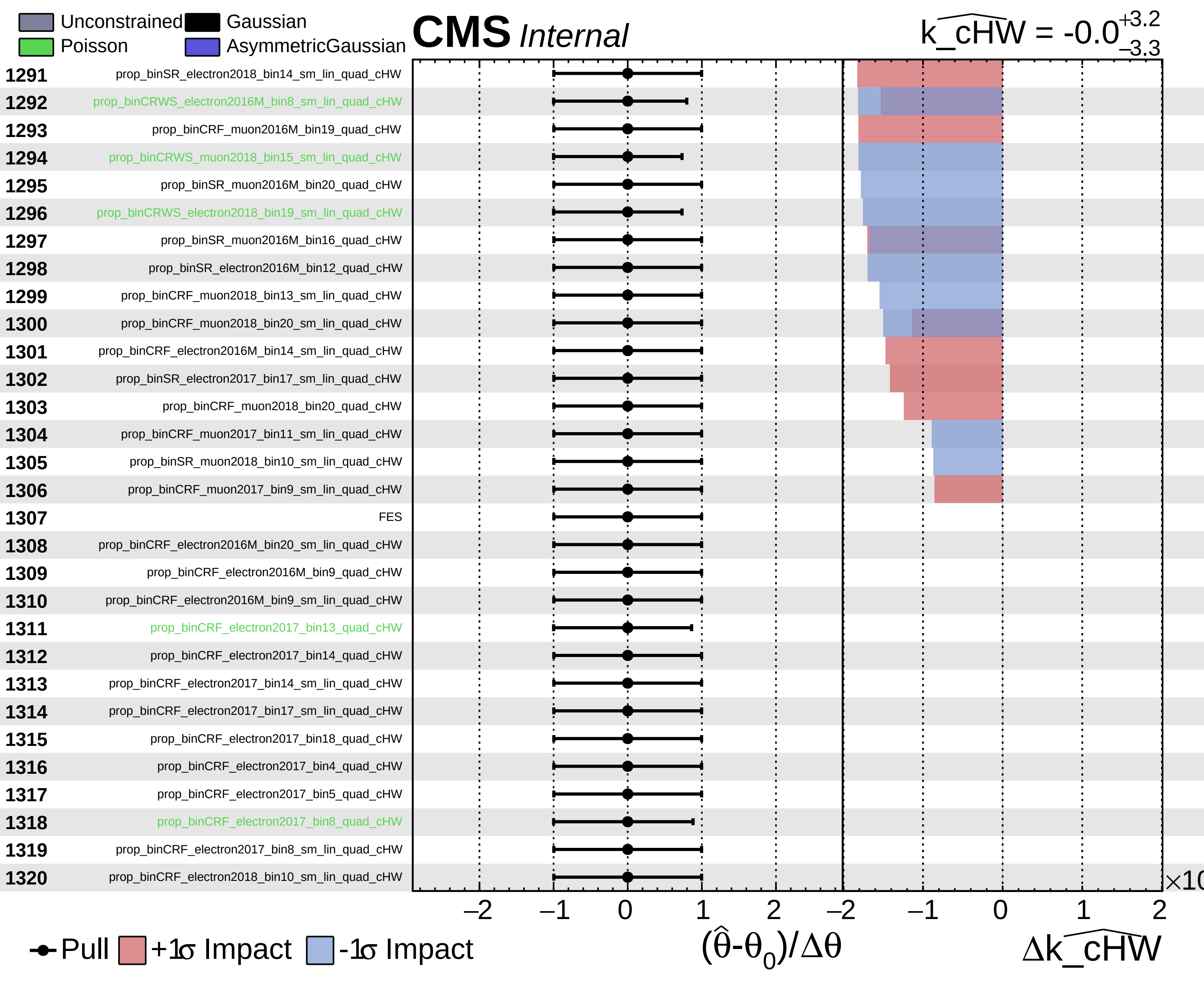
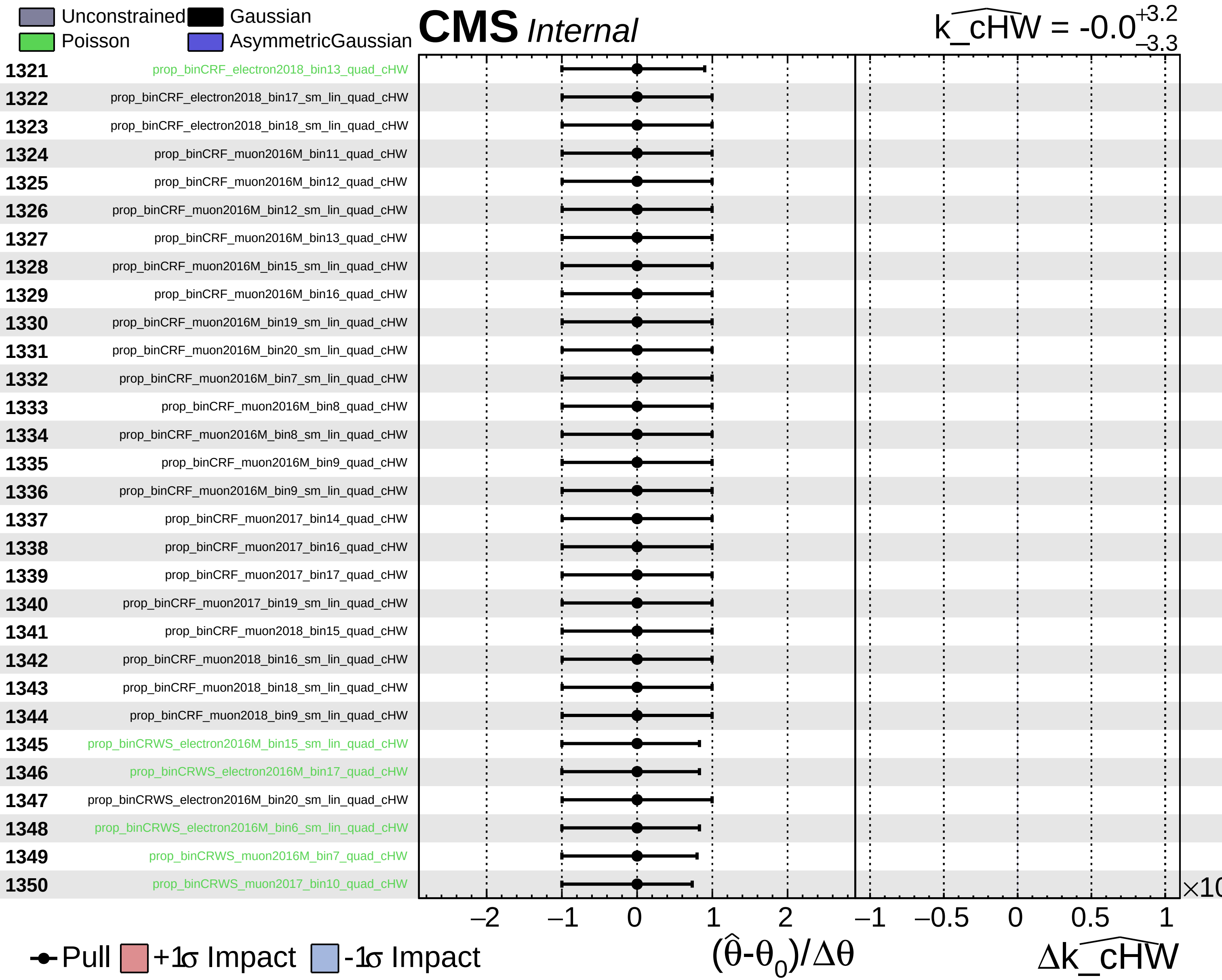
1202 prop_binCRF_muon2016M_bin5_quad_cHW
1203 prop_binSR_electron2016M_bin17_quad_cHW
1204 prop_binCRWS_electron2016M_bin20_quad_cHW
1205 prop_binCRF_electron2016M_bin13_sm_lin_quad_cHW
1206 prop_binCRF_muon2016M_bin6_quad_cHW
1207 prop_binCRWS_electron2016M_bin17_sm_lin_quad_cHW
1208 prop_binCRF_electron2016M_bin6_quad_cHW
1209 prop_binCRF_muon2016M_bin7_quad_cHW
1210 prop_binCRF_electron2017_bin7_sm_lin_quad_cHW
1211 prop_binCRF_electron2016M_bin13_quad_cHW
1212 prop_binSR_electron2017_bin17_quad_cHW
1213 prop_binCRWS_muon2017_bin18_sm_lin_quad_cHW
1214 prop_binCRWS_electron2016M_bin16_quad_cHW
1215 prop_binCRWS_electron2017_bin14_sm_lin_quad_cHW
1216 prop_binCRWS_muon2017_bin18_quad_cHW
1217 prop_binSR_muon2016M_bin15_quad_cHW
1218 prop_binCRF_muon2017_bin9_quad_cHW
1219 prop_binCRWS_electron2016M_bin18_quad_cHW
1220 prop_binCRF_muon2016M_bin10_quad_cHW
1221 prop_binCRF_electron2016M_bin3_sm_lin_quad_cHW
1222 prop_binCRF_muon2016M_bin13_sm_lin_quad_cHW
1223 prop_binCRF_electron2017_bin18_sm_lin_quad_cHW
1224 prop_binCRF_electron2016M_bin10_sm_lin_quad_cHW
1225 prop_binSR_electron2016M_bin11_sm_lin_quad_cHW
1226 prop_binSR_electron2017_bin18_sm_lin_quad_cHW
1227 prop_binCRF_muon2016M_bin18_sm_lin_quad_cHW
1228 prop_binCRF_muon2016M_bin6_sm_lin_quad_cHW
1229 prop_binSR_electron2016M_bin17_sm_lin_quad_cHW
1230 prop_binCRF_muon2018_bin11_sm_lin_quad_cHW
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
1231 prop_binCRF_electron2017_bin5_sm_lin_quad_cHW
1232 prop_binCRF_muon2017_bin10_sm_lin_quad_cHW
1233 prop_binCRF_electron2017_bin12_sm_lin_quad_cHW
1234 prop_binCRF_electron2018_bin8_sm_lin_quad_cHW
1235 prop_binCRF_muon2018_bin3_sm_lin_quad_cHW
1236 prop_binCRF_muon2017_bin17_sm_lin_quad_cHW
1237 prop_binSR_electron2018_bin16_sm_lin_quad_cHW
1238 prop_binSR_electron2017_bin9_sm_lin_quad_cHW
1239 prop_binSR_muon2016M_bin7_sm_lin_quad_cHW
1240 prop_binSR_muon2017_bin18_quad_cHW
1241 prop_binCRF_electron2018_bin8_quad_cHW
1242 prop_binCRF_electron2016M_bin14_quad_cHW
1243 prop_binCRF_electron2018_bin17_quad_cHW
1244 prop_binCRF_muon2017_bin11_quad_cHW
1245 prop_binSR_muon2016M_bin10_quad_cHW
1246 prop_binCRF_muon2017_bin13_quad_cHW
1247 prop_binCRWS_muon2018_bin20_sm_lin_quad_cHW
1248 prop_binCRF_muon2018_bin11_quad_cHW
1249 prop_binCRF_muon2016M_bin18_quad_cHW
1250 prop_binCRF_muon2017_bin19_quad_cHW
1251 prop_binCRF_electron2016M_bin11_sm_lin_quad_cHW
1252 prop_binCRF_electron2017_bin11_quad_cHW
1253 prop_binCRF_muon2016M_bin5_sm_lin_quad_cHW
1254 prop_binCRF_electron2017_bin12_quad_cHW
1255 prop_binCRF_electron2017_bin17_quad_cHW
1256 prop_binCRF_muon2018_bin18_quad_cHW
1257 prop_binCRF_electron2016M_bin10_quad_cHW
1258 prop_binCRF_electron2016M_bin5_quad_cHW
1259 prop_binSR_muon2017_bin13_quad_cHW
1260 prop_binCRF_electron2017_bin7_quad_cHW
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

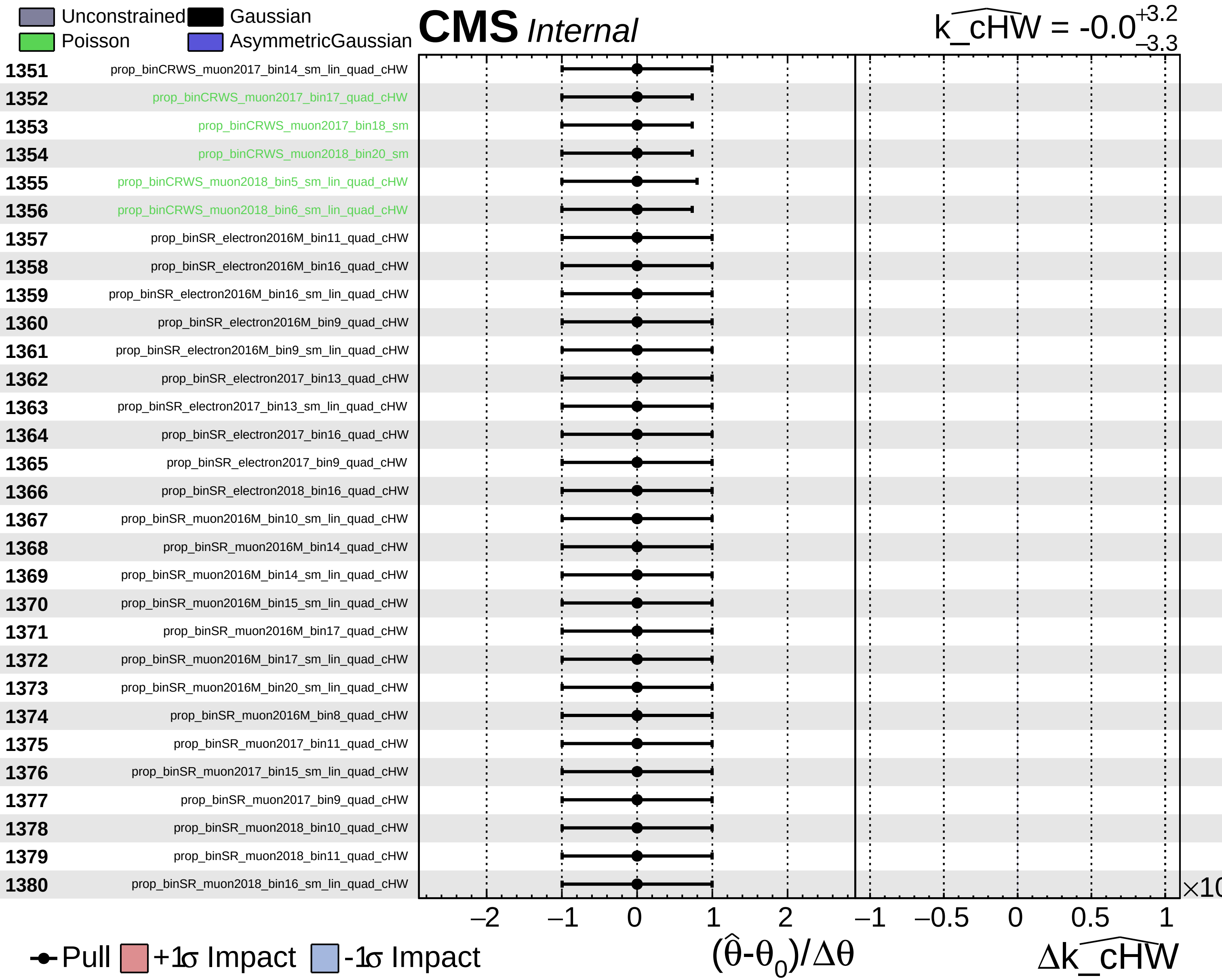

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
1261 prop_binSR_electron2018_bin18_quad_cHW
1262 r
1.0 +0 −0
1263 tau_vsele_2016M
1264 tau_vsele_2017
1265 tau_vsele_2018
1266 tau_vsmu_2016M
1267 tau_vsmu_2017
1268 tau_vsmu_2018
1269 prop_binSR_muon2017_bin11_sm_lin_quad_cHW
1270 prop_binCRF_muon2017_bin7_sm_lin_quad_cHW
1271 prop_binSR_muon2018_bin11_sm_lin_quad_cHW
1272 prop_binSR_electron2017_bin5_sm_lin_quad_cHW
1273 prop_binCRWS_muon2017_bin8_sm_lin_quad_cHW
1274 prop_binSR_electron2017_bin10_sm_lin_quad_cHW
1275 prop_binCRF_electron2017_bin11_sm_lin_quad_cHW
1276 prop_binSR_muon2016M_bin13_sm_lin_quad_cHW
1277 prop_binSR_electron2017_bin11_sm_lin_quad_cHW
1278 prop_binCRWS_muon2016M_bin5_quad_cHW
1279 prop_binCRWS_muon2017_bin10_sm
1280 prop_binCRWS_muon2017_bin17_sm_lin_quad_cHW
1281 prop_binCRF_electron2016M_bin18_quad_cHW
1282 prop_binSR_electron2018_bin14_quad_cHW
1283 prop_binSR_muon2016M_bin11_quad_cHW
1284 prop_binCRF_muon2016M_bin11_sm_lin_quad_cHW
1285 prop_binCRF_muon2017_bin13_sm_lin_quad_cHW
1286 prop_binCRWS_electron2018_bin19_quad_cHW
1287 prop_binCRF_muon2017_bin18_quad_cHW
1288 prop_binCRF_electron2018_bin10_quad_cHW
1289 prop_binSR_muon2016M_bin11_sm_lin_quad_cHW
1290 prop_binCRF_electron2017_bin4_sm_lin_quad_cHW
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
1291 prop_binSR_electron2018_bin14_sm_lin_quad_cHW
1292 prop_binCRWS_electron2016M_bin8_sm_lin_quad_cHW
1293 prop_binCRF_muon2016M_bin19_quad_cHW
1294 prop_binCRWS_muon2018_bin15_sm_lin_quad_cHW
1295 prop_binSR_muon2016M_bin20_quad_cHW
1296 prop_binCRWS_electron2018_bin19_sm_lin_quad_cHW
1297 prop_binSR_muon2016M_bin16_quad_cHW
1298 prop_binSR_electron2016M_bin12_quad_cHW
1299 prop_binCRF_muon2018_bin13_sm_lin_quad_cHW
1300 prop_binCRF_muon2018_bin20_sm_lin_quad_cHW
1301 prop_binCRF_electron2016M_bin14_sm_lin_quad_cHW
1302 prop_binSR_electron2017_bin17_sm_lin_quad_cHW
1303 prop_binCRF_muon2018_bin20_quad_cHW
1304 prop_binCRF_muon2017_bin11_sm_lin_quad_cHW
1305 prop_binSR_muon2018_bin10_sm_lin_quad_cHW
1306 prop_binCRF_muon2017_bin9_sm_lin_quad_cHW
1307 FES
1308 prop_binCRF_electron2016M_bin20_sm_lin_quad_cHW
1309 prop_binCRF_electron2016M_bin9_quad_cHW
1310 prop_binCRF_electron2016M_bin9_sm_lin_quad_cHW
1311 prop_binCRF_electron2017_bin13_quad_cHW
1312 prop_binCRF_electron2017_bin14_quad_cHW
1313 prop_binCRF_electron2017_bin14_sm_lin_quad_cHW
1314 prop_binCRF_electron2017_bin17_sm_lin_quad_cHW
1315 prop_binCRF_electron2017_bin18_quad_cHW
1316 prop_binCRF_electron2017_bin4_quad_cHW
1317 prop_binCRF_electron2017_bin5_quad_cHW
1318 prop_binCRF_electron2017_bin8_quad_cHW
1319 prop_binCRF_electron2017_bin8_sm_lin_quad_cHW
1320 prop_binCRF_electron2018_bin10_sm_lin_quad_cHW
−2 −1 0 1 2
−2 −1 0 1 2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = -0.0^{+3.2}_{-3.3}$
1321 prop_binCRF_electron2018_bin13_quad_cHW
1322 prop_binCRF_electron2018_bin17_sm_lin_quad_cHW
1323 prop_binCRF_electron2018_bin18_sm_lin_quad_cHW
1324 prop_binCRF_muon2016M_bin11_quad_cHW
1325 prop_binCRF_muon2016M_bin12_quad_cHW
1326 prop_binCRF_muon2016M_bin12_sm_lin_quad_cHW
1327 prop_binCRF_muon2016M_bin13_quad_cHW
1328 prop_binCRF_muon2016M_bin15_sm_lin_quad_cHW
1329 prop_binCRF_muon2016M_bin16_quad_cHW
1330 prop_binCRF_muon2016M_bin19_sm_lin_quad_cHW
1331 prop_binCRF_muon2016M_bin20_sm_lin_quad_cHW
1332 prop_binCRF_muon2016M_bin7_sm_lin_quad_cHW
1333 prop_binCRF_muon2016M_bin8_quad_cHW
1334 prop_binCRF_muon2016M_bin8_sm_lin_quad_cHW
1335 prop_binCRF_muon2016M_bin9_quad_cHW
1336 prop_binCRF_muon2016M_bin9_sm_lin_quad_cHW
1337 prop_binCRF_muon2017_bin14_quad_cHW
1338 prop_binCRF_muon2017_bin16_quad_cHW
1339 prop_binCRF_muon2017_bin17_quad_cHW
1340 prop_binCRF_muon2017_bin19_sm_lin_quad_cHW
1341 prop_binCRF_muon2018_bin15_quad_cHW
1342 prop_binCRF_muon2018_bin16_sm_lin_quad_cHW
1343 prop_binCRF_muon2018_bin18_sm_lin_quad_cHW
1344 prop_binCRF_muon2018_bin9_sm_lin_quad_cHW
1345 prop_binCRWS_electron2016M_bin15_sm_lin_quad_cHW
1346 prop_binCRWS_electron2016M_bin17_quad_cHW
1347 prop_binCRWS_electron2016M_bin20_sm_lin_quad_cHW
1348 prop_binCRWS_electron2016M_bin6_sm_lin_quad_cHW
1349 prop_binCRWS_muon2016M_bin7_quad_cHW
1350 prop_binCRWS_muon2017_bin10_quad_cHW
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = -0.0 +3.2 −3.3
1351 prop_binCRWS_muon2017_bin14_sm_lin_quad_cHW
1352 prop_binCRWS_muon2017_bin17_quad_cHW
1353 prop_binCRWS_muon2017_bin18_sm
1354 prop_binCRWS_muon2018_bin20_sm
1355 prop_binCRWS_muon2018_bin5_sm_lin_quad_cHW
1356 prop_binCRWS_muon2018_bin6_sm_lin_quad_cHW
1357 prop_binSR_electron2016M_bin11_quad_cHW
1358 prop_binSR_electron2016M_bin16_quad_cHW
1359 prop_binSR_electron2016M_bin16_sm_lin_quad_cHW
1360 prop_binSR_electron2016M_bin9_quad_cHW
1361 prop_binSR_electron2016M_bin9_sm_lin_quad_cHW
1362 prop_binSR_electron2017_bin13_quad_cHW
1363 prop_binSR_electron2017_bin13_sm_lin_quad_cHW
1364 prop_binSR_electron2017_bin16_quad_cHW
1365 prop_binSR_electron2017_bin9_quad_cHW
1366 prop_binSR_electron2018_bin16_quad_cHW
1367 prop_binSR_muon2016M_bin10_sm_lin_quad_cHW
1368 prop_binSR_muon2016M_bin14_quad_cHW
1369 prop_binSR_muon2016M_bin14_sm_lin_quad_cHW
1370 prop_binSR_muon2016M_bin15_sm_lin_quad_cHW
1371 prop_binSR_muon2016M_bin17_quad_cHW
1372 prop_binSR_muon2016M_bin17_sm_lin_quad_cHW
1373 prop_binSR_muon2016M_bin20_sm_lin_quad_cHW
1374 prop_binSR_muon2016M_bin8_quad_cHW
1375 prop_binSR_muon2017_bin11_quad_cHW
1376 prop_binSR_muon2017_bin15_sm_lin_quad_cHW
1377 prop_binSR_muon2017_bin9_quad_cHW
1378 prop_binSR_muon2018_bin10_quad_cHW
1379 prop_binSR_muon2018_bin11_quad_cHW
1380 prop_binSR_muon2018_bin16_sm_lin_quad_cHW
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$